

Proposal

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Painting of the clipper ship Thermopylae at sea, Aberdeen behind. From Hook, Line, Sink documentary of fishing industry in Aberdeen: **'insistent, clamorous, that is the voice of Aberdeen whose harvests are from the sea'**

Aberdeen

Preface

Acknowledgements

Aberdeen is at a crossroads. For decades following the discovery of North Sea oil, Aberdeen bucked the trend set by other Northern and Scottish cities of falling into economic malaise following the phasing out of their secondary sectors. Oil is a finite resource and the economic buoyancy provided by the oil fields in the North Sea lie at a third of their peak in the 90s. How can Aberdeen transition out of oil reliance in the next 50 years and how can the built fabric of the city support its inhabitants in this period of transition?

This masterplan is the result of a collaborative effort, and we extend our deepest gratitude to all who contributed to its development. The final product has been made possible owing to the knowledge and insight from the following:

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To be read in conjunction with:
Volume 1 - Analysis
Volume 2 - Process
Volume 3 - Proposal



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- Flooding
- Cultural Deficit

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- Decommissioning of the harbour
- Building end of life

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- City centre, re-wilding
- Transport
- Energy Network
- Locale building retrofit
- Reusing materials from site clearing

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- Golf course biodiversity
- Surface water flooding interventions
- Sponge Streets
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- Traffic calming and shared streets
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Phase 4 - Densification

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- City wide interventions
- Landscape phase 2
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Phase 5 - Sustained Growth

- Barrage
- University Research Cluster

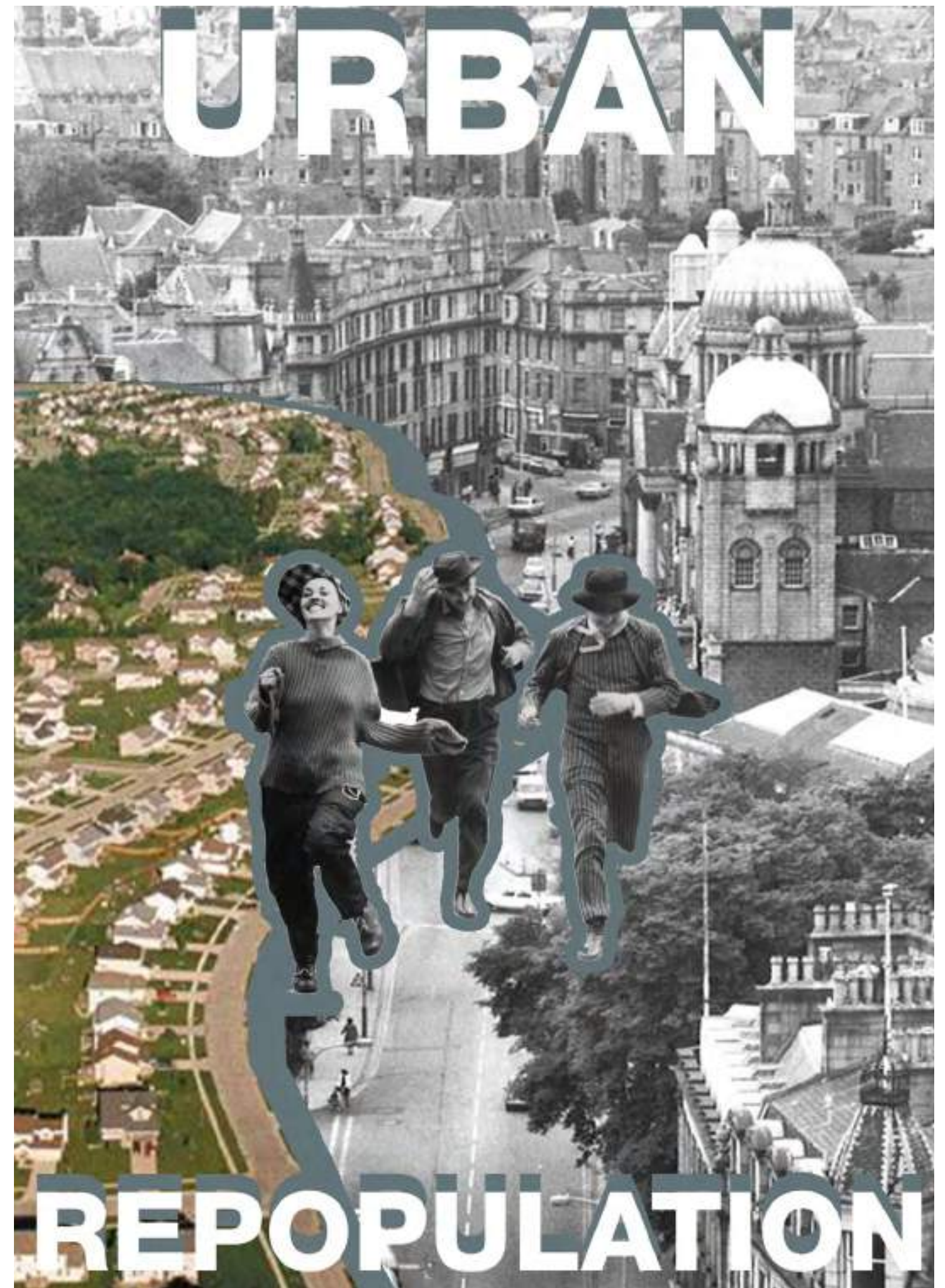
Intr

roduction and Key Issues

Urban Repopulation

Key Issues

Although Aberdeen is less affected by the housing crisis, its slow increase in population and lack of density has meant the city currently lives as a 'friendly ghost-town'. Containing urban sprawl caused by migration to cheaper housing in the Shire outside Aberdeen City, proposing an urban realm which entices migration population (90% of population growth, students) to stay, are therefore what we have identified as one of our key issues.





1925 student dwellings
2695 single/couple dwellings
3080 dwellings for families



1 new secondary school
7 new primary schools



8.23km re-activated waterfront



1 new GP practise



40sqm green space per
person

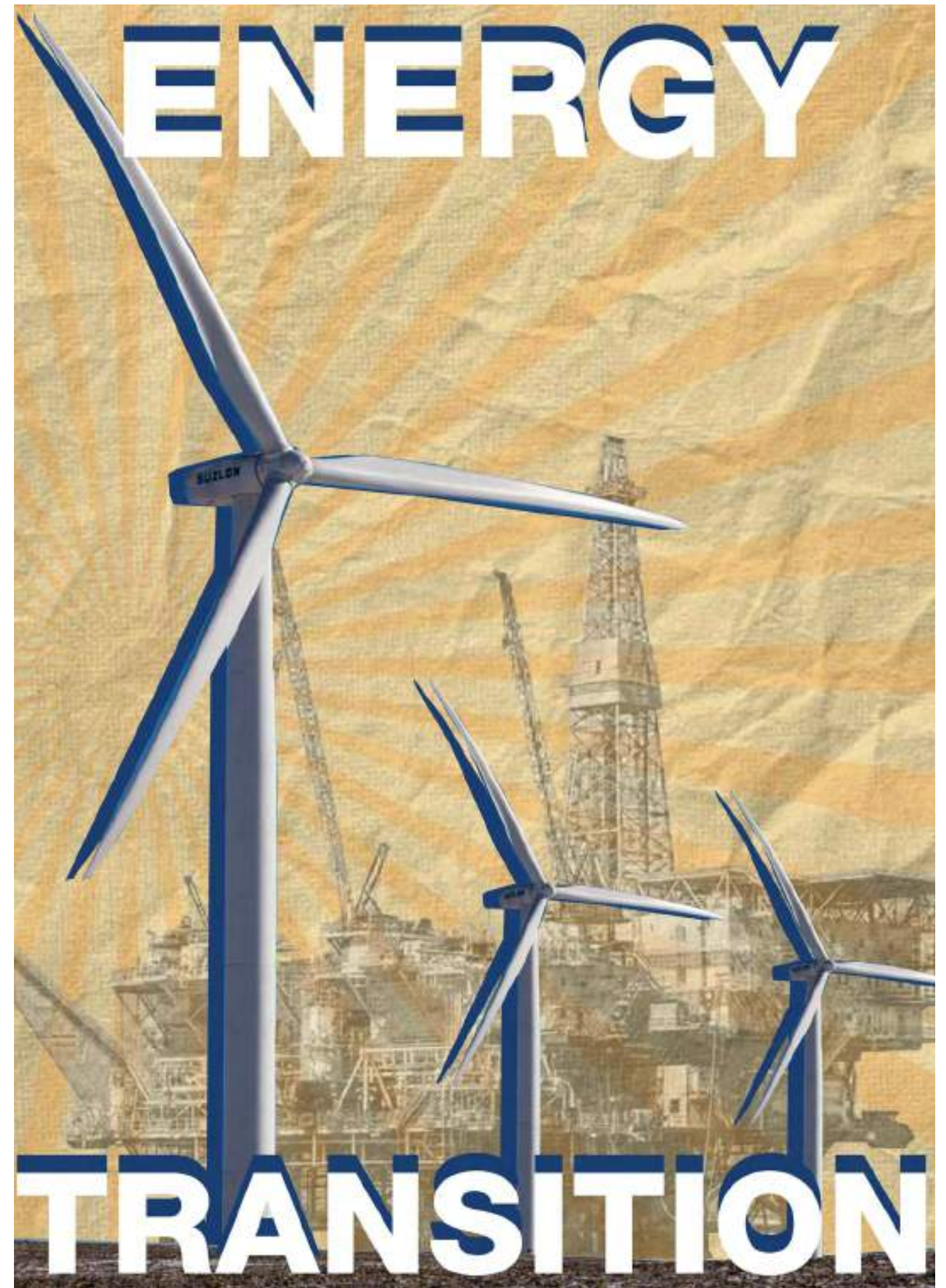


2 new joint campus

Energy Transition

Key Issues

Aberdeen's history in extractive economy, both land and sea, mean that the Energy Transition will have a massive impact on environmental, social and economic sectors. In our masterplan we would like to address this newest upcoming and developing identity as a green energy centre for the UK.





No private vehicles and extensive public transport prevents potential increase of 8000



Public transport moving to 100% hydrogen buses and increased tram network further reduces congestion



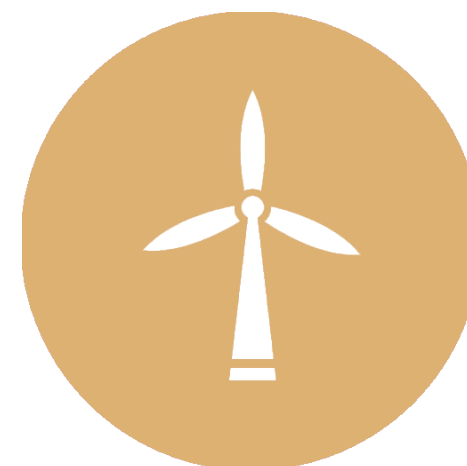
Existing housing stock: 162 kWh/yr per m² (SAP rating band C)



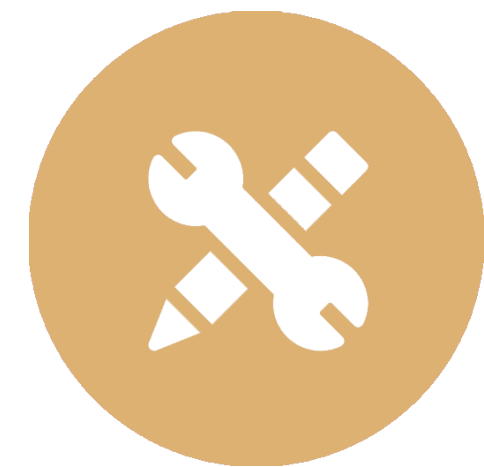
Proposed housing stock: 120 kWh/yr per m² (Passivhaus req)



National grid avg= 30% natural gas



Grid upgrades and transition on-site = 100% renewable (Wind and Hydrogen store)



- 1x skills transition hub
- 80 new developments offering potential employment



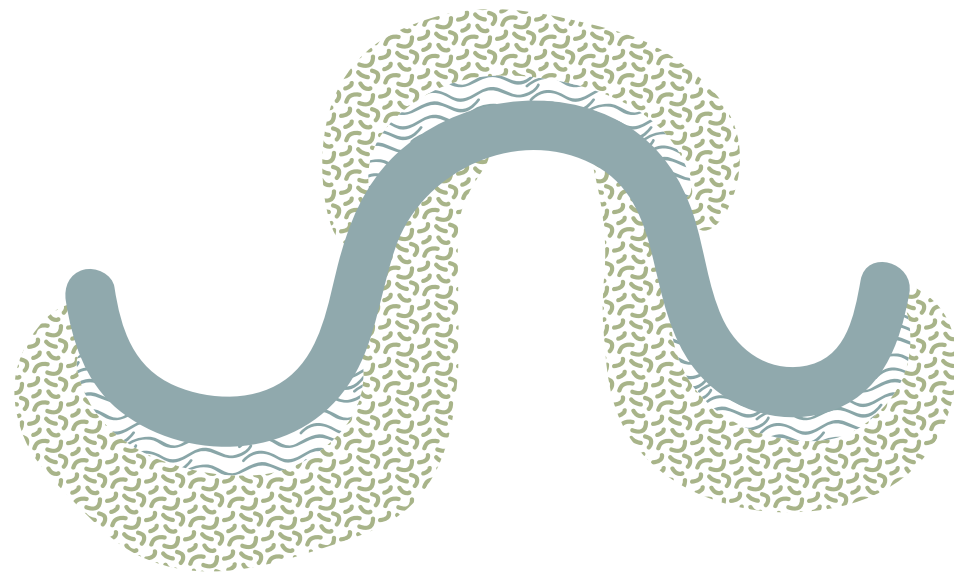
- Provision for increase in working population of 6740 to boost economy

Flooding

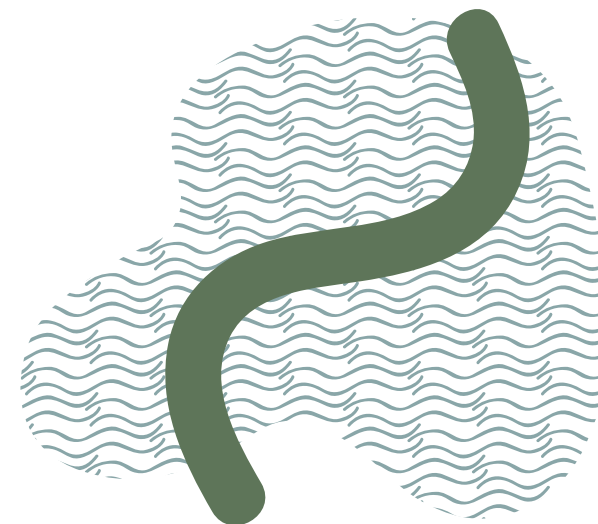
Key Issues

As a coastal port city, we would like to explore strategies for the protection against flood events as currently the sea wall and beach provide the only protection from the coastline. We have paid special attention to the Grade II listed fishing village Fittie at the tip of the port as key cultural heritage that should be protected.

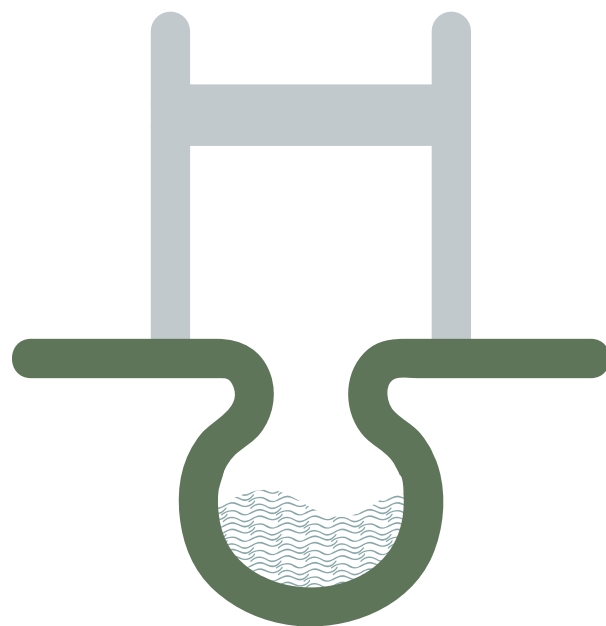




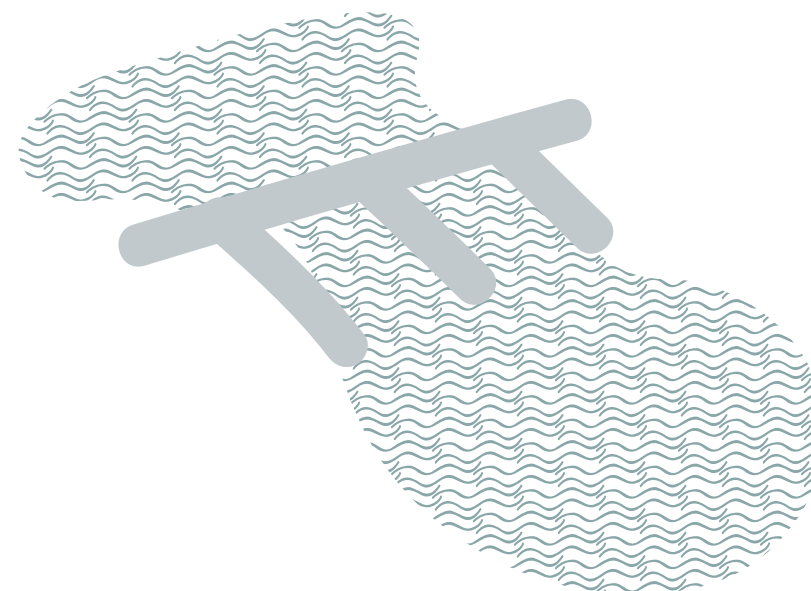
Sponge City



Reintroduce water meadows & peatland



Cloud burst tunnels

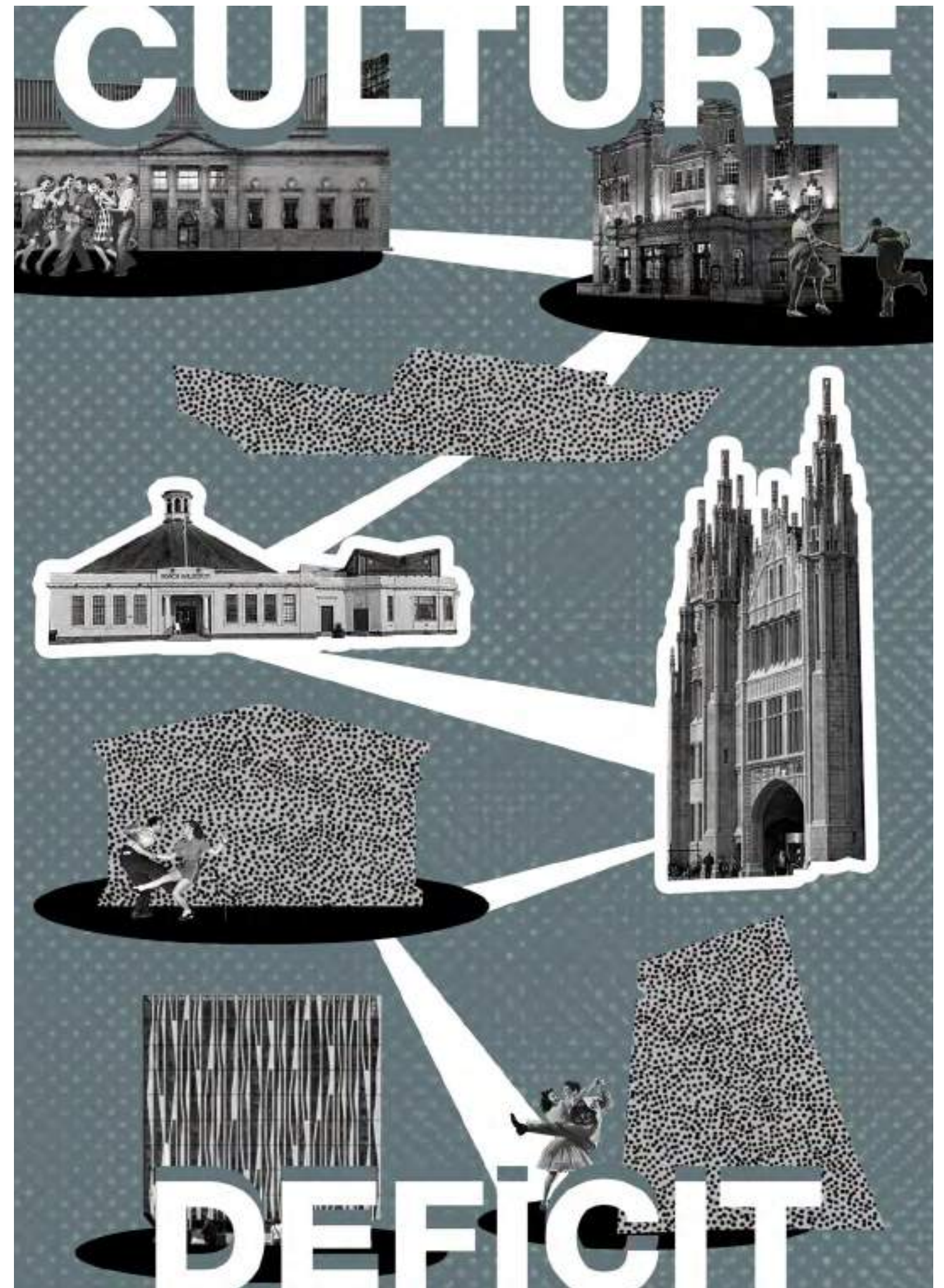


Proposed barrage

Cultural Deficit

Key Issues

The constant flow of the city's economy meant that cultural assets continually changed hands because of industry, a turnover of people, leading to a lack of identity. Memory rich buildings such as the Ballroom or Cordona's Fairground remain unloved and forgotten. In our masterplan we would like to address the reviving of these existing buildings, but also pay tribute to the city's past, present and future, reflecting the heartfelt pride of Aberdonians in their city.





Relocated Cordona's ferris wheel
offers 20 km of views across central
Aberdeen and the north sea



Reused decommissioned infrastructure
50,000 m² from oil rigs
8 km from industrial sheds
190 m from wind turbines



53,000 m² of pocket parks



3 km of rejuvenated wildlife coastal
trails



10 hectares of shared community street
trails



7km of reclaimed waterfront
trails

Objectives

Masterplan Narrative

Key Issues

Fundamentally, our masterplan aims to provide existing communities with affordable housing, local employment opportunities, and an activated waterfront. They equip future communities with resilience to environmental change through regenerative strategies and urban amenities that support a strong sustainable future, including flood defences and initiatives that strengthen collaboration between the city's universities. Scotland's Nature Positive Plan requires that 30% of land and sea be allocated to sustainable green and blue infrastructure by 2030; in our masterplan this figure reaches 61%, covering 86 hectares.

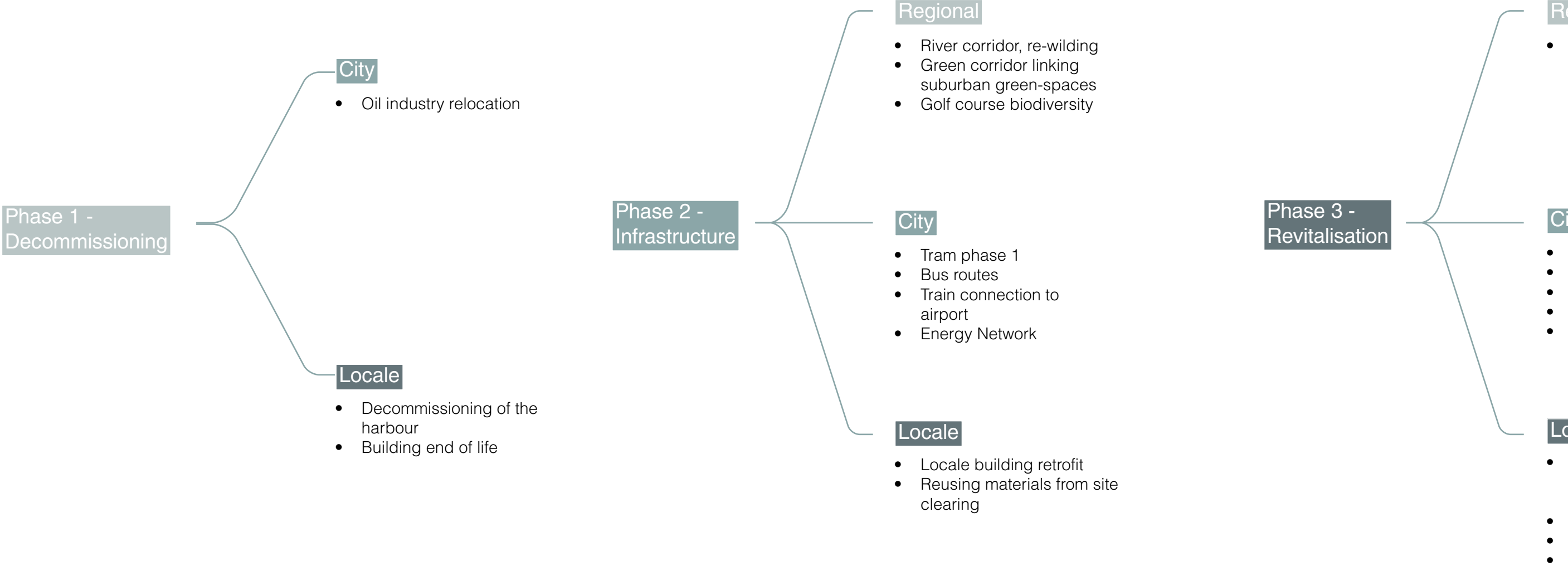
Through an understanding of the city's heritage, we have retained elements of the existing urban fabric and drawn inspiration from historic structures, such as the fish market roof and courtyard typologies as an homage to Aberdeen's past, present, and future. These industrial relics inform a rich material and narrative palette, forming part of our regenerative design principles for the city's future development. This approach supports the transition to net zero while sustaining high-quality, long-term employment. Our mixed-use proposal suggests a sustainable diversification of industry, generating job opportunities that reinforce the growth of green energy and help Aberdeen navigate industrial transition without economic downturn. We aim to address Aberdeen's social, environmental, and economic complexities through a regenerative design that responds to the challenges of today and the needs of tomorrow.





Phasing Strategy

Key Moves



Regional

Surface water flooding interventions

City

Pocket parks
SUDS
Cycle routes
Beachfront promenade
Beachfront landscaping

Locale

Residential phase 1
 Vistas
 Standard residential blocks
Landscape phase 1
Traffic calming and shared streets
Promenade phase 1 - Harbourside
Promenade

Phase 4 -
Densification

City

- Torry residential density
- Tram phase 2
- City wide interventions
 Union Street
 Golden Square
 Retail scene
 Train station residential

Locale

- Landscape phase 2
 Hardscape
 Greenscape
- Residential phase 2
 Waterfront living
- Cultural institutions
- Promenade phase 2 - Northern
 Edge

Phase 5 -
Sustained Growth

City

- Barrage for the River Dee

Locale

- University Research Cluster

Phasing Strategy

Timeline

Phase 1 -
Decommissioning



Phase 2 -
Infrastructure



Phase 3 -
Revitalisation

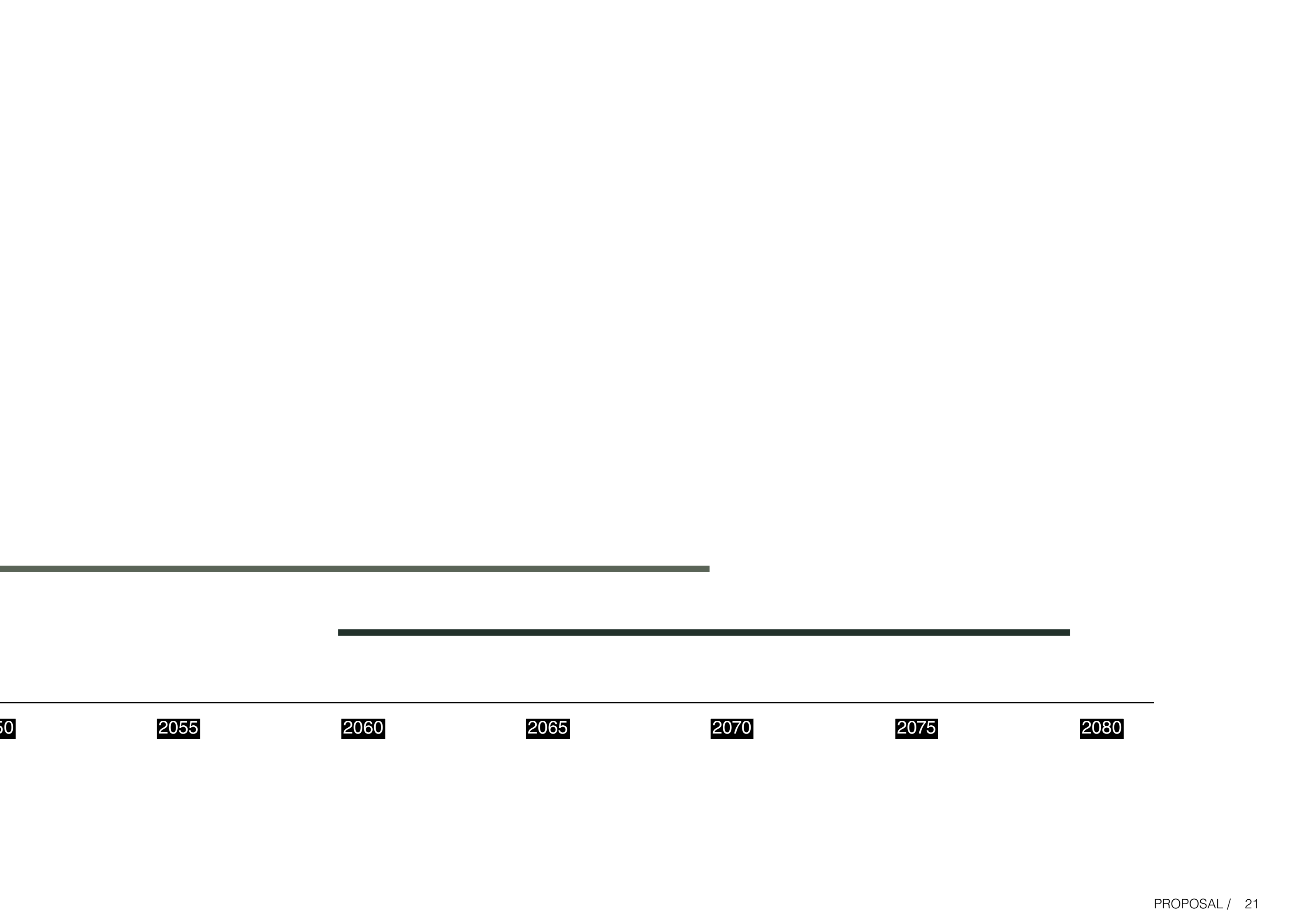


Phase 4 -
Densification



Phase 5 -
Sustained Growth





Regional Strategies

Aberdeenshire

Our regional strategies are rooted in extensive research and respond directly to Aberdeen's environmental challenges. The first addresses urban sprawl by introducing interconnected green corridors that enhance biodiversity and link currently isolated landscapes. This includes rethinking nearby golf courses, diversifying their monocultural planting, and improving their accessibility as part of a wider green network.

The second approach tackles surface and river flooding.

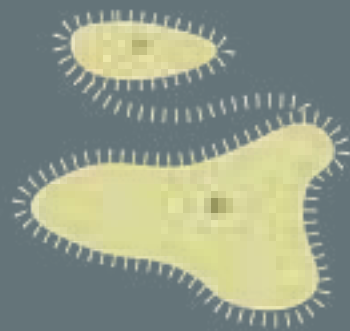
Sponge streets manage surface water by using permeable surfaces, bioswales, and sub-surface storage to slow and absorb runoff. Along the rivers, meadow cultivation stabilises banks, increases infiltration, and provides seasonal water storage while enriching local habitats.

These strategies align with Aberdeen City Council policies, particularly the Aberdeen Local Development Plan 2023, which emphasises green infrastructure, open space protection, and climate-resilient design.





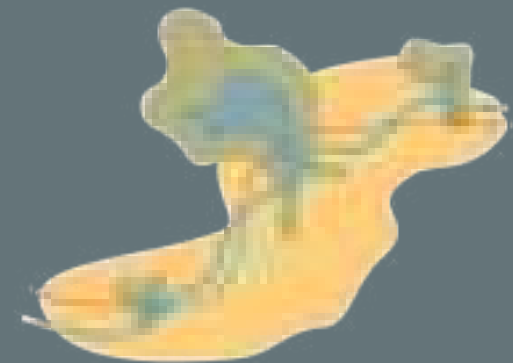
Green Corridor



Golf Course Edge
Transformation



Sponge Street



Meadow Cultivation

City Strategies

Aberdeen

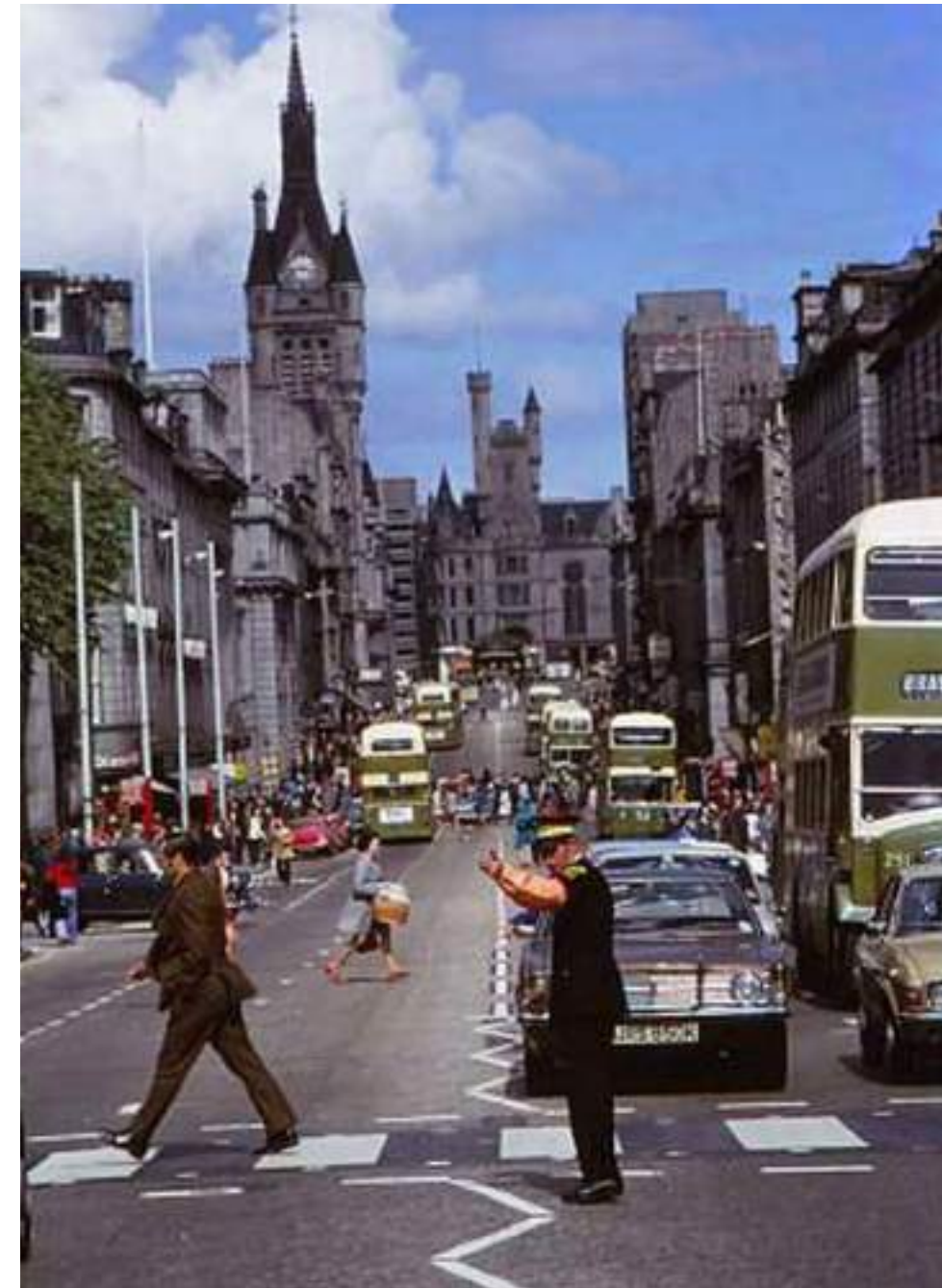
Our city strategy addresses Aberdeen's mobility, flooding, and energy challenges and aligns with the Aberdeen Local Development Plan 2023 (ALDP), National Planning Framework 4 (NPF4).

To improve mobility, we reintroduce a tram network supported by new mobility hubs in the west and north, enabling a Park and Ride model and reducing car reliance. This directly supports ALDP and NPF4 policies on sustainable transport and modal shift.

Flood resilience is strengthened through pocket parks, which replace on-surface parking and manage surface water through natural infiltration, in line with ALDP's blue-green infrastructure policies. Coastal flooding is addressed by a barrage that

remains open under normal conditions but closes during extreme events, protecting the harbour and unlocking redevelopment potential. we revitalise the beachfront through sand dunes, breakwaters, and new amenities.

Our energy strategy proposes cultural buildings that act as anchor loads for the Queen's Link District Heat and Power Plant, alongside an ambient heat loop that draws waste heat from buildings, wastewater, and the harbour. These measures support NPF4's low-carbon heat requirements.





Pocket Parks



Flood Defence



Energy Network



Beachfront Revitalisation



Port Densification



Transport Network

Locale Strategies

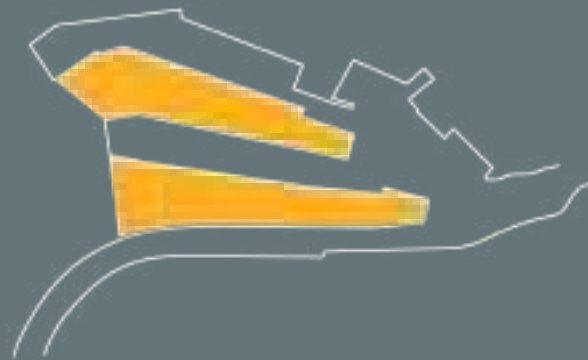
Aberdeen Harbour

Densification of the harbour is a visual statement to identify the new heart of Aberdeen, with relationship to water at its heart. All building typologies activate the waterfront in different ways according to program, public or private. Our cultural buildings are spaced to frame the centre of the harbour, and our vistas cut across the water to ensure that the public once again feel connected to the river and the sea. Taking inspiration of the historic fishing village typology, a landmark to community living in Aberdeen, we have integrated secondary streets and courtyard typologies throughout our masterplan. Together with this, our conversion of car park space to pocket parks

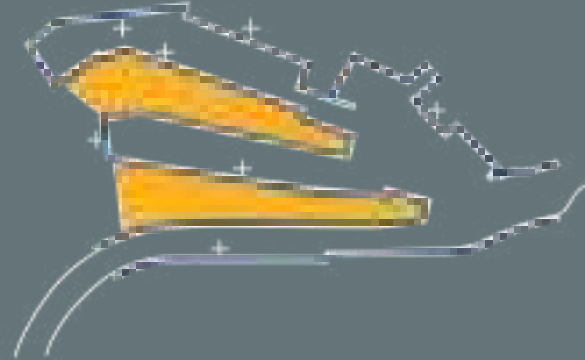
enrich the city's green space as a new public amenity.

Our key strategies directly address Aberdeen's key policy outlines. Namely the Aberdeen Harbour Development Framework Core Objectives, Scotland's National Planning Framework 4 (NPF4), and Aberdeen's 2023 Local Development Plan. These three key documents emphasise the need to optimise the newly available harbour land for mixed use programs, promote green infrastructure, improve connectivity and permeability and enhance public open spaces.

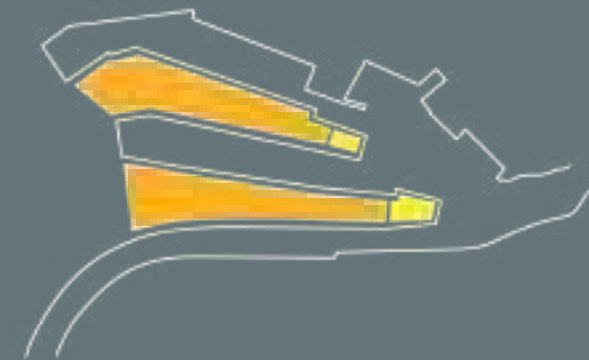




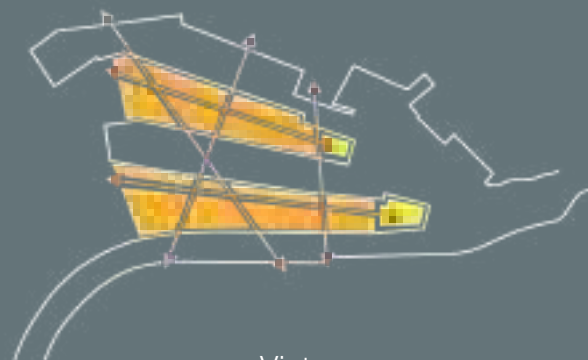
Harbour Density



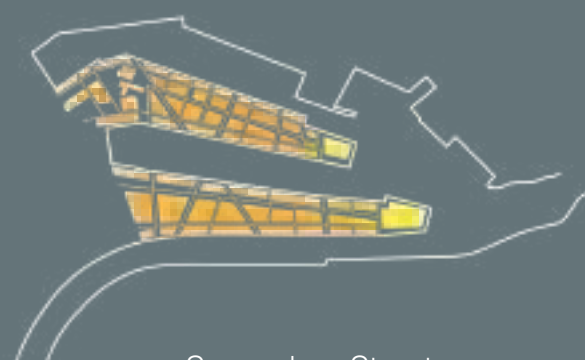
Waterfront Activation



Cultural Asset at the Tip



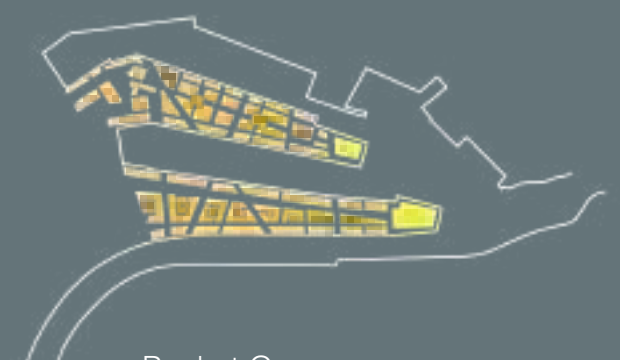
Vistas



Secondary Streets



Courtyard Typology



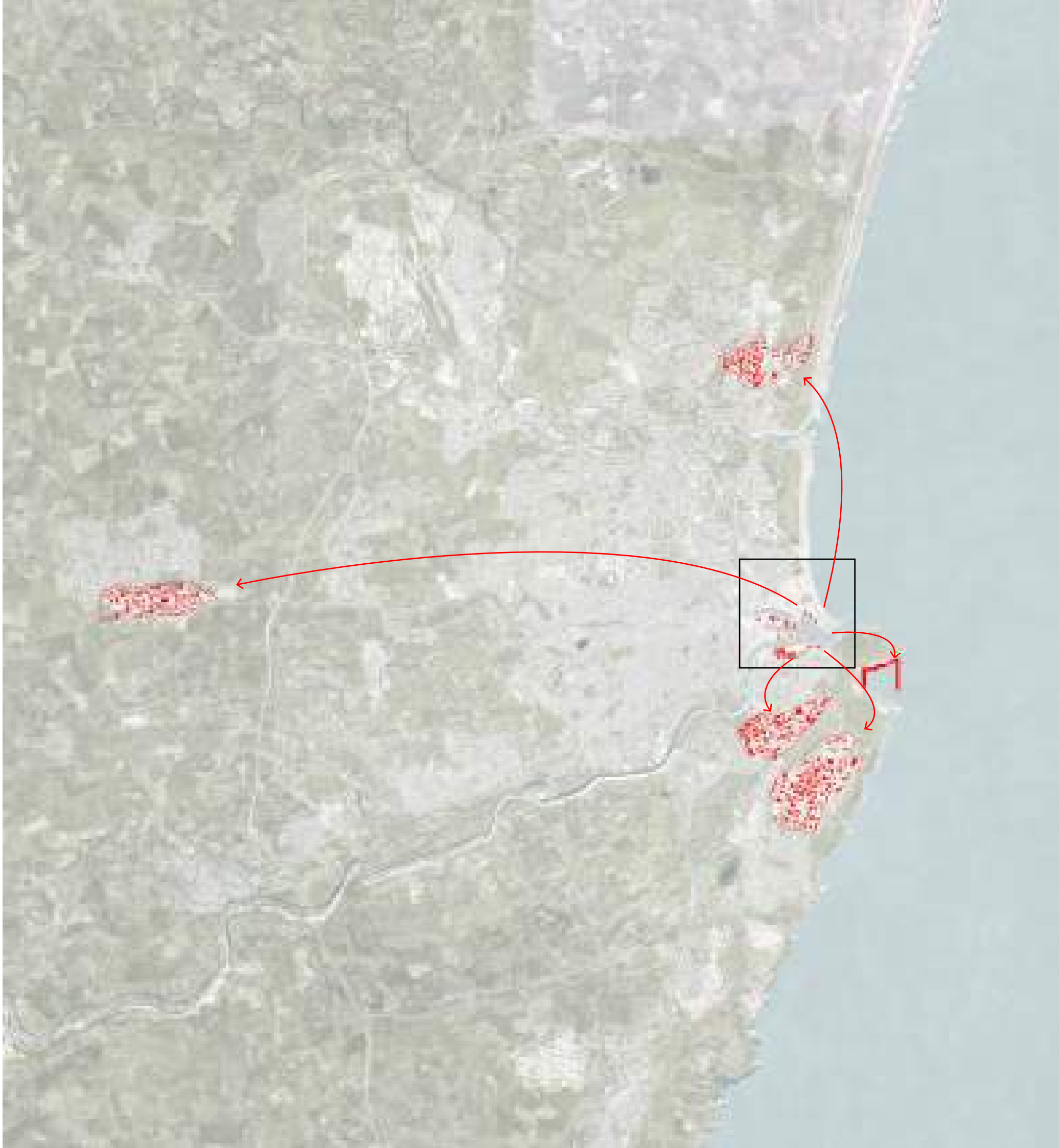
Pocket Greenspace

Phase 1

2025-2030

Decommissioning

- City
- Oil industry relocation
- Locale
- Decommissioning of the harbour
- Building end of life

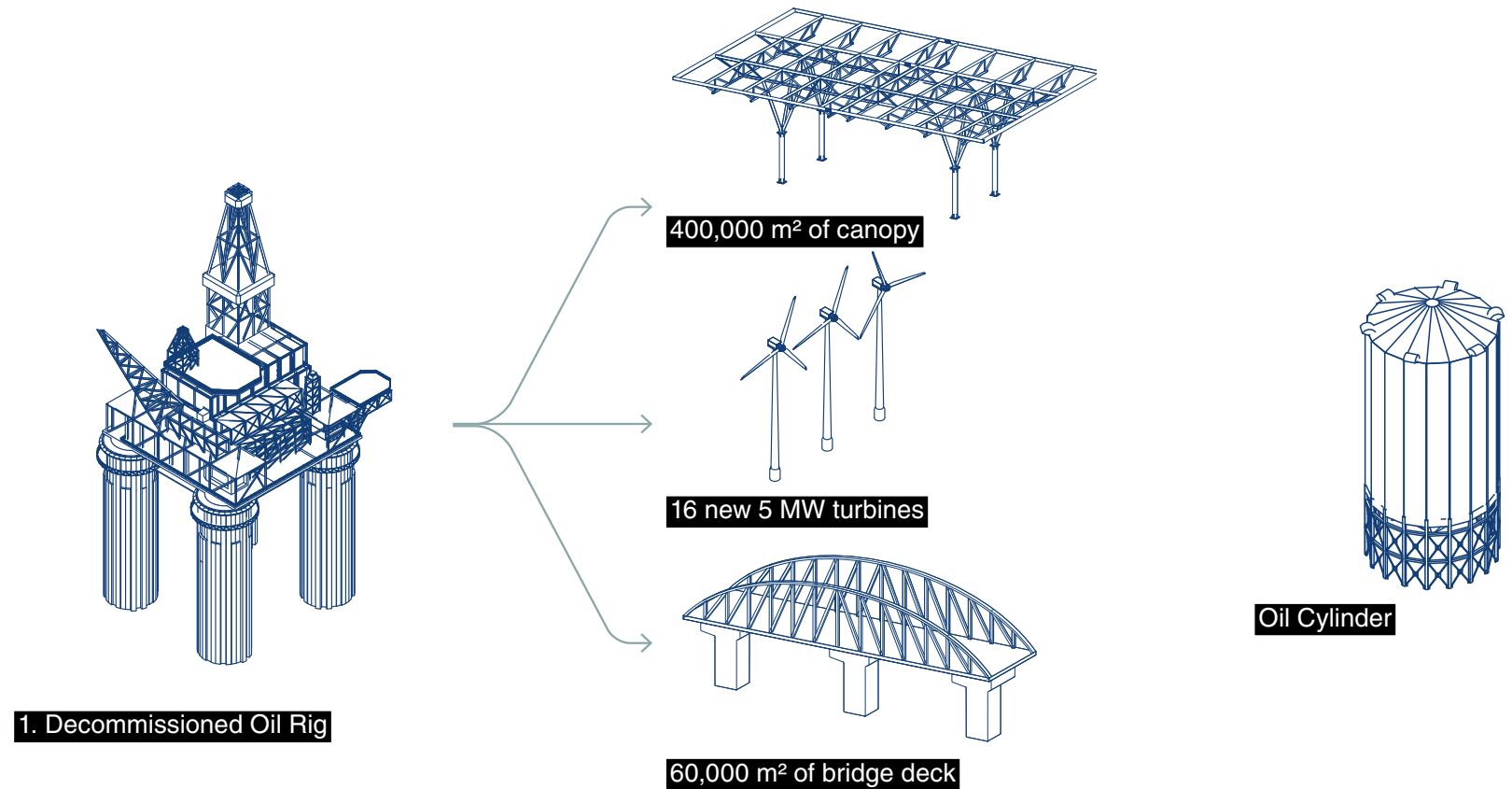


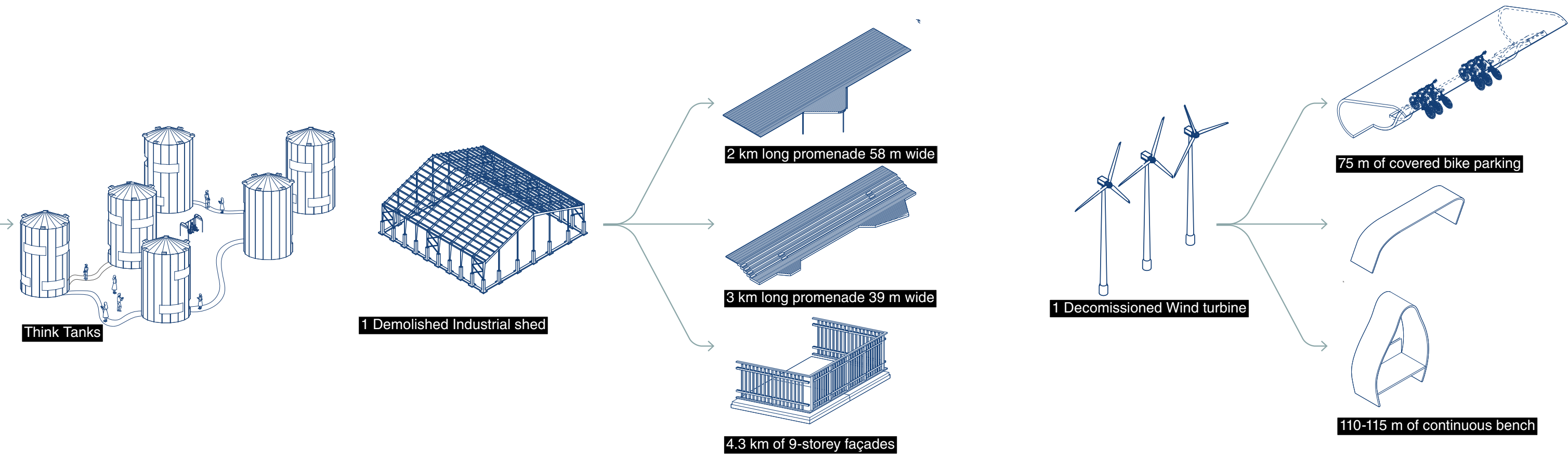
Decommission

Phase 1

Aberdeen is facing a significant transition as it moves from a highly lucrative oil-extraction economy toward a more sustainable, renewable-energy future. This shift raises an important question: what happens to the extensive industrial infrastructure that will become obsolete?

Our masterplan addresses this challenge by envisioning how decommissioned assets can be transformed rather than abandoned. Through principles of circular economy, upcycling, and recycling, we propose new ways to repurpose materials and structures, creating opportunities for innovative building components, public infrastructure, and site-specific interventions. This approach not only reduces waste but also positions Aberdeen as a model for regenerative, low-carbon urban development.





Relocation

Oil Relocation from Harbour



From Harbour to Westhill

“Oil has no loyalty.”

As land values decline, oil companies are relocating their offices from central Aberdeen particularly the harbour area, to Westhill.



From Harbour to Bridge of Don Industrial Estate

Similarly to the shift toward Westhill, many oil-related offices and service companies have also relocated north to the Bridge of Don Industrial Estate. This move is driven primarily by lower land and rental costs, larger industrial units, and better access for logistics.



From Harbour to South Harbour

South Harbour, completed in 2023, provides Aberdeen with new deep-water berths and facilities capable of handling much larger vessels. This allows a greater share of industrial and service ships to operate from the new site, easing pressure on the existing harbour. As a result, the main Harbour gains freed capacity and flexibility, opening opportunities for redevelopment, cleaner industries, and a more civic, public-facing waterfront.

Decontamination

Brownfield Sites

Addressing the contamination of the harbour through extensive hydrocarbon storage will require a combination of methods:

Whole site

Point-source removal (off-site):
Removal of large objects like the oil tanks

Harbourside

Thermal desorption (in-situ):
Heating the soil contaminants and collecting the volatile vapours, Aberdeen has investment in carbon capture off-shore so may be opportunity there.

Immobilisation (in-situ):

Chemical or physical process fixing polluting heavy metals to soil to limit further contamination.

Torry Riverside

Phytoremediation:

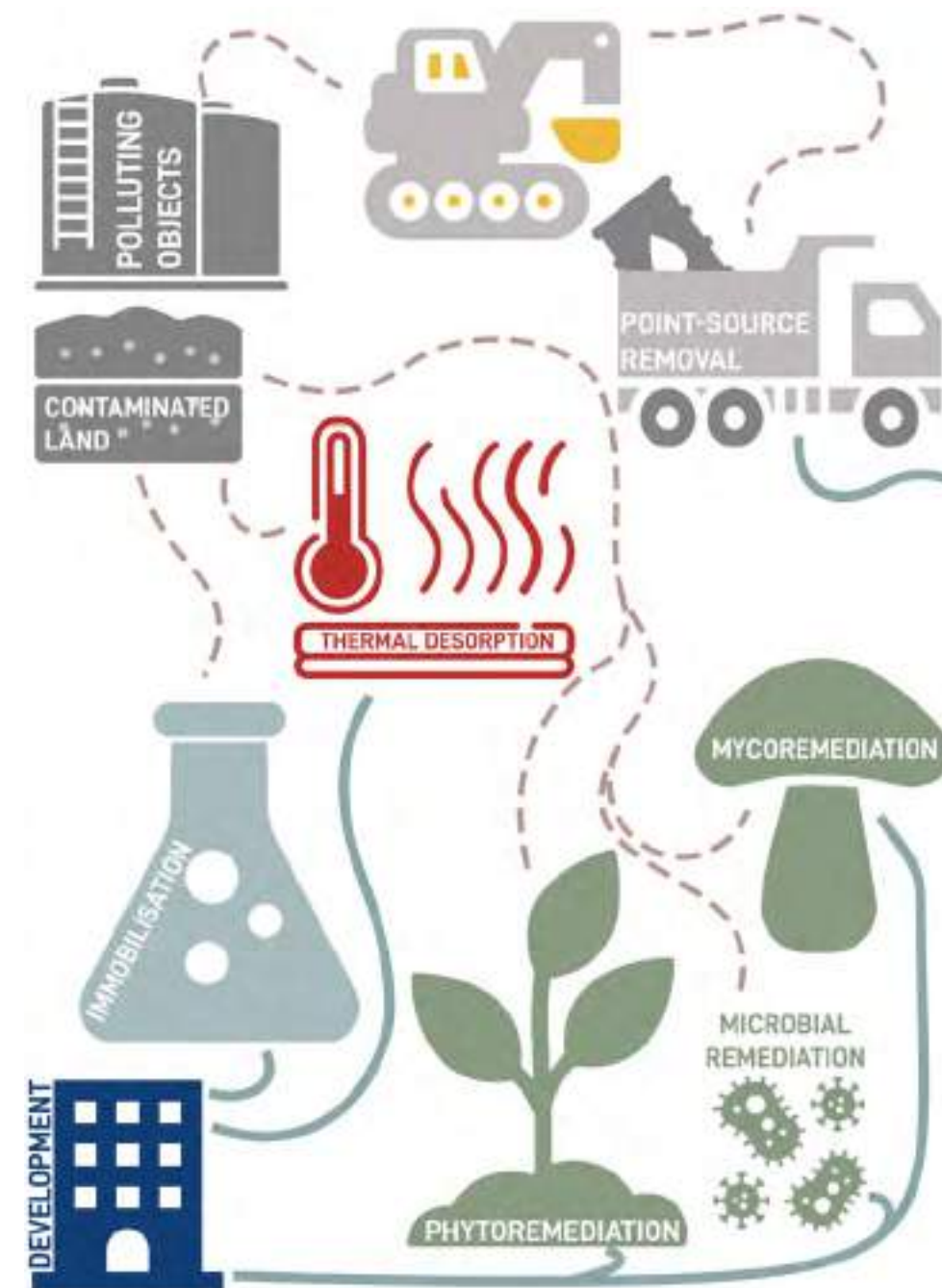
Plants that produce chemicals which degrade contaminants

Microbial re-mediation:

Use microbes to speed the degradation of contaminants

Mycoremediation:

Use of fungus to speed the degradation of contaminants



Phase 2

2030-2045

Infrastructure

Regional

- River corridor, re-wilding
- City centre, re-wilding

City

- Transport
 - Tram Zone 1
 - Train to Airport
- Energy Network

Locale

- Locale building retrofit
- Reusing materials from site clearing



Many sections along the River Dee has 10% chance of flooding every year. Banchory, Aboyne, Ballater and Inverurie are defined as 'potential vulnerability area' where highest flood risk exists now or is likely to occur in the future. These are geographically prone to inundation or have a history of flood damage.



Upstream Floodplains

Meadow Cultivation

At Upstream Regions



Nature Connection



Preserve Habitats

Meadows are introduced on the underused green slopes and plots of the targeted regions: Banchory, Aboyne, Ballater, and Inverurie. Three along the River Dee and one along the River Don.

These natural interventions absorb significant amounts of water, helping to reduce river flooding. Riverside boardwalks for walking and cycling are incorporated to create attractive outdoor spaces that foster a closer connection with nature.

8,100 residential and 2,000 non-residential properties in the Aberdeen Deeside zone are at risk from river-flooding.



Riparian Buffer Zones

At Agricultural Water Edges



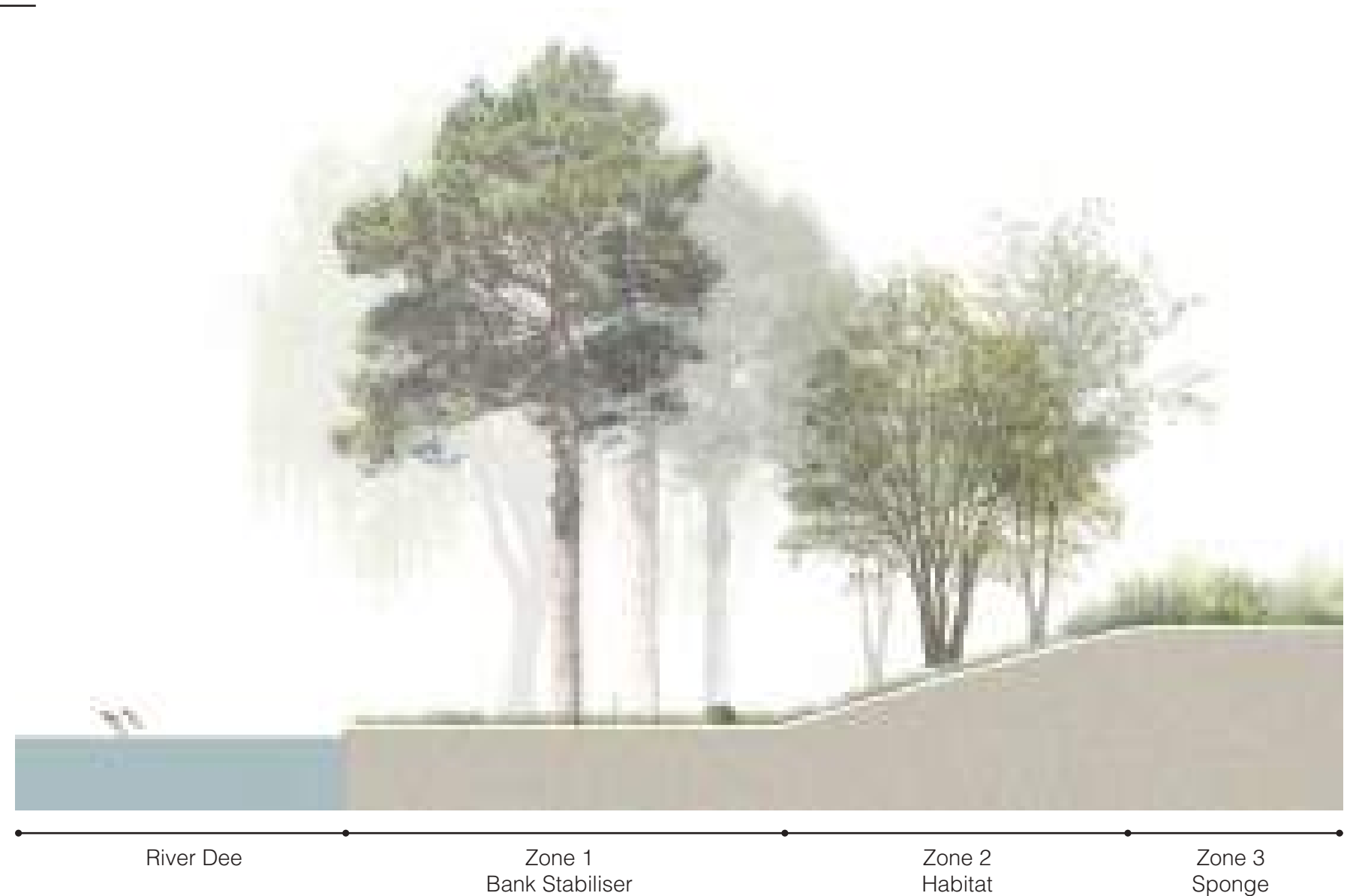
Filter off Chemicals



Reduce Water Temperature

Heavy agricultural practises at Aberdeenshire reduces the soil's ability to absorb rain and lead to increased water running off, particularly during storm events.

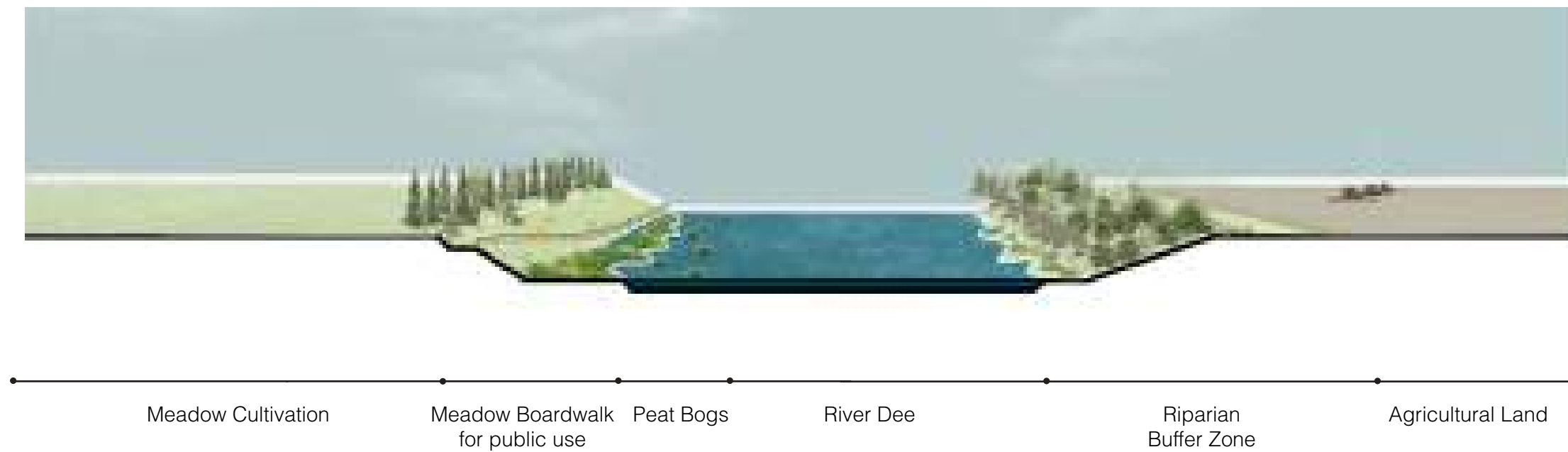
Riparian Buffer Zones are introduced at edges of agricultural land to decrease speed of surface water runoff, and filter off pesticides from water before entering the river. This not only address river flooding problem, but also helps preserve wildlife.



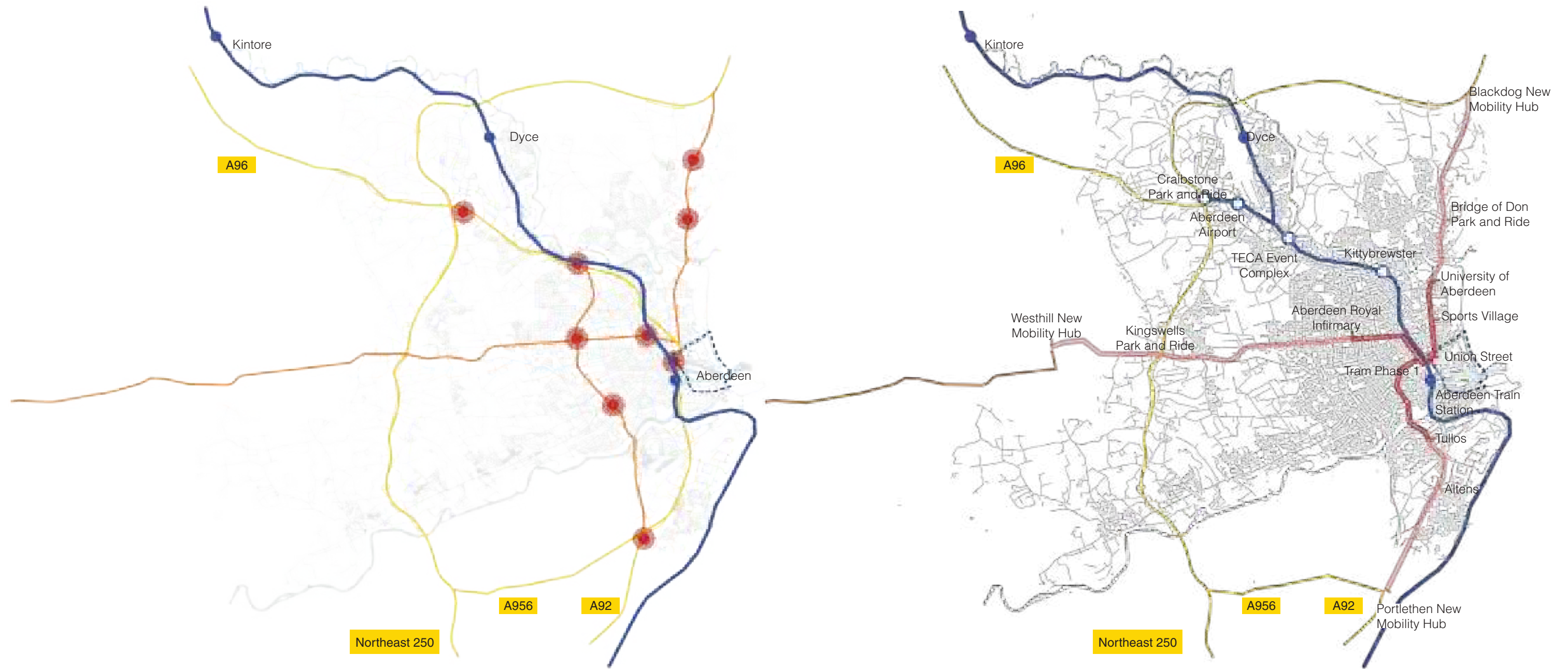
Before



After



River Corridor Overview



Existing

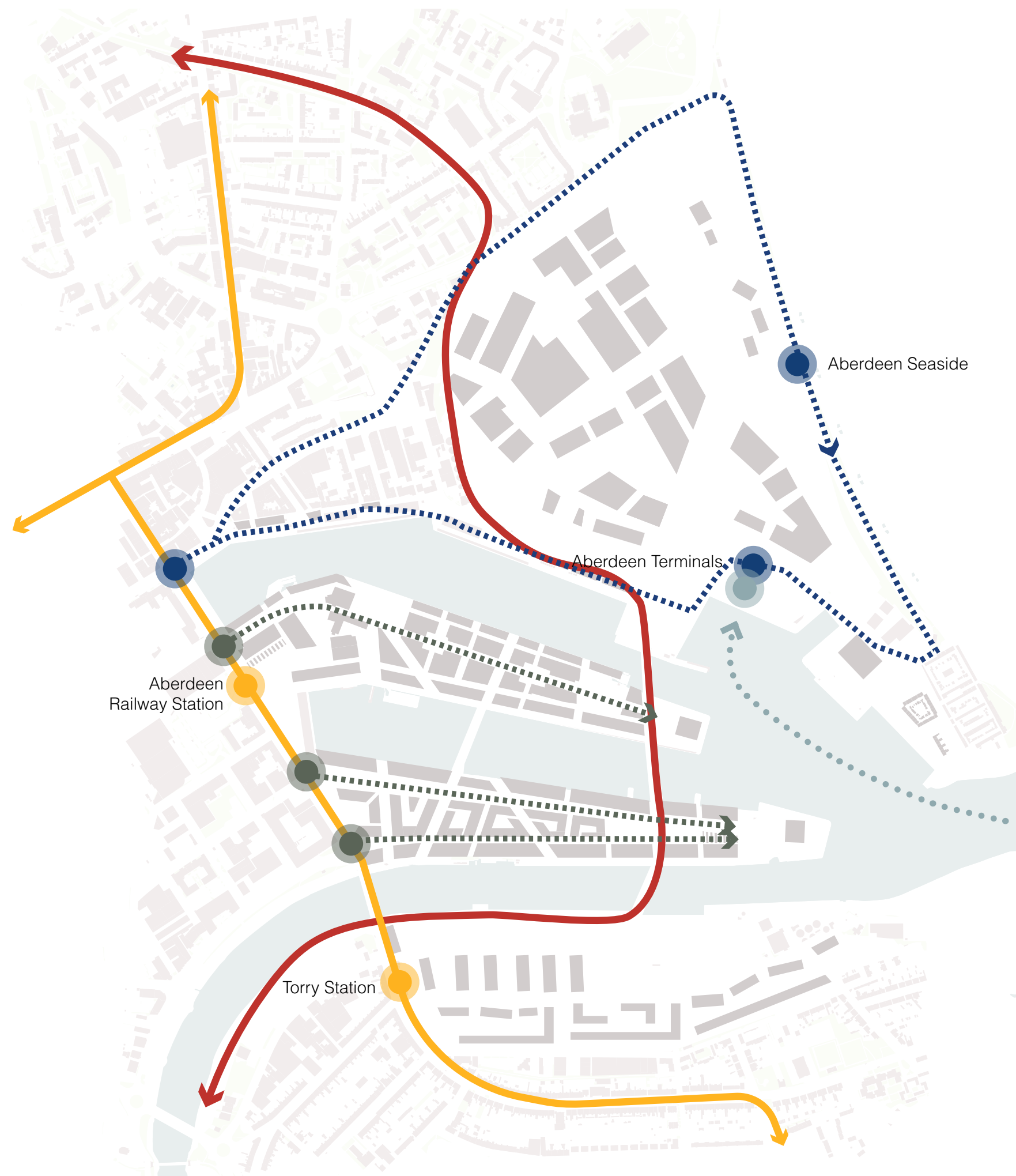
Proposed

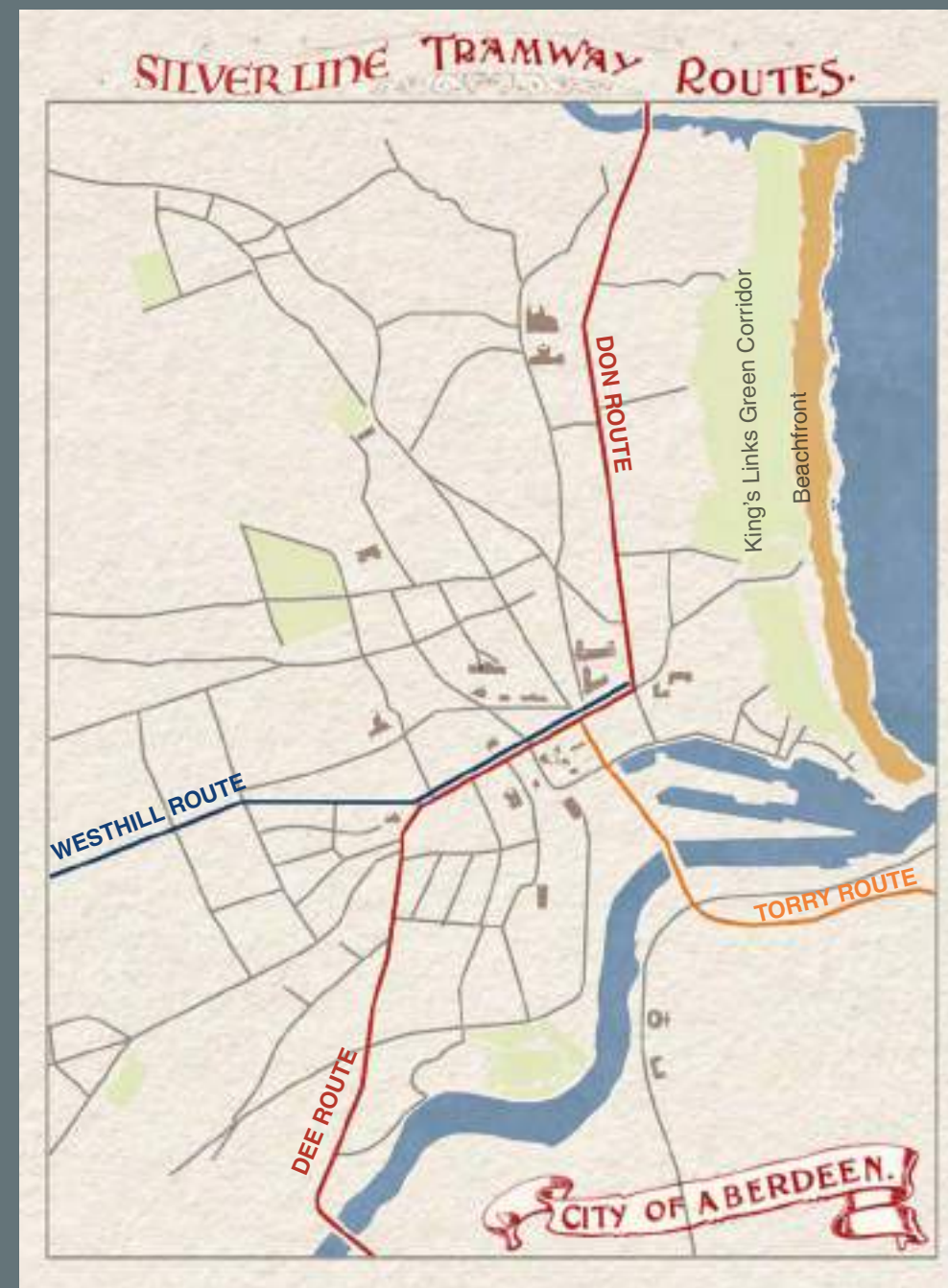
Transport, Infrastructure

- Tram Phase 1
- Tram Phase 2
- Rail
- Car
- Congestion Points
- Locale

Transport, Locale

-  Cars Do not Go past this point
-  Bus Stop
-  Ferry Terminal
-  Tram Stop
-  Service, Emergency vehicles, Mobility Bus route
-  Bus Route
-  Ferry Route
-  Tram Route
-  Cyclist's Route





Historic Tramways In 1870 the Tramways Act was passed to Regulate Street Tramway Construction on those who had gained Parliamentary Approval. The Aberdeen District Tramways Act of 1872 Authorised several Lines through Aberdeen; all the Tracks were to be Single Track with Passing Loops. 2-Routes were built from King Street to Queen's Cross & St Nicholas Street to Kittybrewster & Opened in 1874 as Horse Tramways.

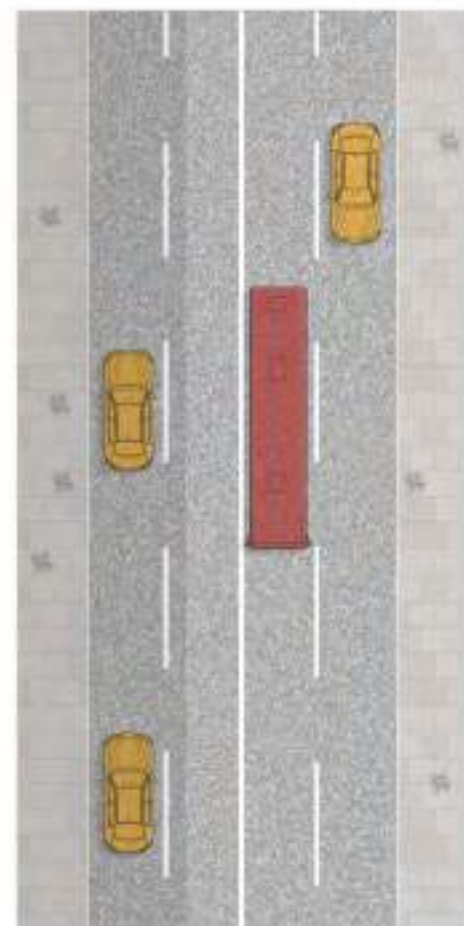
Reinstate Tram Network for Sustainable Travel Tramways bridging the two rivers of Aberdeen, River Don and River Dee will be reinstated, with the benefit of bringing people from the two major universities to the shared campus and reseach institutes developed within the harbour. Better connections with Torry is also proposed together with links with the new oil industry zone of Westhill just outside of Aberdeen.



Transport, Reinstate Historic Tramways



3.5m 14.5m 3.5m
4-Lane Traffic



Existing



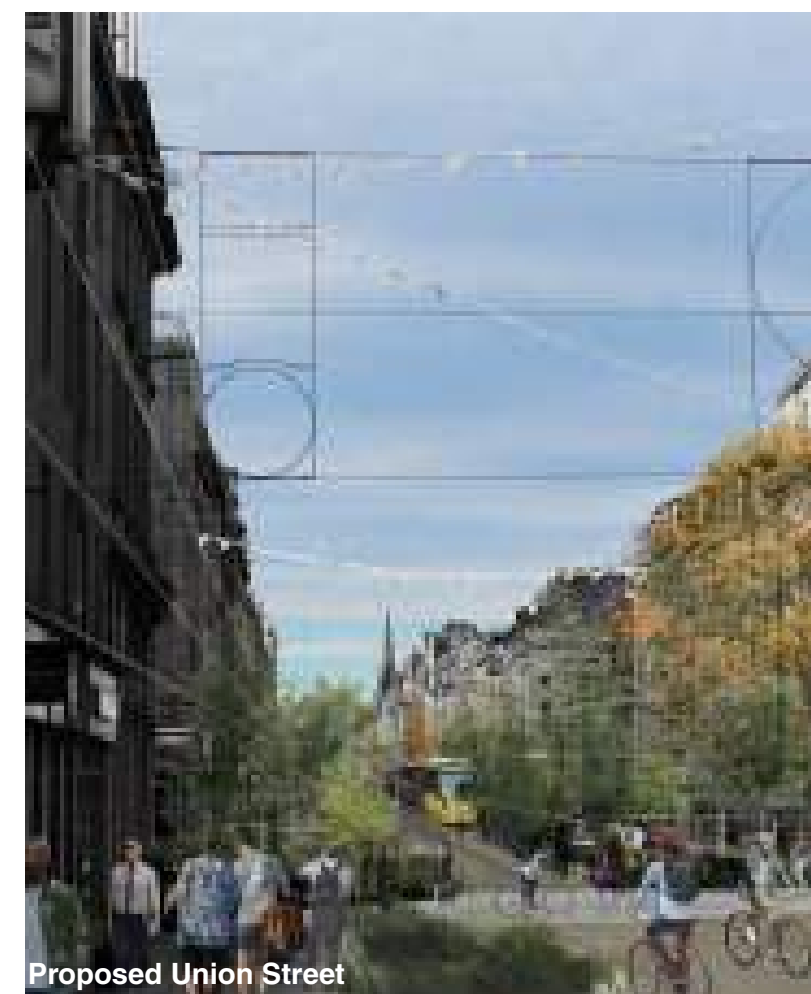
7.7m 7.0m 7.7m
Tram/Bus/Car



Proposed



Existing Union Street



Proposed Union Street

Mobility

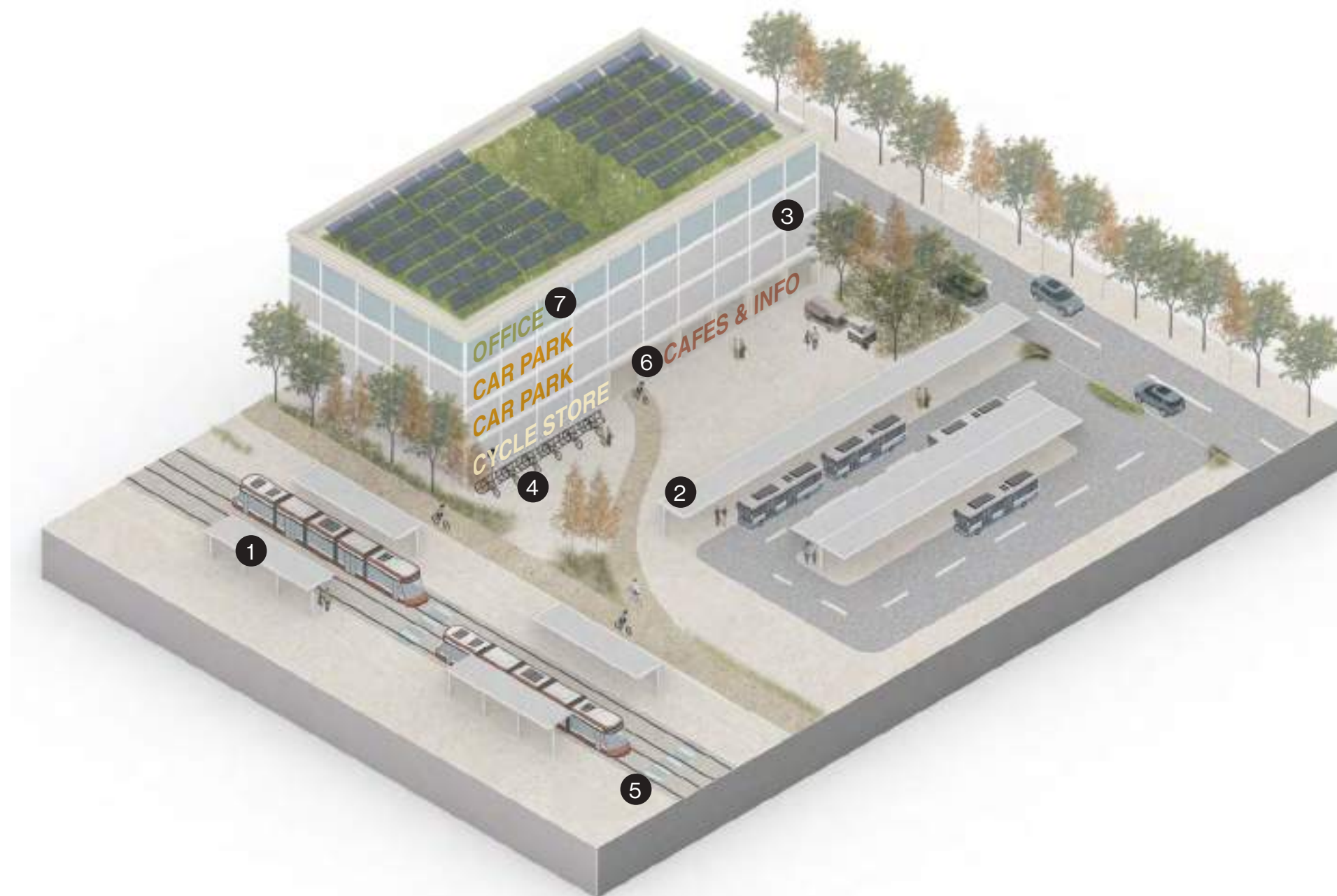
Mobility Hub Design Guide

Aberdeen is very car-dominated. As the industries relocate to areas outside of Aberdeen, more infrastructure connections are needed to facilitate commute and transportation of goods. Mobility Hubs will be strategically placed to concentrate parking in designated areas and promote public transport. Most of the hubs will be introduced along the outskirts of the city, in order to, first serve the new industrial estates, second, encourage private vehicles users adopt greener alternative transport means to travel into Aberdeen.

A smaller scale city centre mobility hub is also going to be introduced near the train station to serve the passengers utilising the railway network and the ferry.

Together with Park & Ride hubs introduced along the Northeast 250 routes, the mobility scheme also brings more tourists to Aberdeen, making the city a more vibrant place.

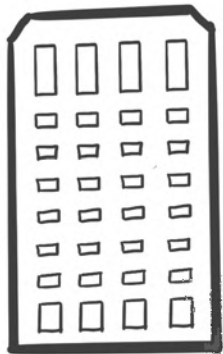




- ① **Tram Terminal**
Positioned in adjacent to facilitate commute.
- ② **Bus Terminal**
Positioned in adjacent to facilitate commute.
- ③ **Multi-storey Carpark**
Provide concentrated carparking facilities to liberate more land for public use and encourage alternative means of travel.
- ④ **Cycle Parking and Rental**
Promote flexibility and encourage cycling within the city.
- ⑤ **Designated Safe Pedestrian Crossing Area**
Safe area is introduced to ensure safety of pedestrians when navigating around the mobility hub.
- ⑥ **Cafes & Information Point**
Waiting areas and information points will be implemented to facilitate smoother journeys.
- ⑦ **Office & Development Opportunities**
As many mobility hubs are located close to new industrial estates, upper floor development opportunities are provided.

Repurpose Analysis

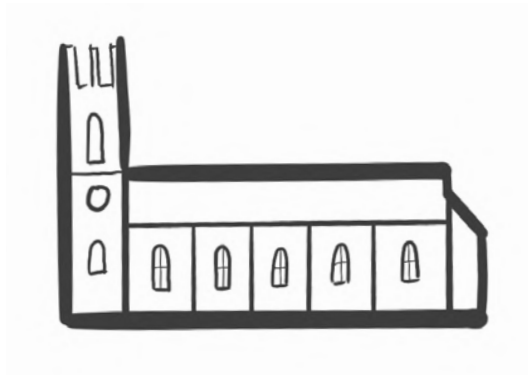
Retrofit



Life Span: 30+ Years
Design: Adequate
Typology: Offices/ Commercial

Buildings are recently built and are in good condition. They will be repurposed to suit the typologies of new masterplan.

Upgrade



Life Span: <20 years or Unknown
Design: Adequate with historic/social value
Typology: Religion/ leisure

These houses has a short remaining lifespan but they would have high social contribution to the new masterplan, and some of them has high architectural value. Structures are reinforced, insulation is improved and solar panels could be installed to upgrade the buildings.

Demolish



Life Span: <20 years
Design: Functional
Typology: Industrial related to oil

Many warehouses built in the 80s to serve the oil boom will fall into disuse. The carbon footprint to convert industrial warehouse to buildings that accommodate offices and homes is high so we can replace them with new built.

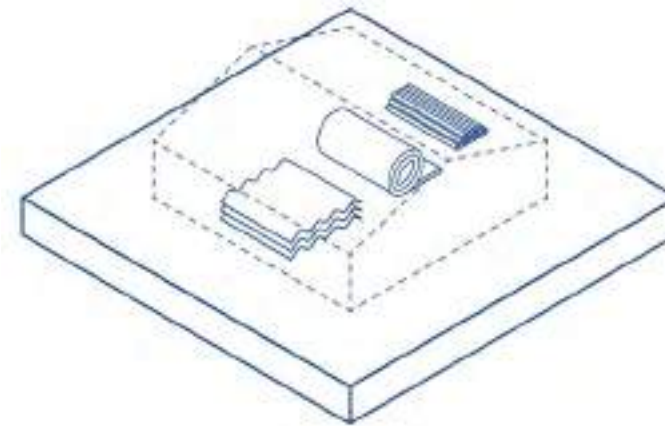
4 Typologies

Retain|Demolish|Volume|Footprint

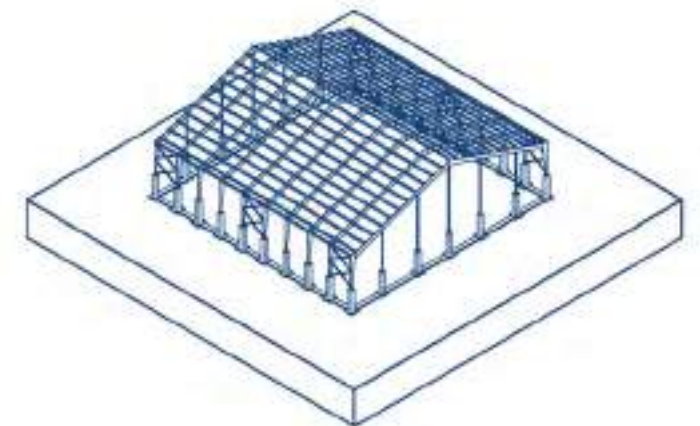
Four strategies are applied across the site to guide how existing industrial buildings are treated: Demolish, Volume, Footprint, and Keep.

1. **Demolish:** Remove poor-condition buildings to open space for public use and new development.
2. **Volume:** Rebuild using the former massing to retain the site's industrial rhythm.
3. **Footprint:** Trace demolished outlines in the landscape to preserve memory.
4. **Retain:** Retain and adapt existing structures for new uses.

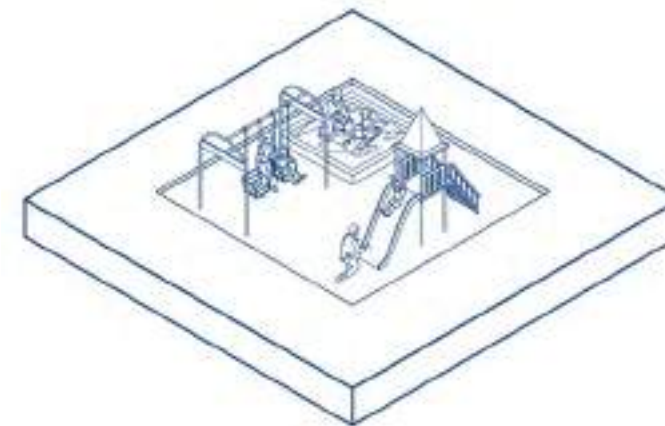
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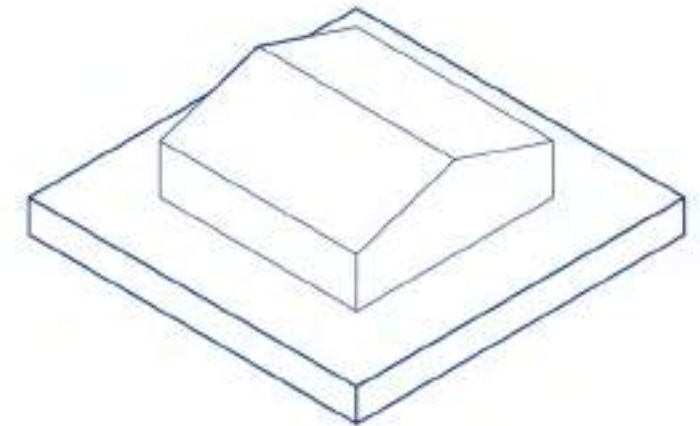
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3



4



Residential Building

Design Guide

Replace Windows

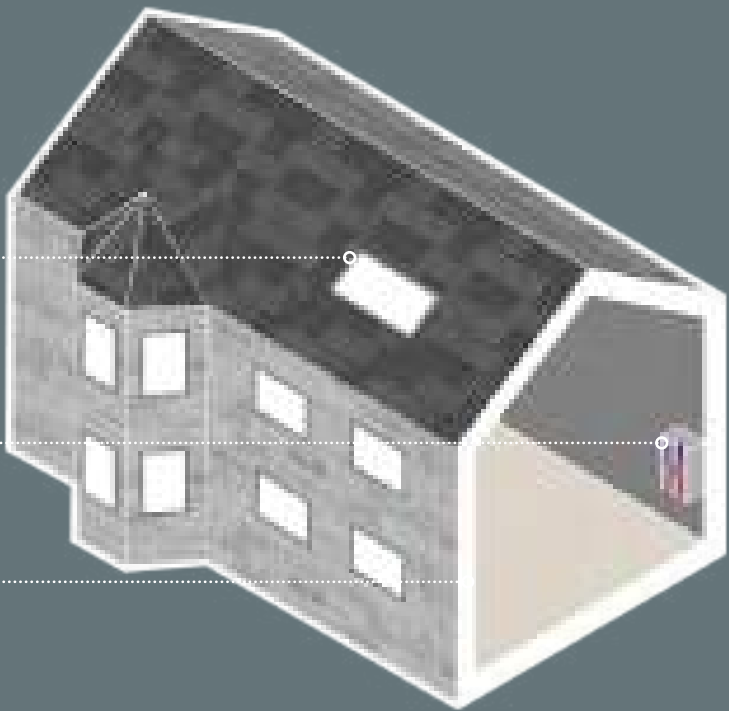
High-performance windows with 0.85 W/(m²K) U-values and ensuring airtightness at construction to significantly reduce heat loss

Installation of MVHR

Recover heat to increase energy saving and improve indoor air quality through filtered fresh air.

Upgrade insulation

To increase user comfort by reducing heat loss and risk of surface condensation and mould growth in fabric build-up



Solar Panels

Optimum angle to capture most sunlight and generate energy for building use

Green Roof

Absorb rainwater to alleviate flooding on ground level

Installation of MVHR

Recover heat to increase energy saving and improve indoor air quality through filtered fresh air.

Offset Building front

To ensure user comfort by preventing overheating on south facade

Over 12,000 flats in granite tenements built in Victorian or Edwardian times, which are now over 100 years old.

They will be refurbished to comply to EnerPHit standards, with a heating demand of 25kWh/ annum or below.



Insulation with U-value ≤ 0.15 W/(m²K) for walls, ≤ 0.12 W/(m²K) for roofs



Triple glazed window



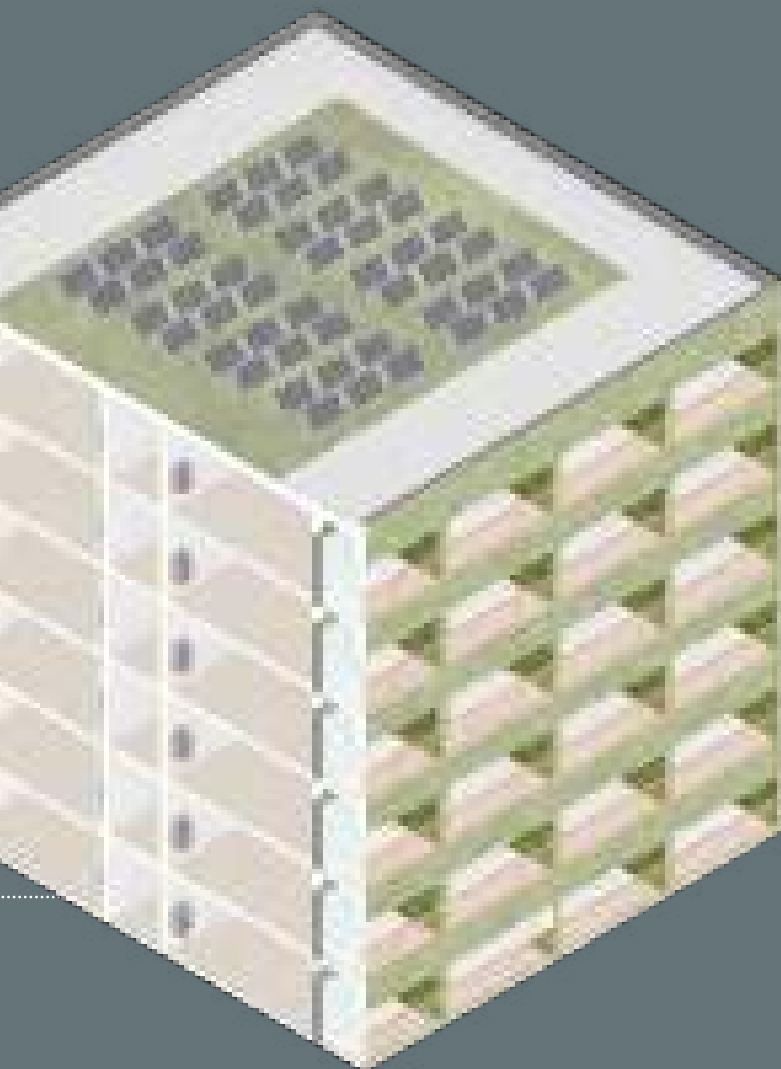
Mechanical Ventilation and Heat Recovery

The residential blocks will follow the Passivhaus Standard, to achieve heating demand of 15kWh/ annum or below.

- Minimise form factor
- Use high thermal mass material
- Appropriate form and thickness of insulation to achieve desired U-value
- Minimise thermal bridging in fabric build up



Offset So



Solar Panels

Optimum angle to capture most sunlight and generate energy for building use

Integrated Ceiling

Chilled Beams to control ventilation and provide cooling in hot summer days and LED installed to minimise energy usage

Green Roof

Absorb rainwater to alleviate flooding on ground level

Shading Devices

To increase user comfort by avoiding glaring and overheating of indoor environment.



South front



Triple glazed window



Mechanical Ventilation and Heat Recovery

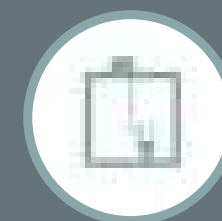
Commercial/ Campus buildings are designed to achieve BREEAM 'outstanding' standard. They will have more state-of-the-art environmental features that support the functions and identity of an innovation hub.



Rooftop Garden at south orientation



Chilled Beam



Rainwater collection for toilet flushing



Shading Devices at South Facade

Energy Strategy

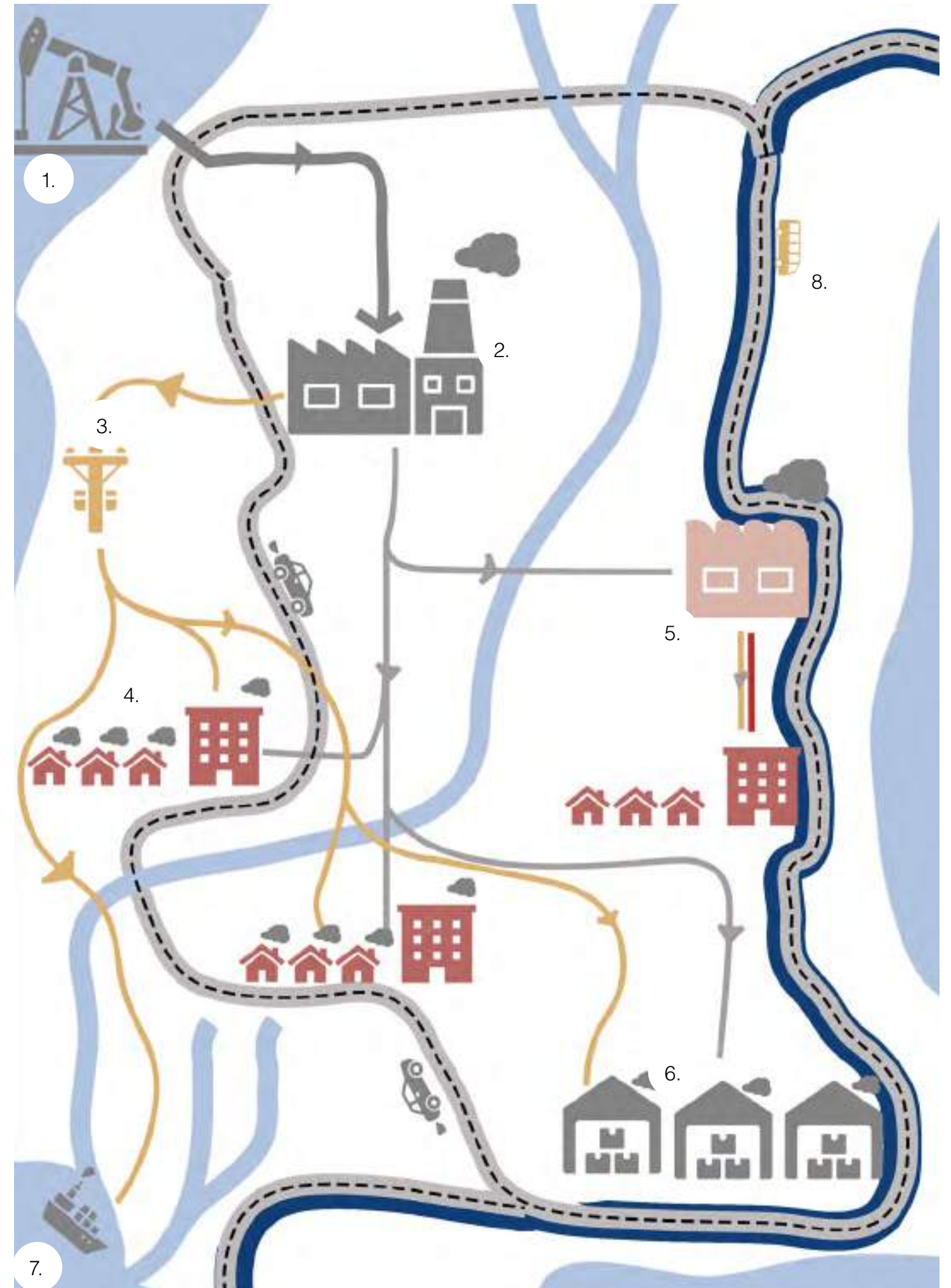
Making the most of the water

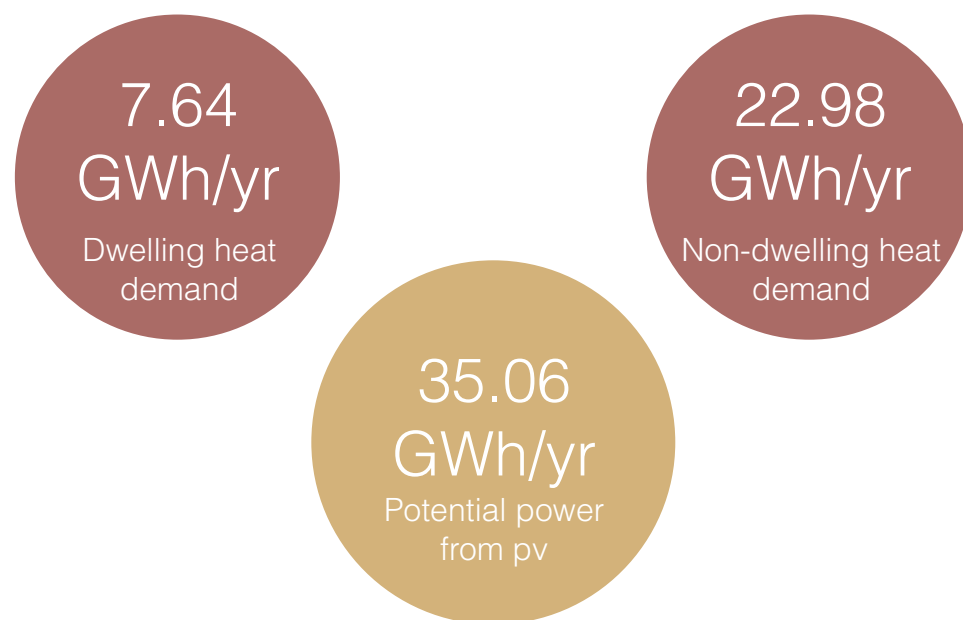
33.35
GWh/yr

Existing heat
demand

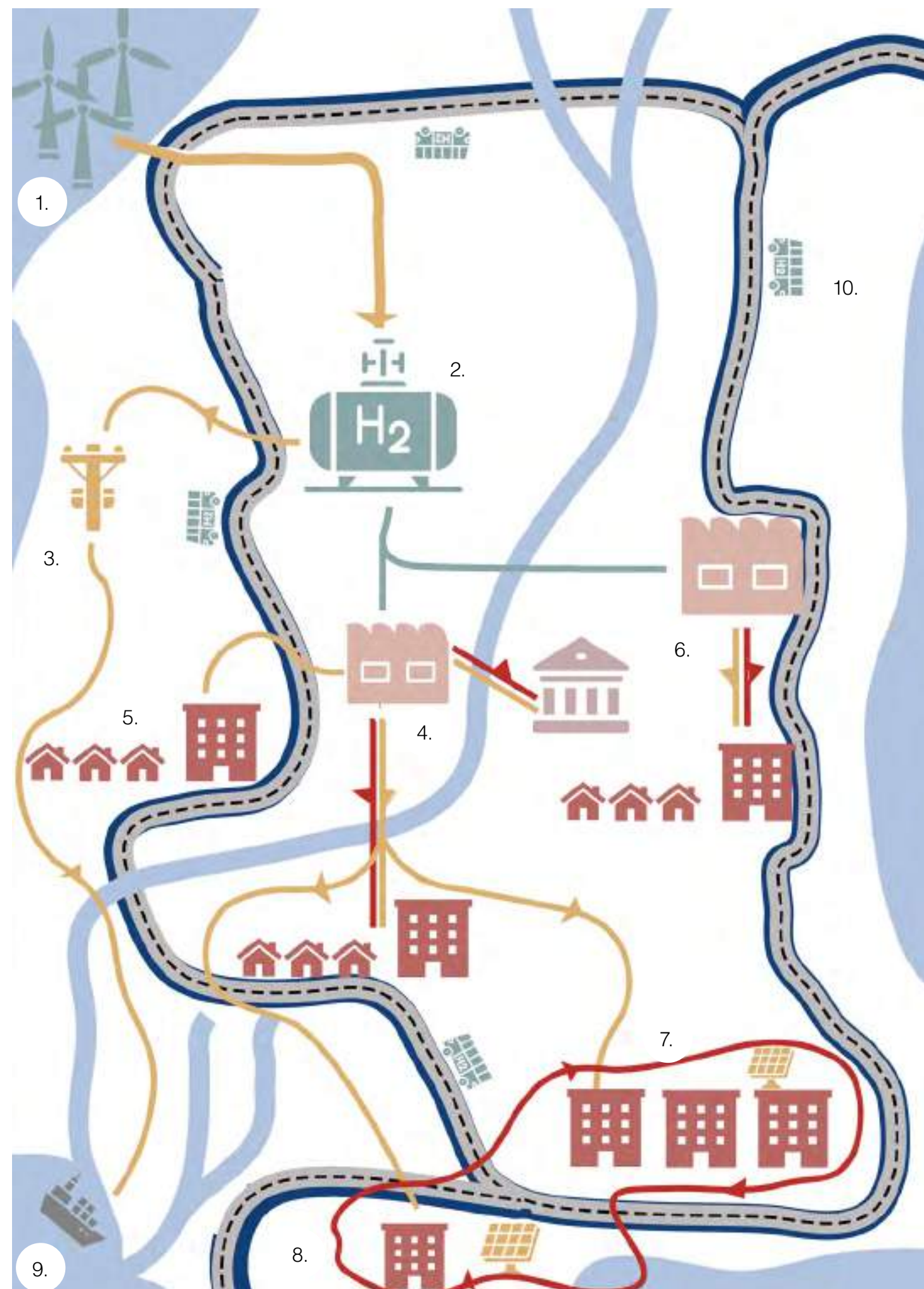


1. The finite supply of natural gas is drilled.
 2. It is refined and burnt to power the grid and provide electricity, and is also sent directly to homes.
 3. The national grid takes power from a mixture of sources, including non-renewables, to provide electricity.
 4. The majority of UK homes use boilers that burn natural gas to provide heating.
 5. In Aberdeen, there are several district heat networks
 6. Existing warehousing stock
 7. Ships in the port in the process of being electrified
 8. Bus routes are limited, though some are hydrogen-powered
- which provide heat and power to homes, but they still rely on electricity and boilers and thus non-renewables still. They also require anchor buildings, which restricts the amount of residential areas they can supply





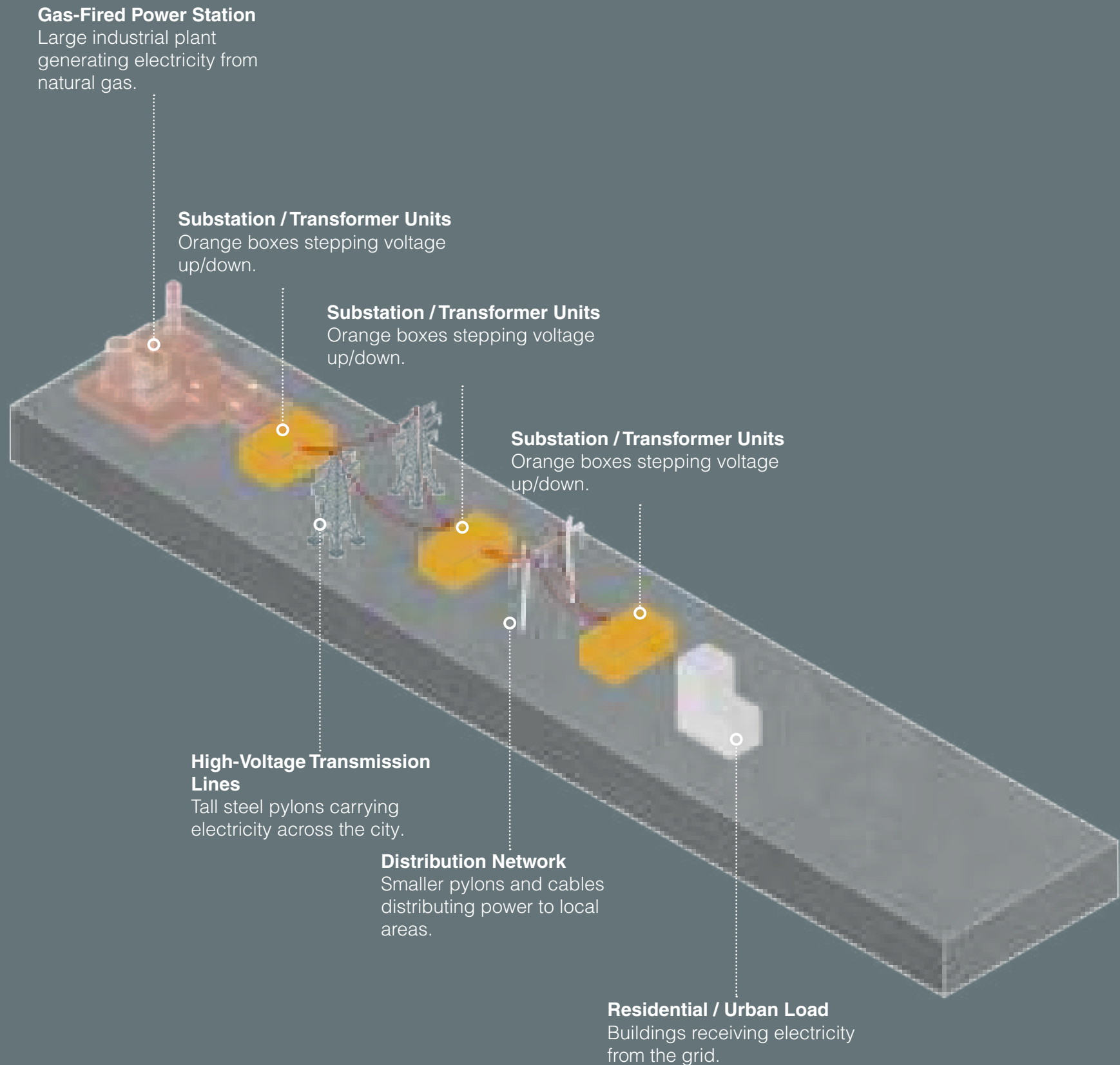
1. EOWDC/Kincardine floating wind farm will supply a renewable source of energy
2. This comes in at the Energy transition zone, where it is converted into hydrogen through electrolysis.
3. This power goes straight into the grid, which supplies the remaining port plug-in hubs
4. . Queens link DHN, currently proposed just North of the site, will be able to open as the new cultural centres will provide the base heating load required for it to operate.
5. This allows further adjacent residential developments to convert to sustainable energy.
6. Hydrogen will also supply the existing Seaton DHN at 100% capacity instead of 20:80 with natural gas.
7. Proposed residencies and commercial buildings in the locale will have a reduced heat energy demand through their Passivhaus compliance, whilst PVs, when exceeding demand, can be stored.
8. An ambient heat loop fed from waste heat from Scottish water waste heat, waste heat from the DHN and potentially the docs
9. Electrified port for ferry
10. Fully hydrogen bus network with a wider route network



Energy Network

Existing

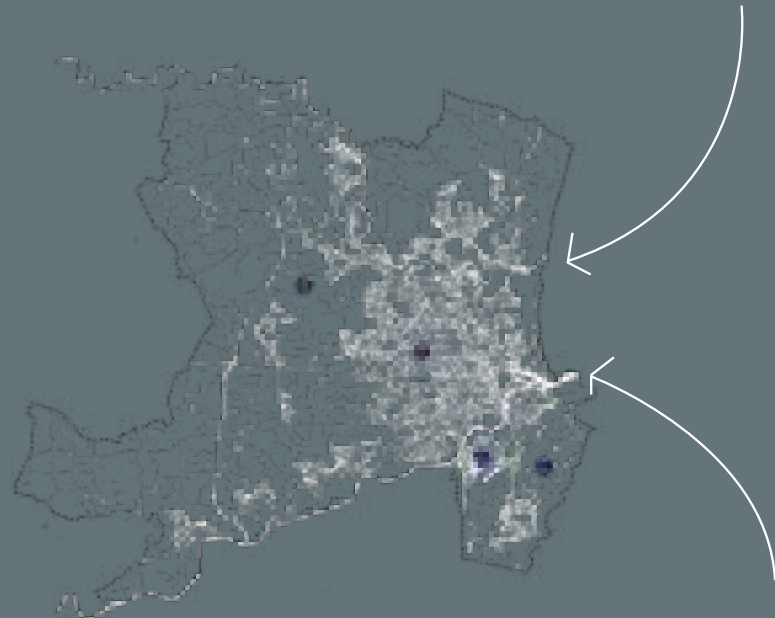
Although the national grid is being decarbonised, around 32% of power is still provided by natural gas. However, even with the use of wind and solar being adopted extensively throughout Scotland, there is still the question of unstable supply and demand. As a result, hydrogen was increasingly being adopted as a means of storing energy through electrolysis and also as a transitional opportunity as we move away from natural gas. Hydrogen can be used in a mixed ratio through existing networks until infrastructure can be upgraded for 100% hydrogen.



Energy Network

Proposed

Locations of accepted proposed hydrogen hubs
Locations of accepted proposed wind farm energy supply



Aberdeen has had Aberdeen Heat and Power operating since 2002, with a large network of DHNs across the city and more being proposed. Finding ways to maximise the efficiency of these networks as well as make the most of the hydrogen being produced across the city has been a high priority. However, on top of this, there is the opportunity for making better use of the wastewater heat and waste building heat across the city with more ambient loop systems in districts. For instance, the border of the locale has an identified Scottish wastewater heat supply ready to be used along with a series of SEPA waste heat points, which can be adopted to create the stable base temp needed for the proposed loop.

Wind Turbines
Generate renewable electricity feeding directly into the substation network.

Transformer Units
Transformer units stepping voltage up or down to manage wind-generated power, EfW electricity, and distribution across the district.

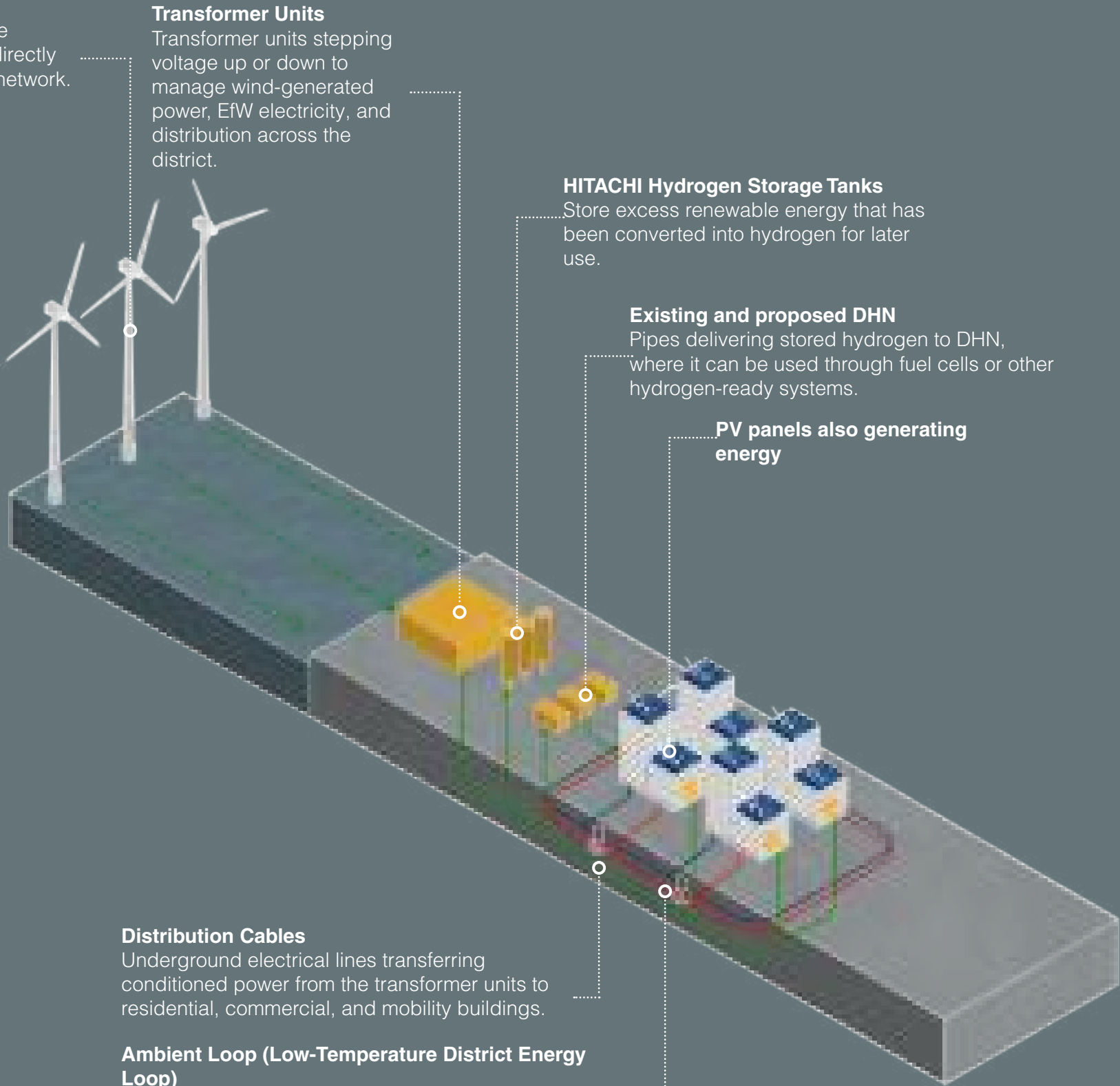
HITACHI Hydrogen Storage Tanks
Store excess renewable energy that has been converted into hydrogen for later use.

Existing and proposed DHN
Pipes delivering stored hydrogen to DHN, where it can be used through fuel cells or other hydrogen-ready systems.

PV panels also generating energy

Distribution Cables
Underground electrical lines transferring conditioned power from the transformer units to residential, commercial, and mobility buildings.

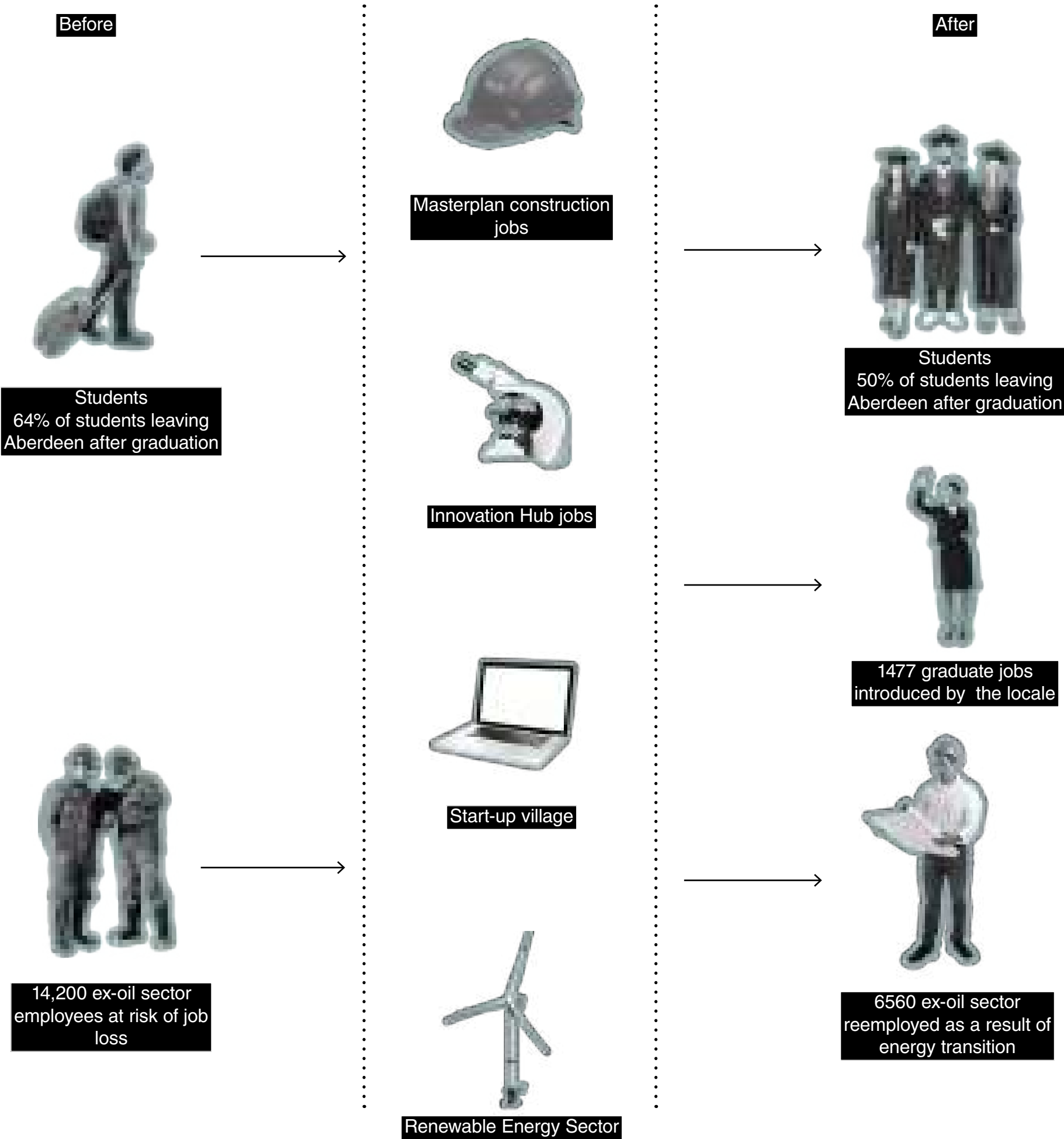
Ambient Loop (Low-Temperature District Energy Loop)
A shared 5th-generation low-temperature loop circulating water through the site; buildings use heat pumps to extract heating or provide cooling as needed. The loop is fed by EfW heat, renewable sources, and rejected building heat.



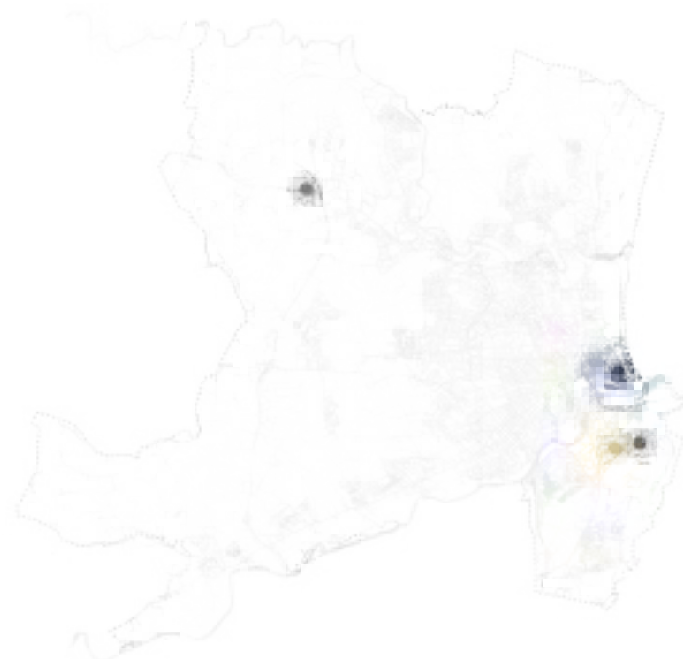
Skills Transition

Employment

Aberdeen faces challenges: 64% of students leave after graduation, and 14,200 oil-sector workers risk job loss as the industry declines. Our masterplan redirects this trajectory by creating new employment pathways in construction, innovation, start-ups, and renewable energy, thus reducing the graduate “Brain drain” to 50%, which is in line with London universities, that lead on the graduate retention aspect, These interventions help retain talent, offering housing for 4926 students as well as 7816 young people and couples. They will benefit the locale through future spending. Locale will also offer jobs for them in the future. The masterplan will also offer opportunities for reskilling of the existing workforce, with 6560 former oil workers transitioning into new green-economy roles.

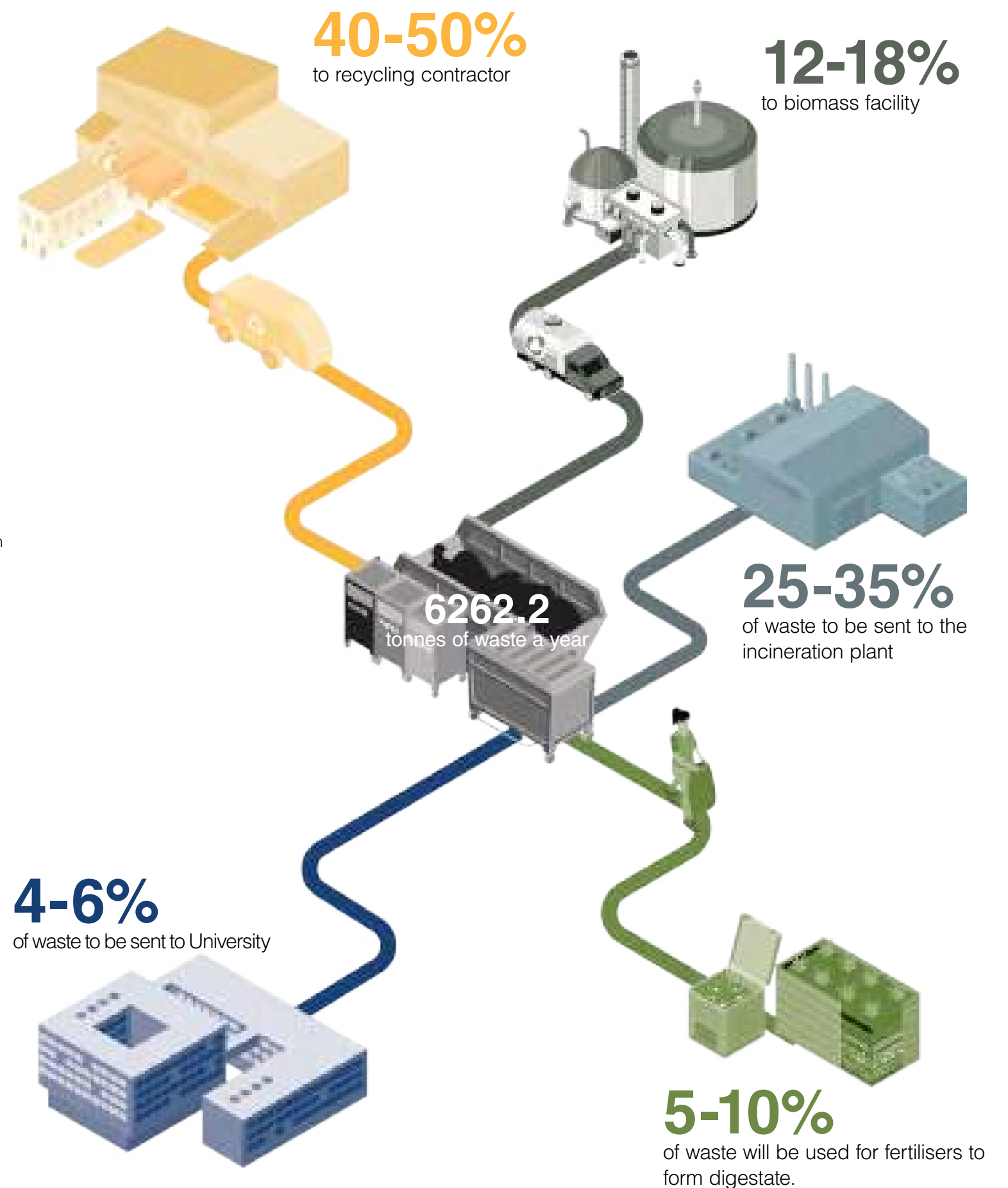


Waste Management



- Tullos Recycling Centre
- NESS Energy from Waste facility
- AD Plant Aberdeen
- Compost Containers through the site
- University Research centre

We will distribute our site's waste across several specialised facilities, prioritising recycling and circular-economy strategies. Recyclable materials go first to the Tullos Recycling Centre for sorting and processing, with some directed to our University Research Hub for material-reuse studies, circular-economy experiments, and design research. A reduced share of residual waste is sent to the NESS Energy from Waste facility, where it generates heat and electricity; metals are recovered and ash is reused as construction aggregate. This remains well below the Scottish incineration benchmark of 45-55%, reinforcing our focus on reuse over disposal. Food waste is sent to AD Plant Aberdeen for anaerobic digestion, producing biogas and nutrient-rich digestate, or used directly as compost/fertiliser.

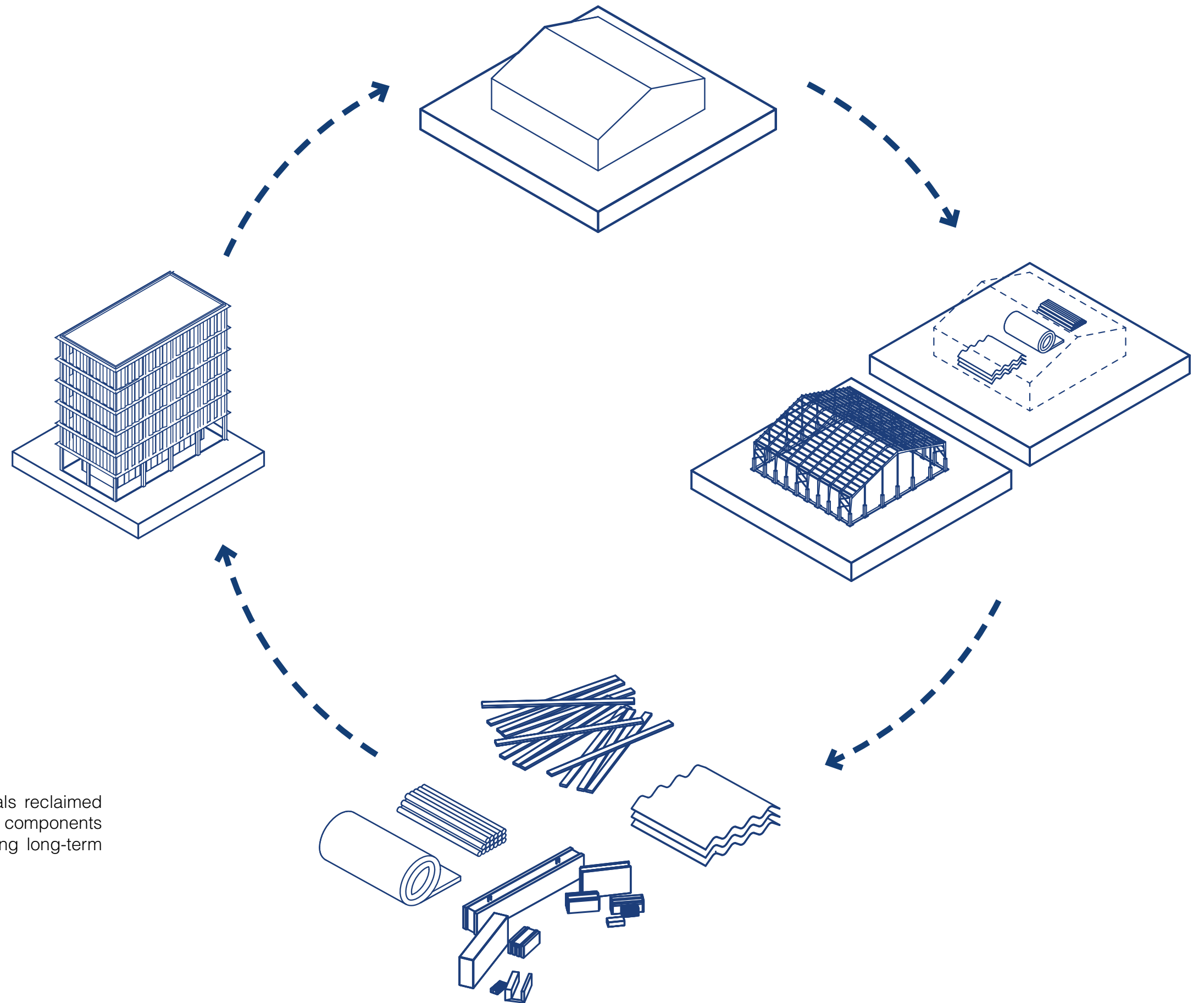


Reuse

Material Strategy

1. Reused Corrugated Steel
Salvaged industrial sheets reused as rainscreens, soffits, canopies and small structures.
2. Reclaimed Slate
Reapplied as cladding and roof accents to reduce carbon and maintain regional character.
3. Reclaimed Timber
Repurposed for façades, linings, kiosks and landscape elements.
4. Reused Slate Shingles
Reapplied as durable façade panels and feature surfaces.
5. Recycled Aluminium Panels
Lightweight aluminium reused or recycled for contemporary façade strips and inserts.
6. Reused Granite
Recovered local granite blocks used for plinths, steps, retaining walls and public-realm edges.





Closed-loop reuse cycle: New buildings use materials reclaimed from on-site demolition and are designed so their own components can be strip-sorted and reused as they age, ensuring long-term circularity and low carbon.

New

Material Strategy

1. Green Roof:
John Chambers Wildflower Seed
(Perthshire)

2. Clay Tiles:
Dunhouse (North East England)

3. Slate Roofing:
Murdoch's Reclaim (Inverurie)

4. Timber Cladding (Scottish Larch
/ Douglas Fir):
James Jones & Sons (Aboyne)

5. Low-Carbon Brick:
Kenoteq / K-Briq (Livingston)
6. Galvanised Metal:
John Lawrie Metals (Aberdeen)

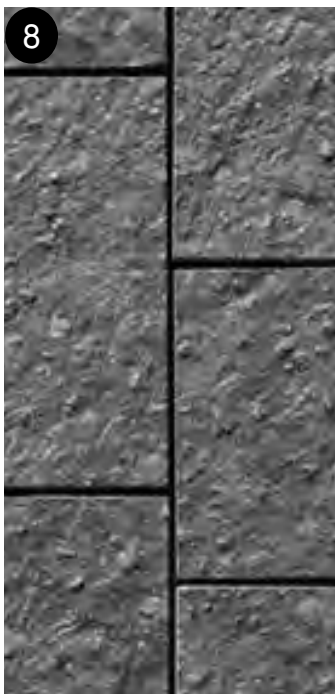
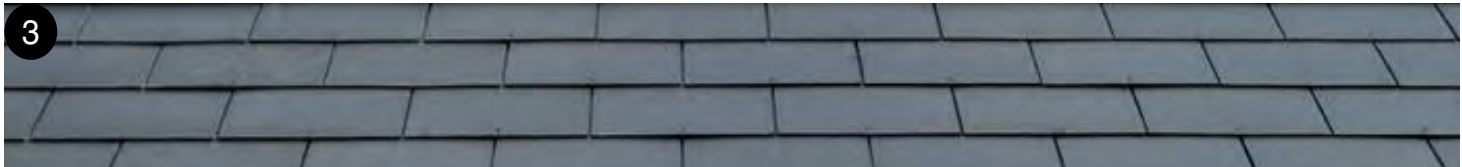
7. Corten Steel:
Murray Steel Products (Westhill)

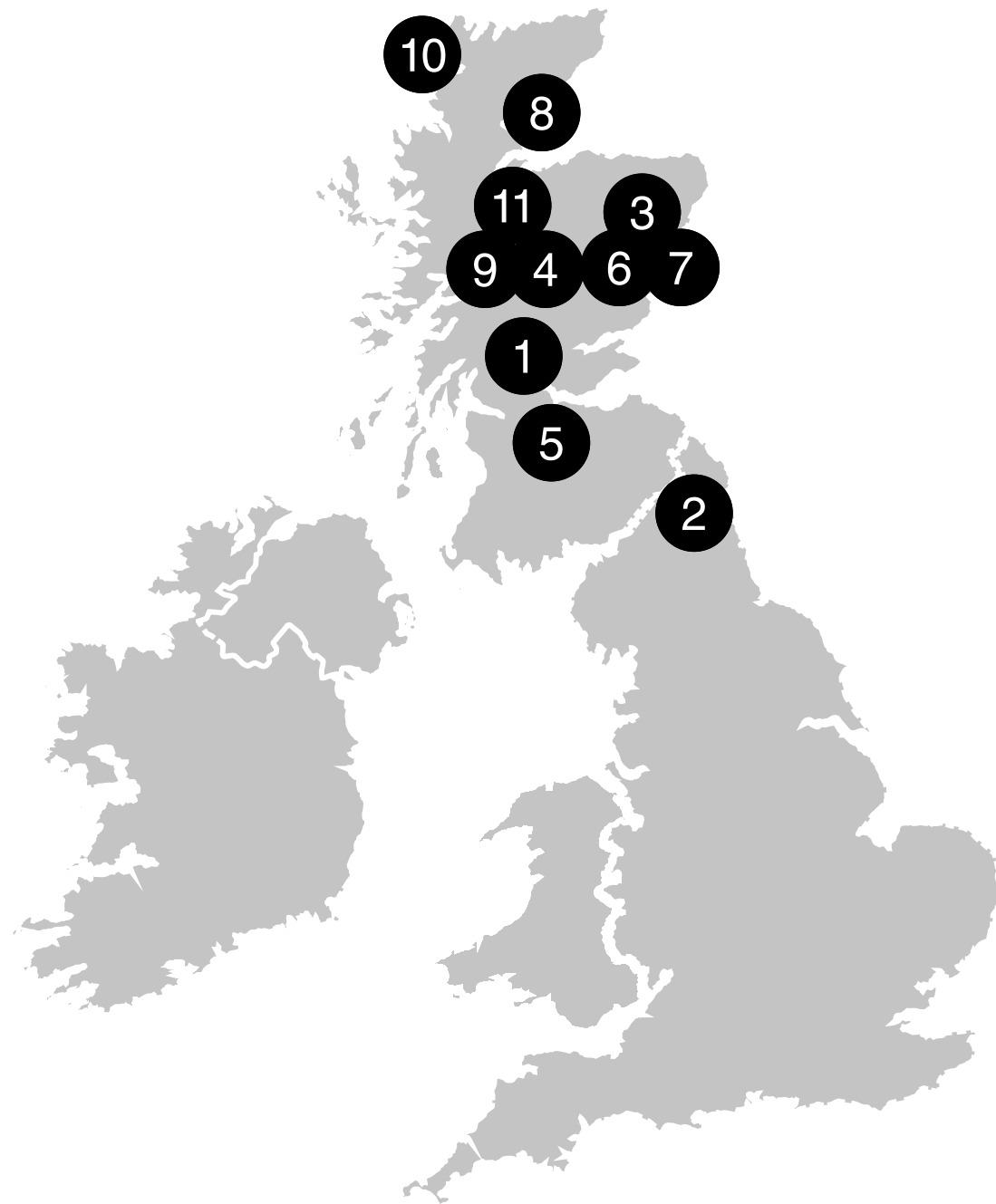
8. Caithness Stone:
A & D Sutherland (Caithness)

9. Whinstone:
Craigienlow Quarry (Dunecht)

10. Green Schist:
Assynt (NW Highlands small
quarries)

11. Silver Grey Granite:
Kemnay Quarry (via Fyfe
Glenrock)





	DISTANCE	CARBON/KM
1. Green Roof: Perthshire to Aberdeen	145 KM	14.5 KG CO ₂
2. Clay Tiles: North East England to Aberdeen	410 KM	41 KG CO ₂
3. Slate Roofing: Inverurie to Aberdeen	26 KM	2.6 KG CO ₂
4. Timber Cladding: Aboyne to Aberdeen	53 KM	5.3 KG CO ₂
5. Low-Carbon Brick: Livingston to Aberdeen	200 KM	20 KG CO ₂
6. Galvanised Metal: Aberdeen to Aberdeen	5-10 KM	1 KG CO ₂
7. Corten Steel: Westhill to Aberdeen	12 KM	1.2 KG CO ₂
8. Caithness Stone: Caithness to Aberdeen	250 KM	25 KG CO ₂
9. Whinstone: Dunecht to Aberdeen	21 KM	2.1 KG CO ₂
10. Green Schist: NW Highlands small quarries to Aberdeen	180 KM	18 KG CO ₂
11. Silver Grey Granite: Kemnay Quarry to Aberdeen	27 KM	2.7 KG CO ₂

Phase 3

2040-2050

Locale Revitalisation

Regional

- Golf course biodiversity
- Surface water flooding interventions

City

- Sponge Streets
- SUDS
- Cycle routes
- Beachfront

Locale

- Mearn's Quay - Residential
- Mearn's Quay - Landscape
- Traffic calming and shared streets
- Mearn's Quay - Promenade



Golf Course Bounds

Before & After

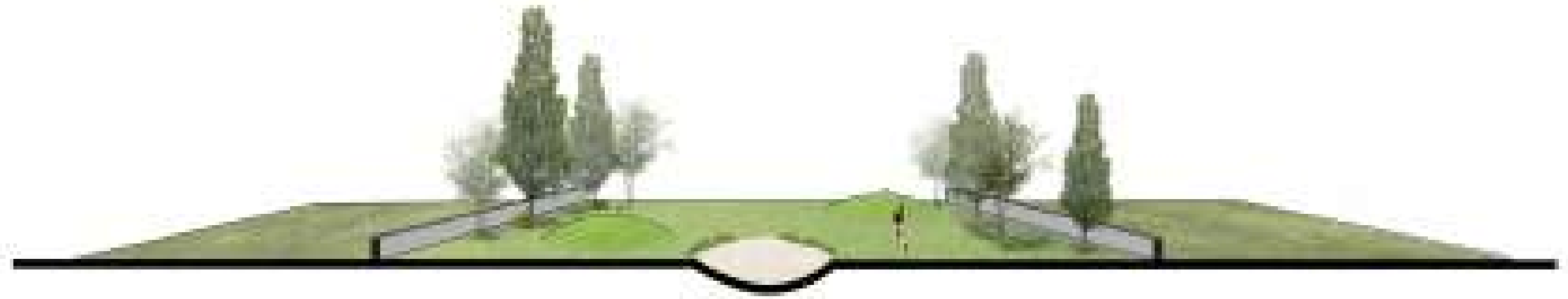


It is proposed that the existing golf courses be integrated with the surrounding public green spaces to establish a continuous ecological and recreational corridor. This intervention forms the first layer of the broader strategic approach aimed at containing urban sprawl while simultaneously enhancing the provision of accessible green space for local communities.

By softening and reconfiguring the currently rigid boundaries of the golf course, an edge condition that often functions

as an exclusionary barrier, the proposal seeks to create more porous, interconnected landscapes. In doing so, it not only improves spatial continuity and ecological performance, but also supports more equitable patterns of urban access and movement across the wider area.





Pedestrians blocked by
boundary wall

Plants that recover
rapidly from traffic

Pedestrians blocked by
boundary wall

Soften the Edge



Boardwalk Swale

Sand Mangrove and
Wild Meadow

Fescus-dominant
sward

Swale Boardwalk

Diversify Golf Course

Before & After

This masterplan outlines a strategic transition for Aberdeen's golf courses, moving them from high-input, low-diversity monoculture spaces dominated by non-native grasses like *Poa annua* and reliance on chemicals, to resilient, species-rich areas rooted in the local ecology. The core strategy is to use the existing sandy links soil as an asset to establish low-maintenance, deep-rooted Fine Fescue turf on playing surfaces while converting large out-of-play areas into diverse, native habitats. This includes creating

wildflower meadows, native scrub (Gorse/Broom) thickets, and functioning bioswales using local water-meadow species. The result is a significant reduction in maintenance costs, chemical use, and water consumption, leading to a substantial increase in flora and fauna biodiversity and positioning the courses as vital, sustainable ecological assets for the region.



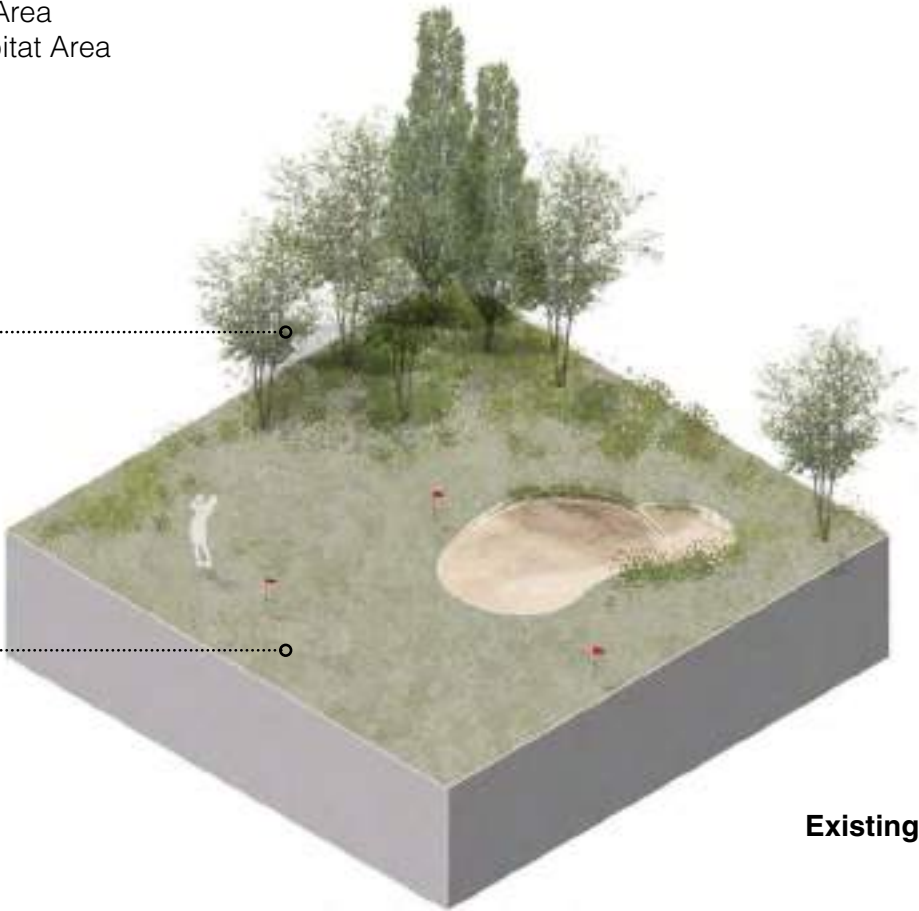
85% Monoculture Area
15% Non-Play Habitat Area

Shelter Belt

Creates dense, dark shade, suppressing most ground flora and native vegetation.

Play Surface

Plants selected for playability, uniformity, and rapid recovery from traffic, typically necessitating higher inputs (fertilisers, water, fungicides).



Existing



60% Monoculture Area
40% Non-Play Habitat Area

Non-Play Habitat Corridor

Shelter Habitat Areas, created in boundaries or unusable corners to provide crucial cover and food sources for birds and mammals.

Structural Diversity

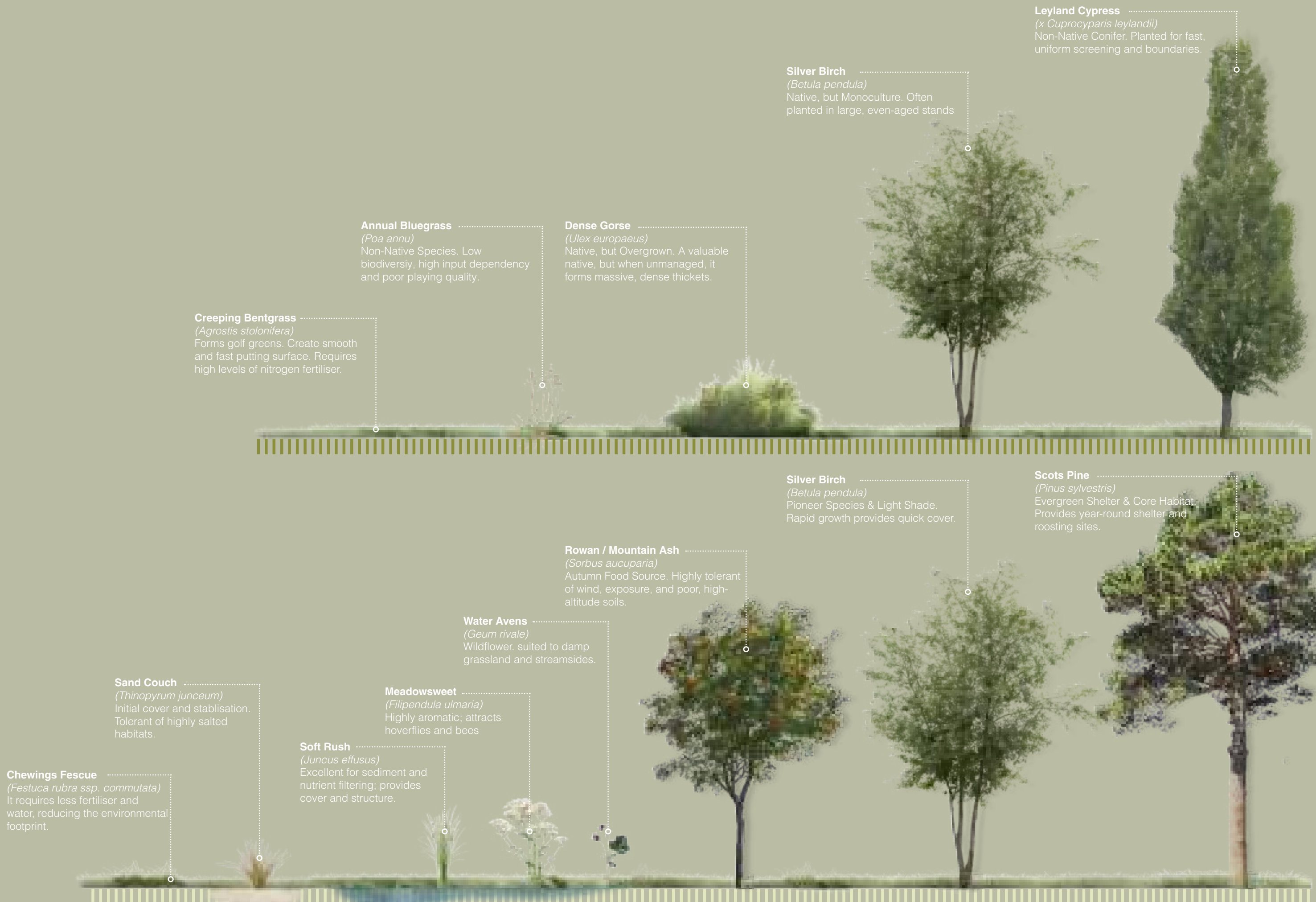
Wild meadow area. These areas only need one cut-and-collect per year (late summer/autumn), vastly reducing maintenance and encouraging wildflowers.

Transitioning Playing Surface

Create a Fescue-dominant sward, which is naturally resilient and requires less input.



Proposed



Chewings Fescue
(Festuca rubra ssp. commutata)
It requires less fertiliser and water, reducing the environmental footprint.

Sand Couch
(Thinopyrum junceum)
Initial cover and stabilisation. Tolerant of highly salted habitats.

Soft Rush
(Juncus effusus)
Excellent for sediment and nutrient filtering; provides cover and structure.

Meadowsweet
(Filipendula ulmaria)
Highly aromatic; attracts hoverflies and bees

Water Avena
(Geum rivale)
Wildflower. suited to damp grassland and streamsides.

Rowan / Mountain Ash
(Sorbus aucuparia)
Autumn Food Source. Highly tolerant of wind, exposure, and poor, high-altitude soils.

Silver Birch
(Betula pendula)
Pioneer Species & Light Shade. Rapid growth provides quick cover.

Scots Pine
(Pinus sylvestris)
Evergreen Shelter & Core Habitat. Provides year-round shelter and roosting sites.

Creeping Bentgrass
(Agrostis stolonifera)
Forms golf greens. Create smooth and fast putting surface. Requires high levels of nitrogen fertiliser.

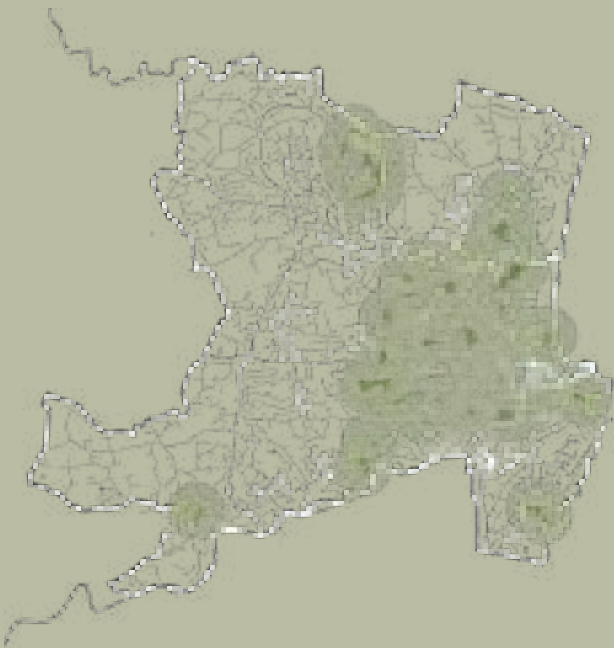
Annual Bluegrass
(Poa annu)
Non-Native Species. Low biodiversity, high input dependency and poor playing quality.

Dense Gorse
(Ulex europaeus)
Native, but Overgrown. A valuable native, but when unmanaged, it forms massive, dense thickets.

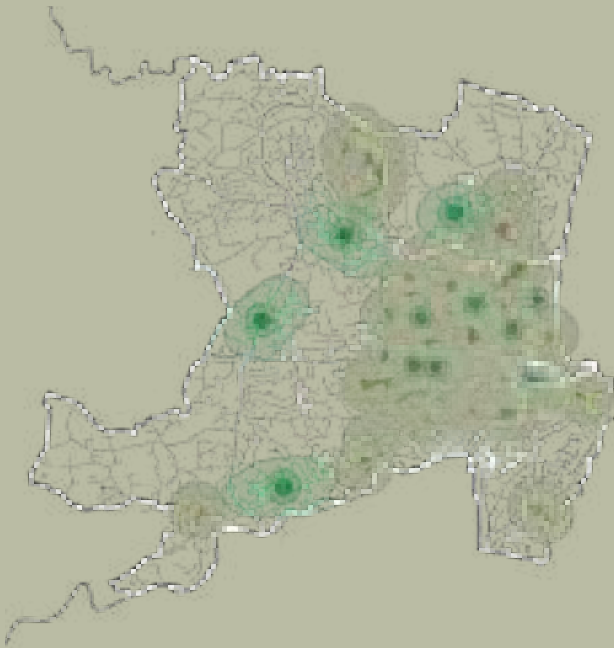
Silver Birch
(Betula pendula)
Native, but Monoculture. Often planted in large, even-aged stands

Leyland Cypress
(x Cuprocyparis leylandii)
Non-Native Conifer. Planted for fast, uniform screening and boundaries.

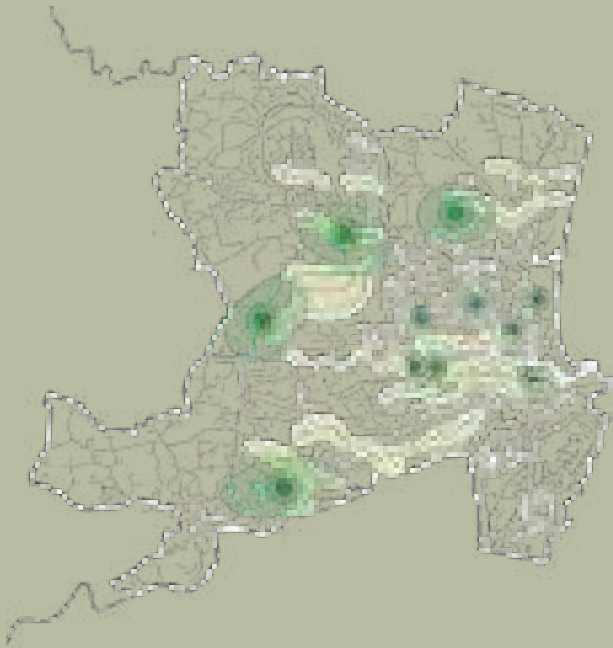
Sponge Parks and Streets



Area less than a 10-minute walk to public parks



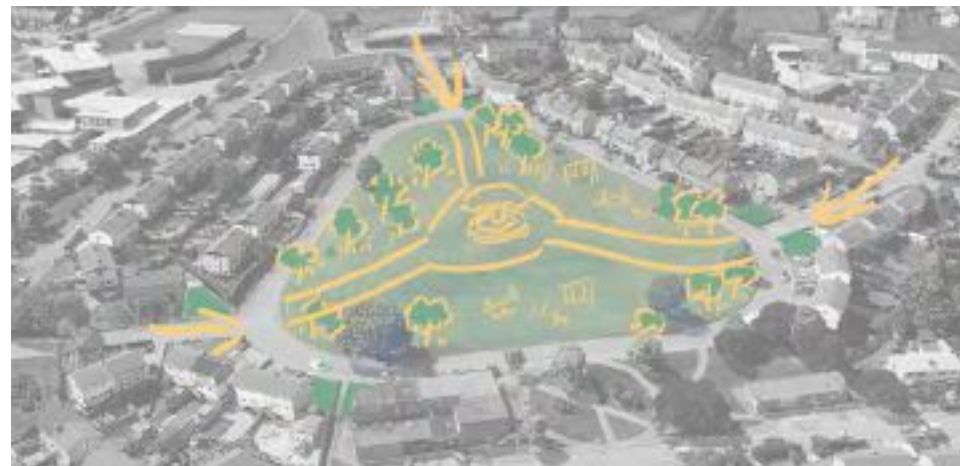
Fill gaps with Parks and provide parks for suburban neighbourhoods



Proposed parks work with Sponge Streets to protect high flood-risk streets

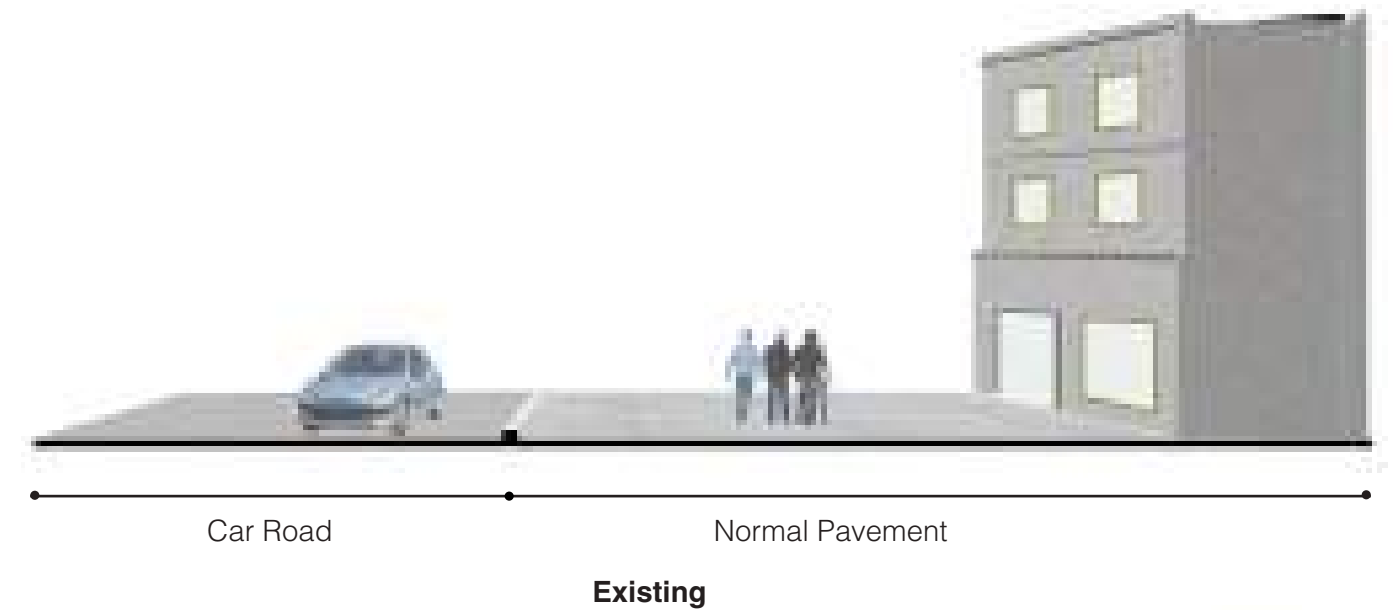
Sponge Design

Surface Water Retention



Transform underused green patches to communal parks for suburban area

Aberdeen has long been affected by severe storms, floodwater levels can reach as high as 1m on the roads. Affected streets are identified and 'Sponge Street' Design is incorporated to alleviate this issue.



Freight Siding repurposed as swale-integrated cycle path

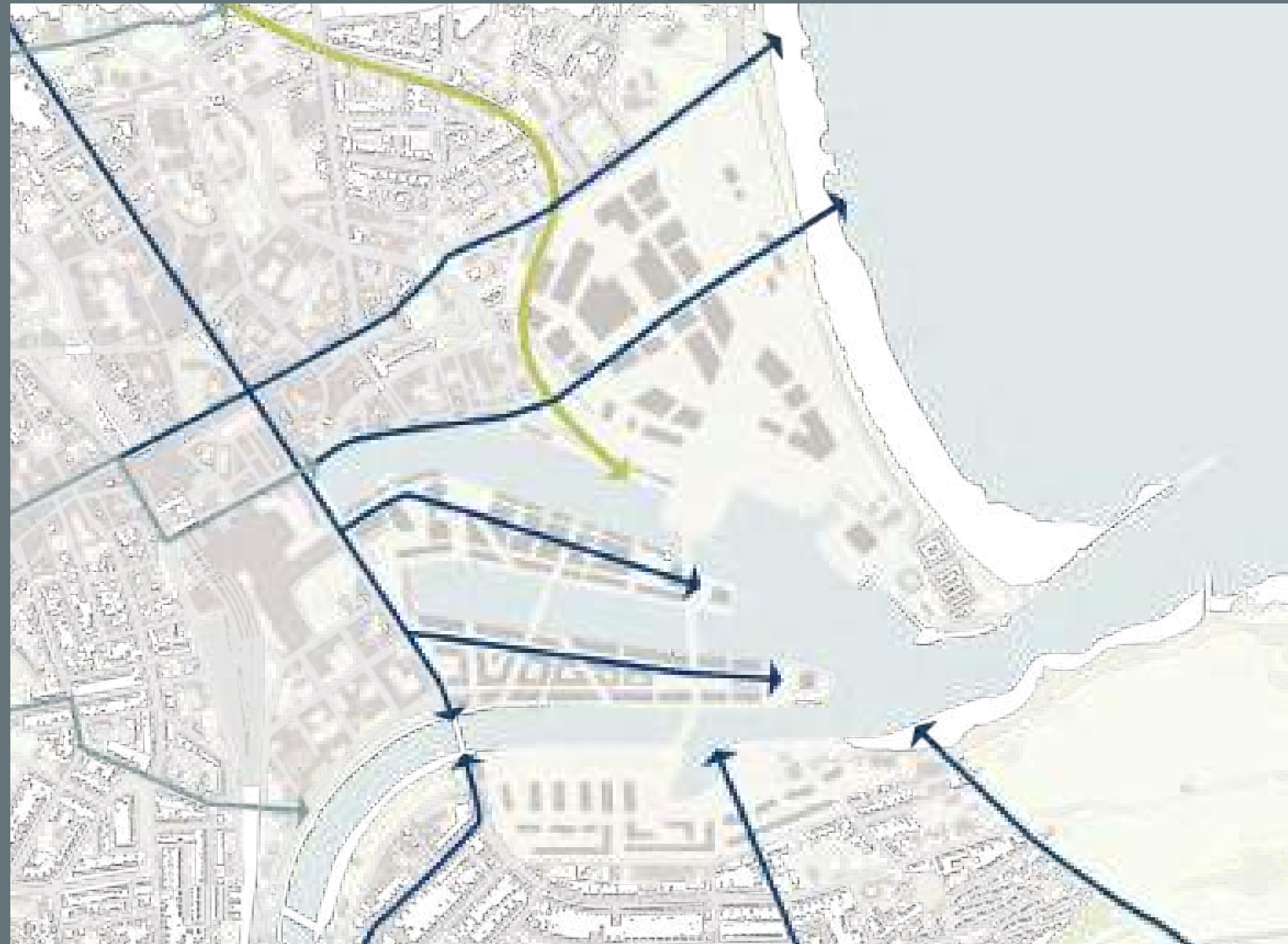
Streets with 1/10 chance of flooding each year, revitalised as sponge streets

Proposed sponge streets at locale



The revitalized and proposed sponge streets have soft landscaping, trees implemented. However they do not only serve as water-retention systems.

The aesthetic of landscape design and climate tiles create inviting urban environments that increase footfall and benefit retail activity.



Locale Sustainable Drainage Systems

Wastewater Treatment

Filtration

To-do List:



Policy Making



Maintain Drainage

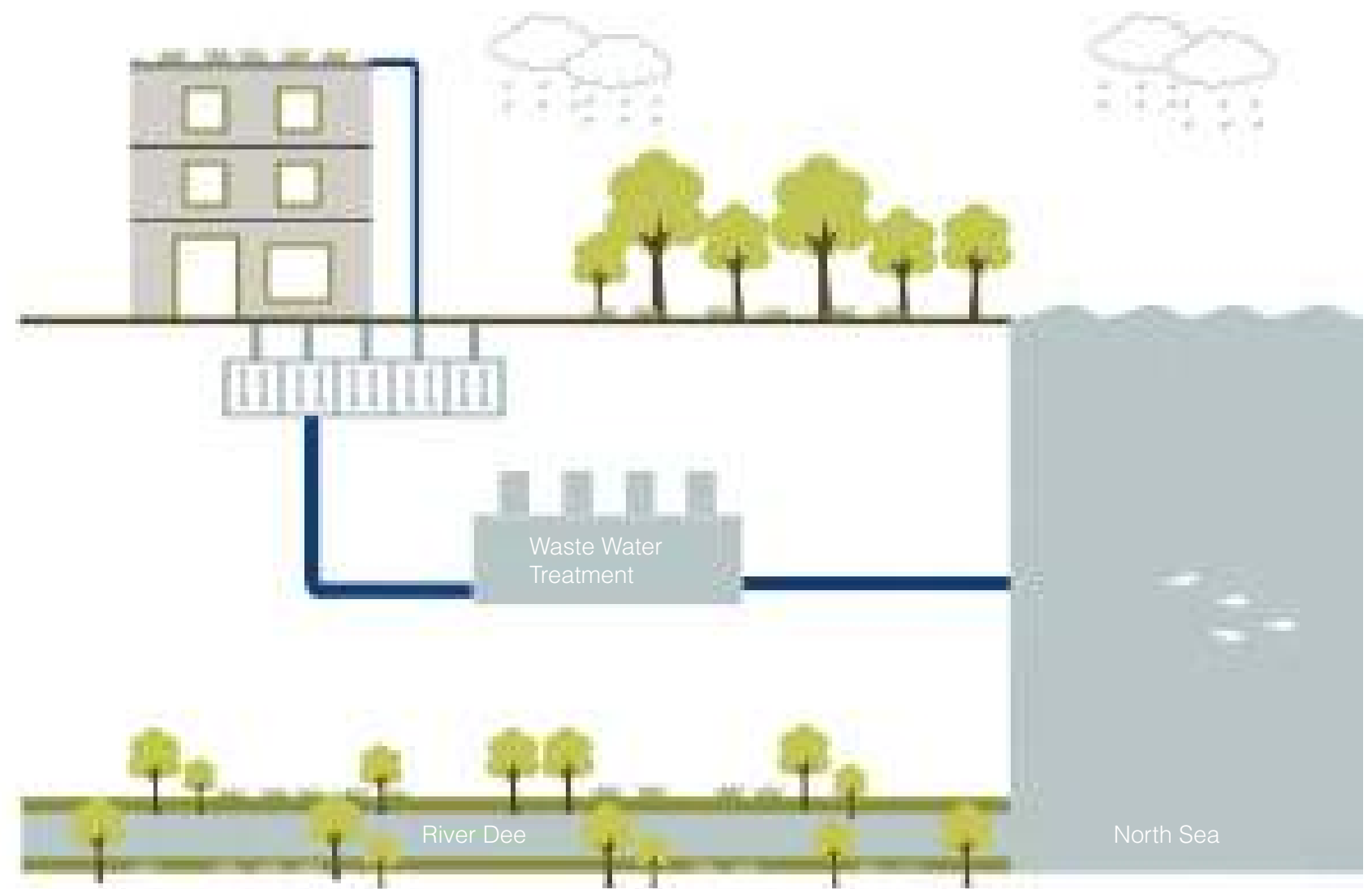


Organic Farming

Storm water is captured by green roofs, sponge streets, and parks, then directed to wastewater filtration facilities before gradually discharged into the sea. Upstream riparian zones and wetlands function as sponge and natural filtration systems, helping to slow down run-offs and discharge cleaner water to the river and sea.

To achieve the above measures, policies must be introduced to regulate agricultural activities, such as limiting the use of fertilisers. Wastewater facilities

and drainage systems need to be properly planned and maintained. Predicted climate change scenario in 2080 by IPCC is considered when determining the volume of floodwater that roads are designed to withstand.



Cycle Route

Before & After

- Expand the low emission zone to encourage more sustainable transport mode to suit the renewable city narrative.
- Vehicles that do not meet the minimum standard will face a penalty if they enter a low emission zone
- Tram and Bus fare control, bike storage to make sustainable travel options more attractive
- Hydrogen fueled bus and tram, reduce air pollution to enhance pedestrian experience

Aberdeen's relatively flat topography might suggest strong potential for active travel, yet the city remains largely unfriendly to cyclists due to a longstanding prioritisation of private vehicles. With the introduction of the Low Emission Zone and the enhancement of public transport infrastructure in the masterplan, cycling is positioned to become a key mode of sustainable mobility within the city.

The proposed cycle route capitalises on the disused

freight siding, repurposing it as a strategic north-south connector that links directly to the existing scenic coastal cycle network. The integrated swales at ground level functioning as part of Aberdeen's surface-water management strategy. Together, these interventions enhance urban permeability, support climate resilience, and encourage a safer and more continuous cycling experience across the city.





Existing



Proposed

- Key**
-  Cycle Route Existing
 -  Cycle Route Proposed
 -  LEZ Existing
 -  LEZ Proposed (Stage 1)
 -  LEZ Proposed (Stage 2)
 -  Locale

South Rail Freight

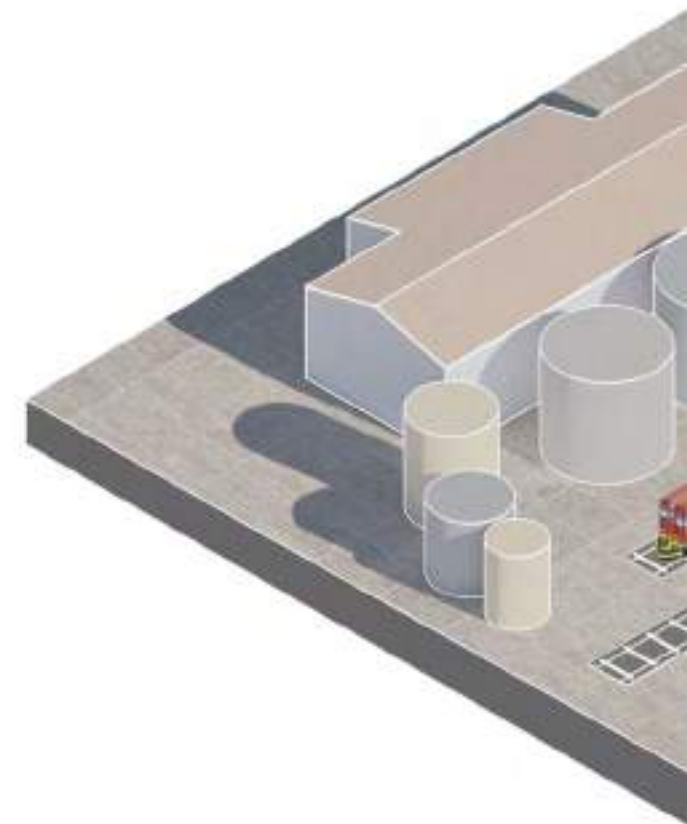


Existing: Disused freight sidings into elevated cycle paths.
This will transform mobility and green infrastructure.

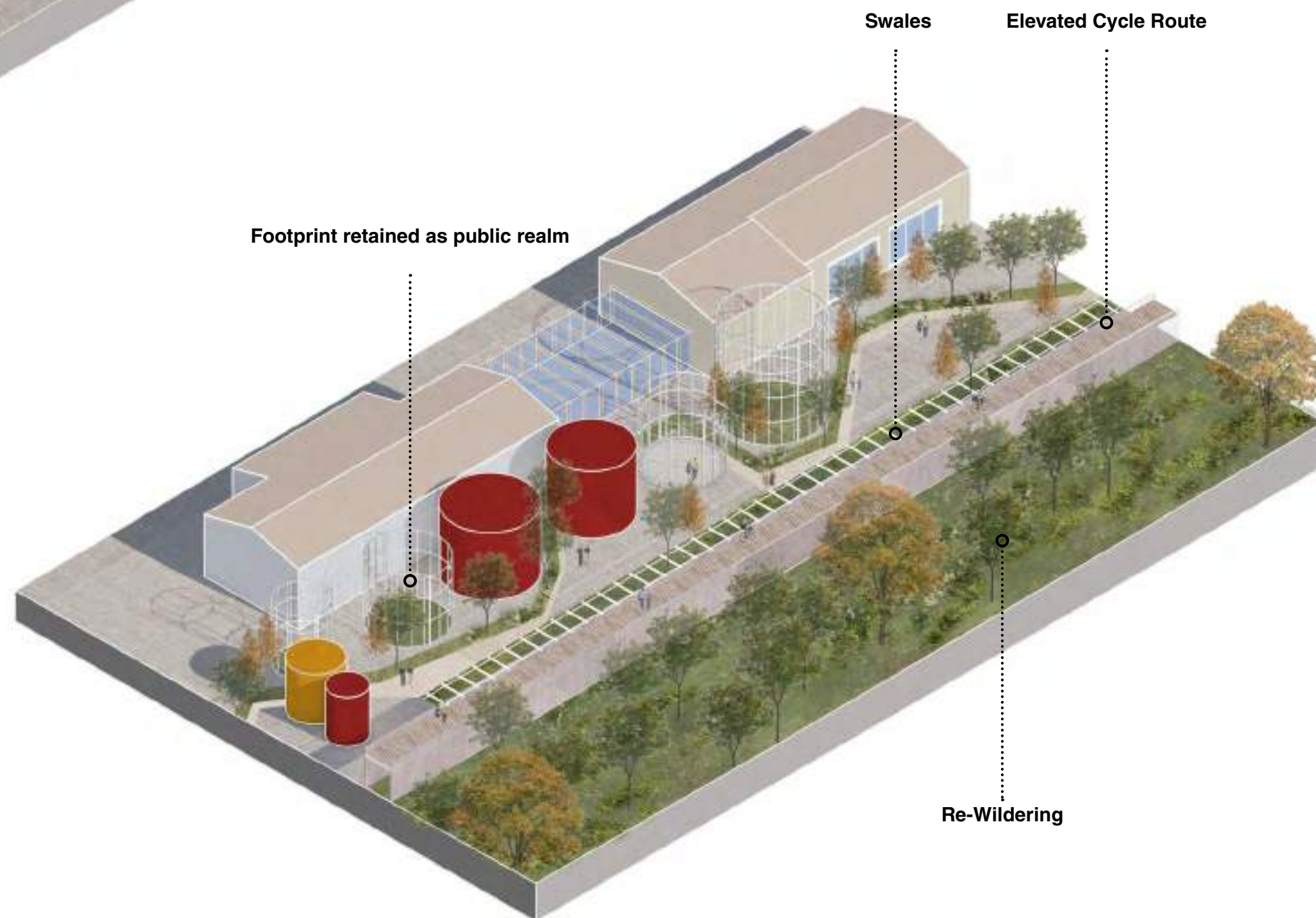
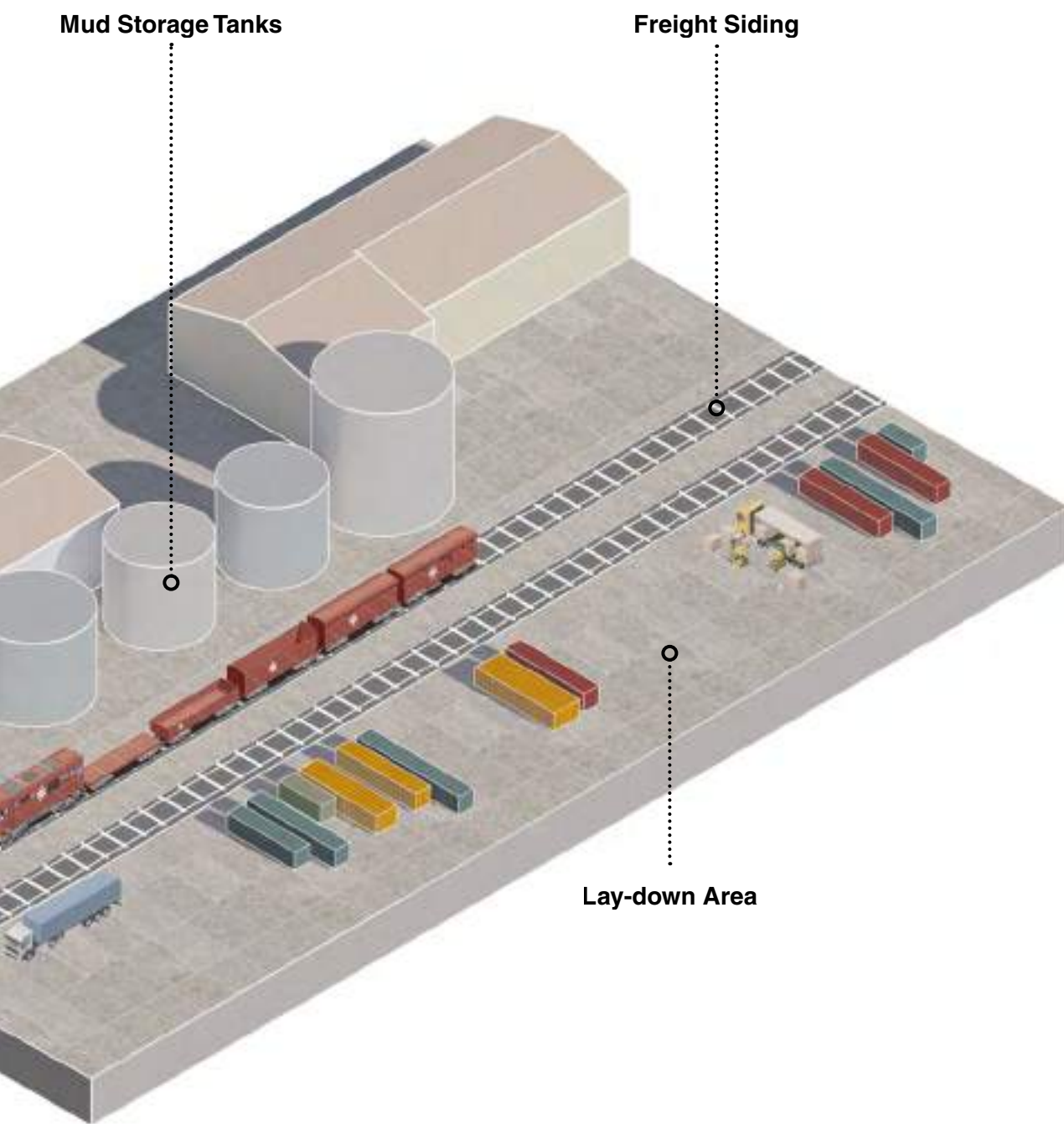
South Rail Freight Cycle



Proposed: Repurposed into Elevated Cycle Path constructed from de-commissioned oil rig structures.
Existing freight rail will be new swale retaining rain water and enhance permeability, reduce risk of surface water flooding.



Cycle Route, Repurpose Freight Siding





Existing: Compromised Pedestrian Experience

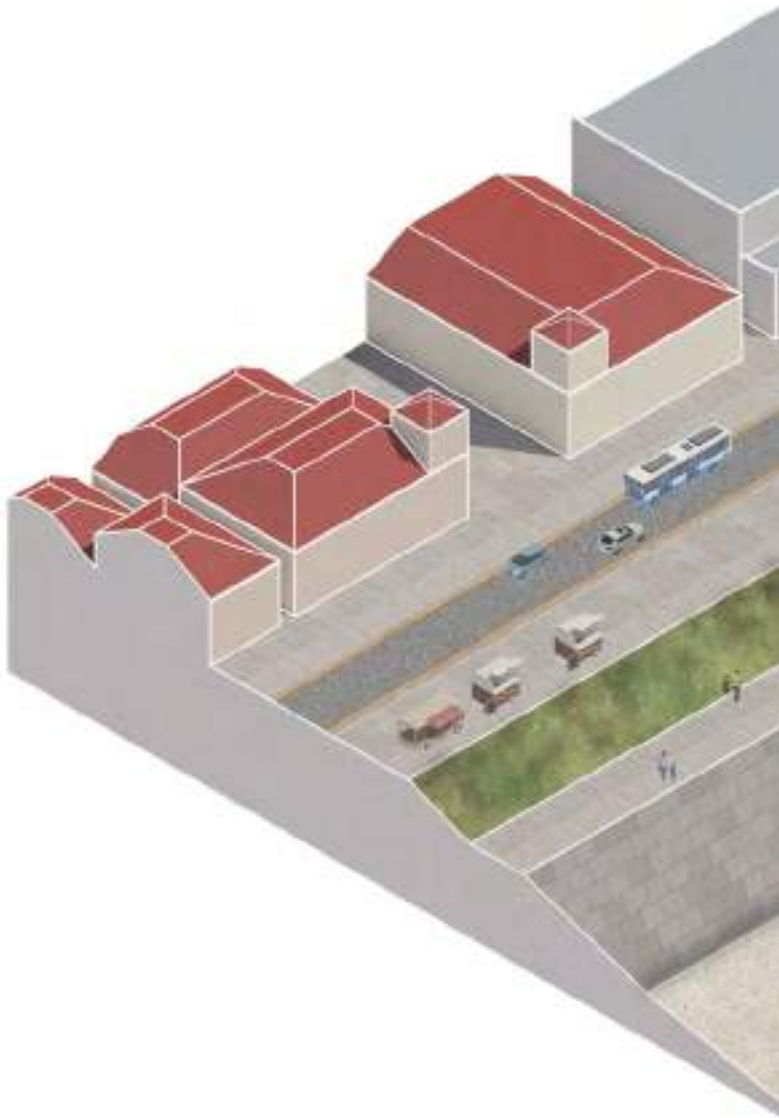
Barren landscape and 80s recreational ground no longer serve the local residents well. Independent businesses like food trucks and sauna pods should be celebrated more.



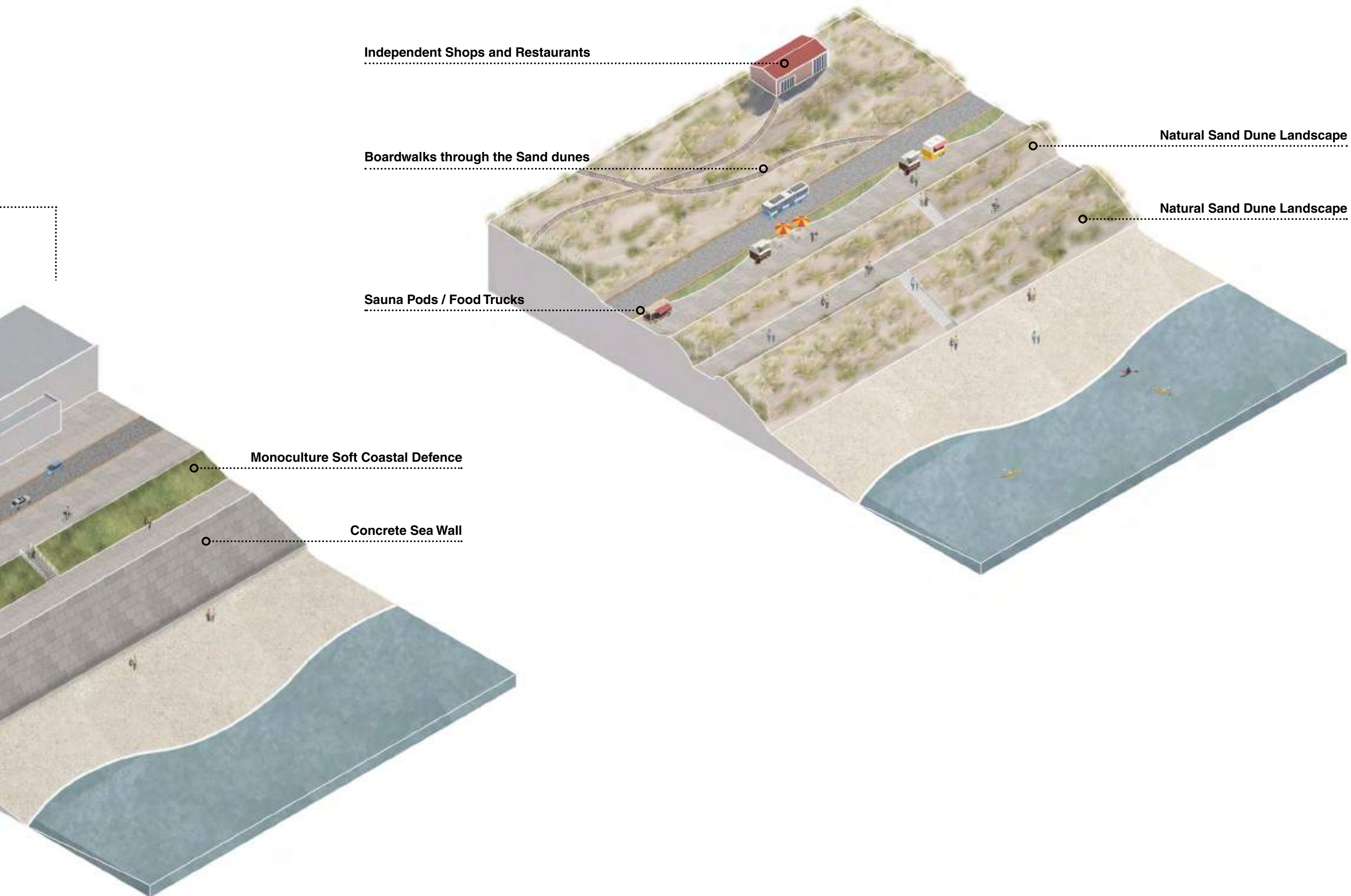
Proposed: Natural Sand Dune Landscape & Beachfront Promenade Introduce the natural sand dune landscape native to north of Aberdeen coastline as soft coastal flooding defence and promote biodiversity. Boardwalks will also be introduced connecting independent businesses dotted around the sand dunes. Pedestrian routes and cycle route will be widened to enhance the beachfront walks.

1980s Retail Park

 Aberdeen Beachfront Leisure Park, constructed in the 1980s with the oil boom. Placeless concrete box filled with restaurants and leisure activities.



Beachfront, Existing and Proposed



Independent Shops and Restaurants

Boardwalks through the Sand dunes

Sauna Pods / Food Trucks

Natural Sand Dune Landscape

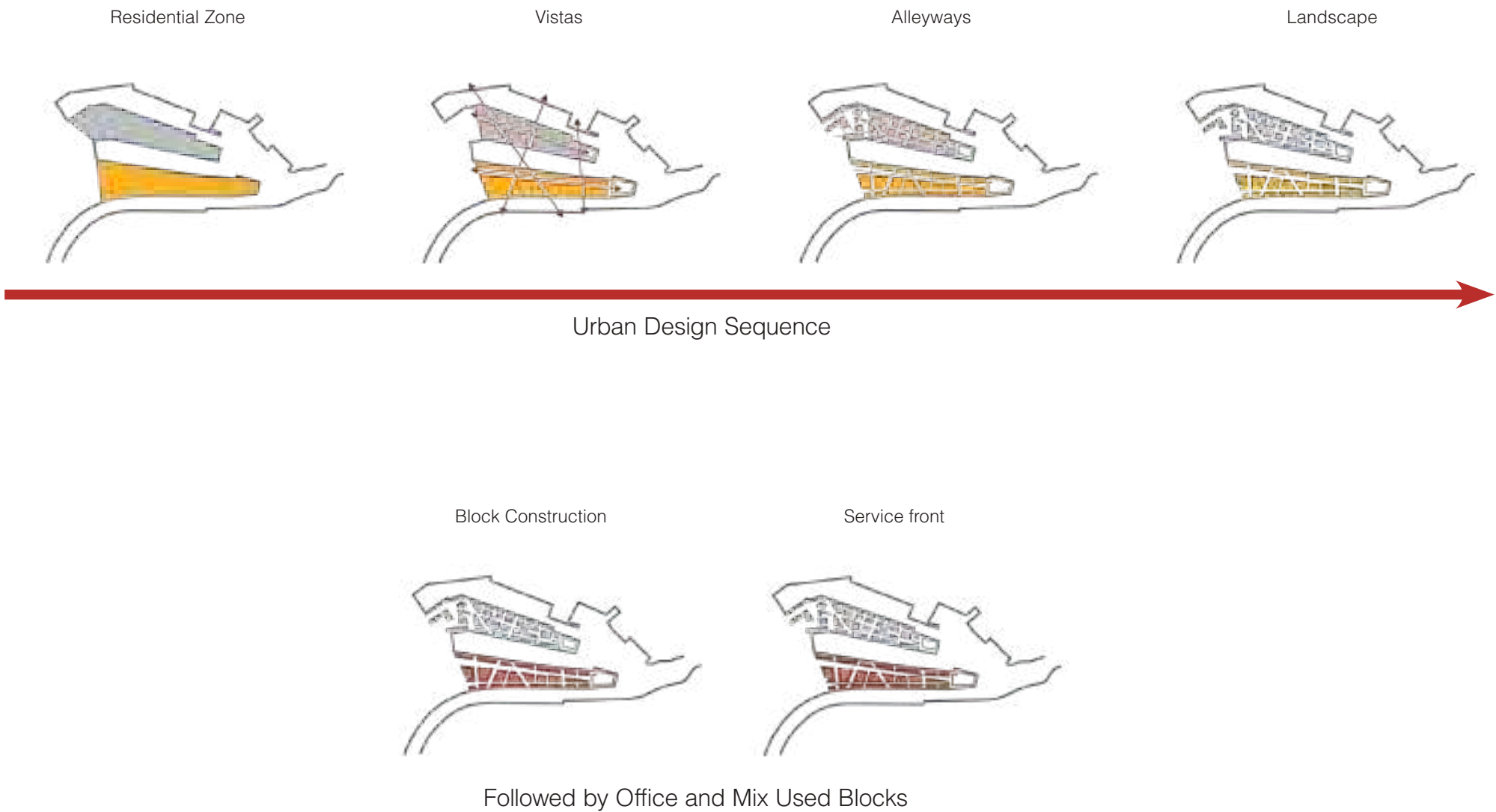
Natural Sand Dune Landscape

Monoculture Soft Coastal Defence

Concrete Sea Wall

Locale - Mearn's Quay

Development Sequence





Mearn's Quay
Within Phase 3

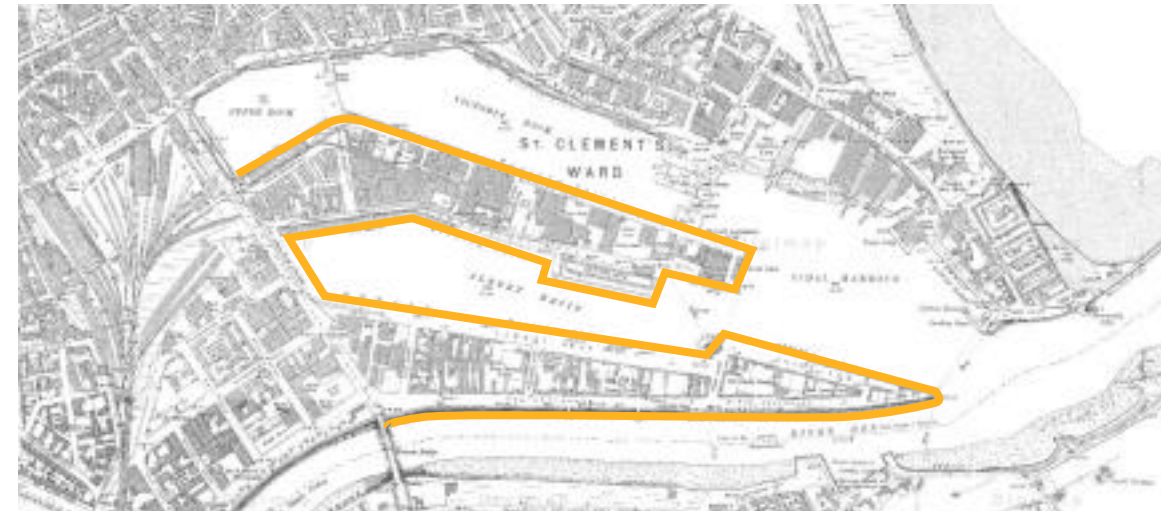
Reinstating History

Historic Footprint of Locale

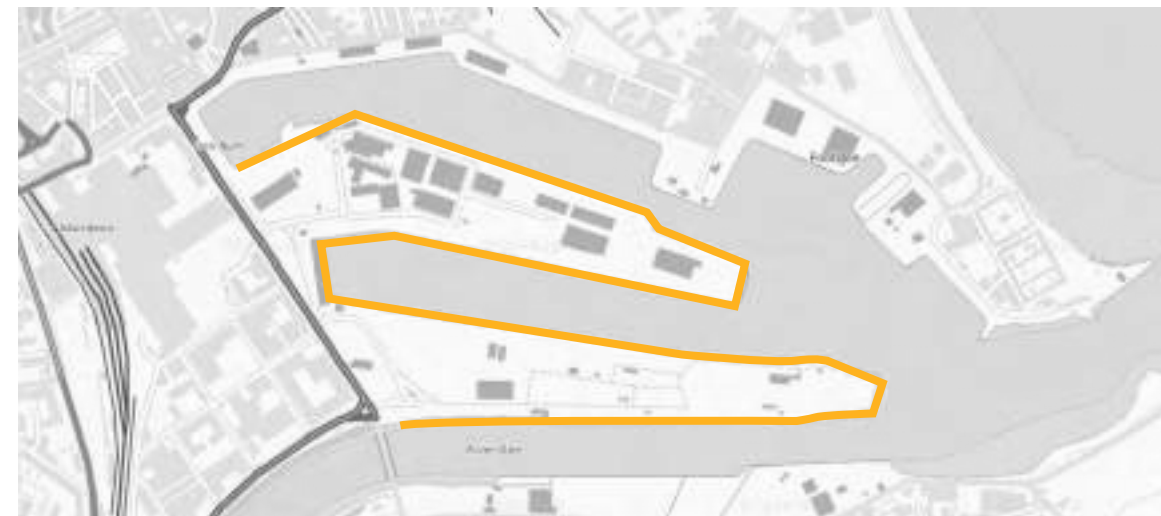


The port is 900 years old, with a long history of shipbuilding and fishing. Following the oil boom of the 1960s, it evolved from serving a fishing fleet to becoming a site where tankers offloaded petroleum into large onshore storage tanks.

This image is taken in 1940s, when Aberdeen was a active fishing port.



Aberdeen Vernacular: Dock Plan in 1900s



Oil Occupation: Existing Dock Plan



Re-densification: Proposed Dock Plan

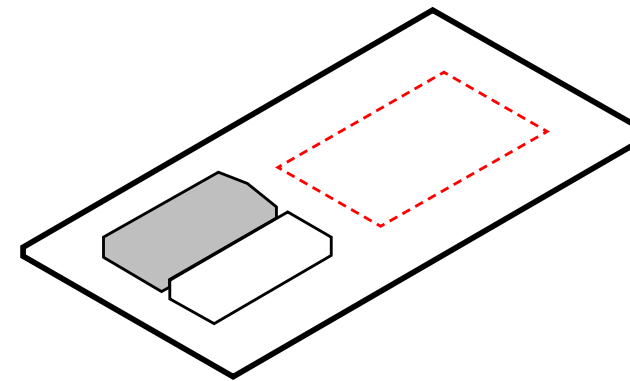
Reinstating History

Reimagine Courtyard Typology

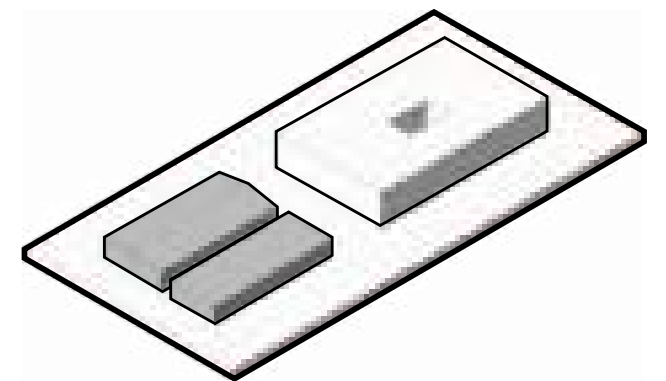


Inspired by the historic fishing village of Footdee, the first example of municipal planned housing in Aberdeen, we offer a reimagined courtyard typology with suitable densities and amenities for modern day living. The varied block heights offer

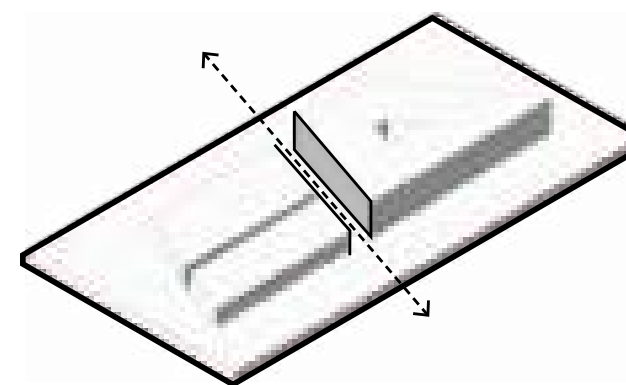
protection from prevailing wind and offer a sense of enclosure.



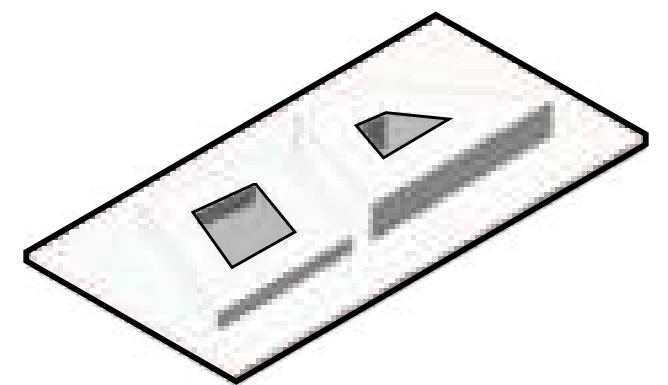
Identify plot and insert mass



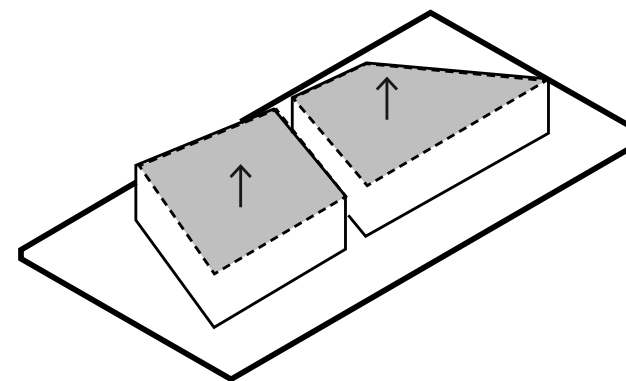
Insert adjacent mass and courtyard



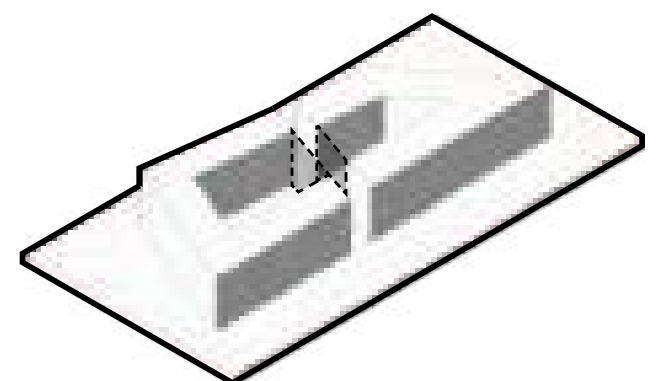
Identify key sight lines



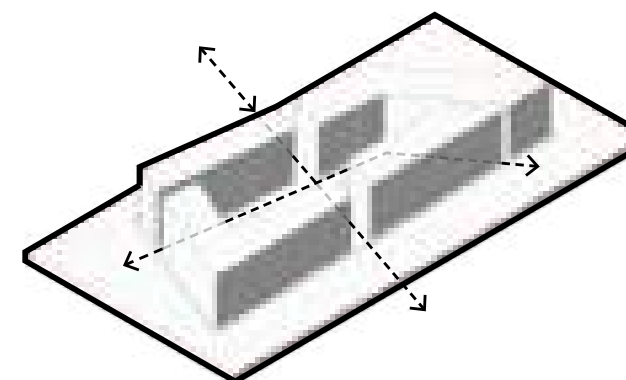
Make virtue of courtyards



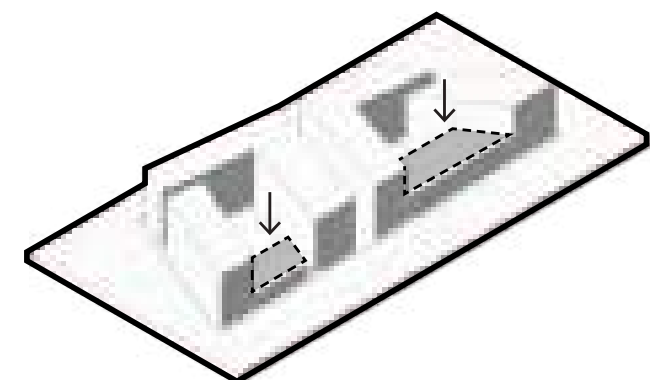
Increase density



Open out courtyards



Reconsider circulation

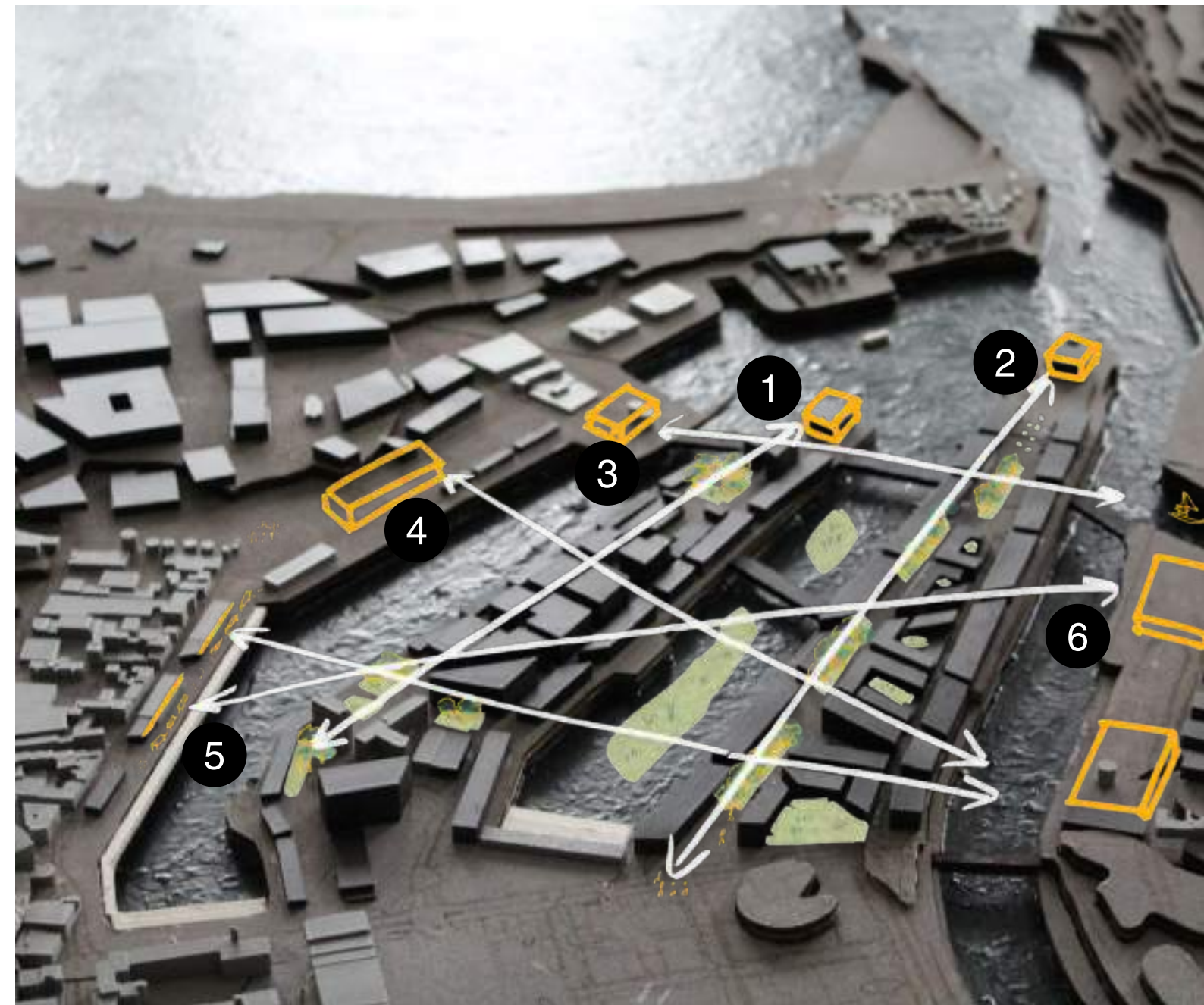


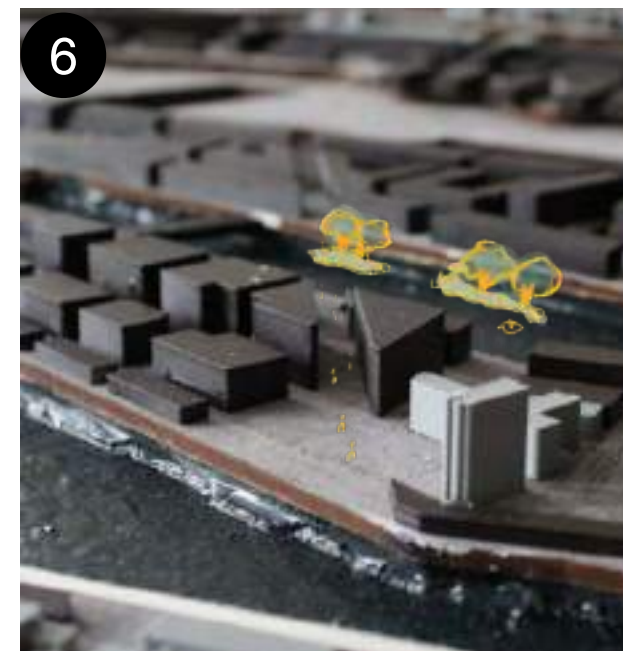
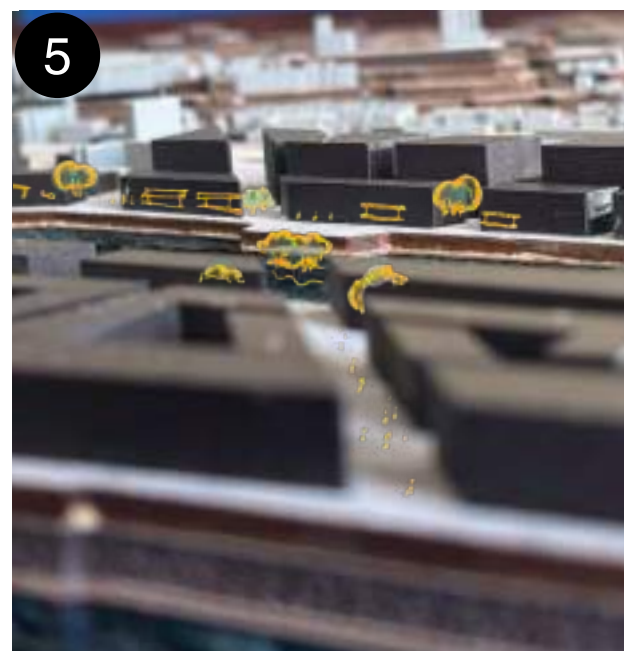
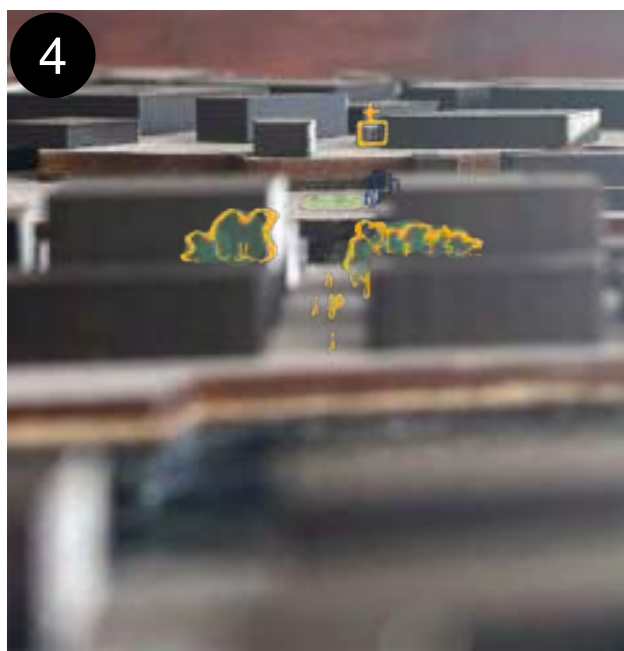
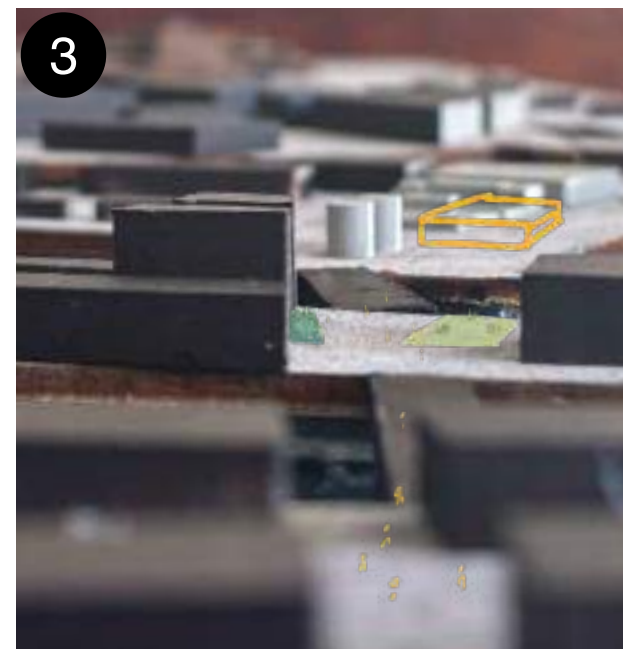
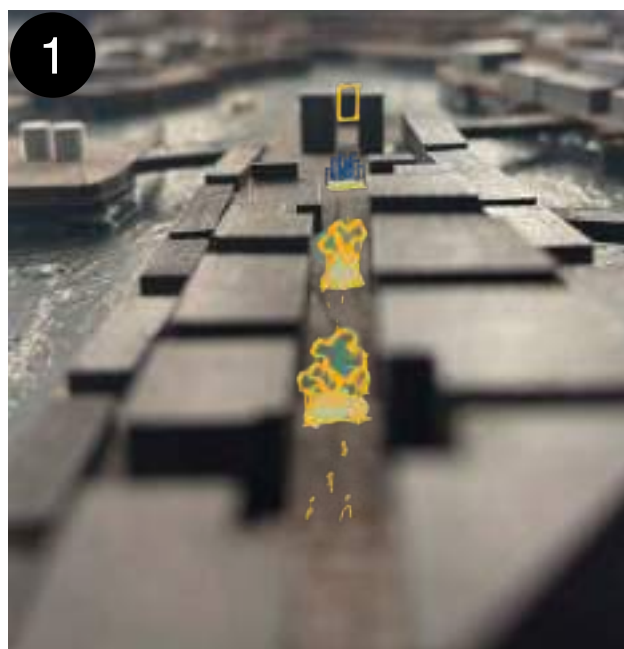
Improve daylighting response

Vistas

View Anchors

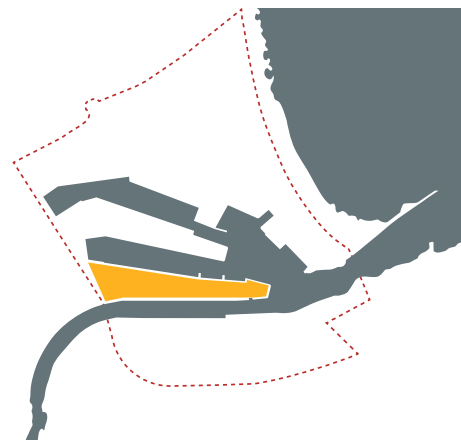
Vistas frame key cultural interventions on the site, currently left mostly untouched for our individual proposals, which will enrich the centre of Aberdeen and will focus on giving back to the public and relationships to water.



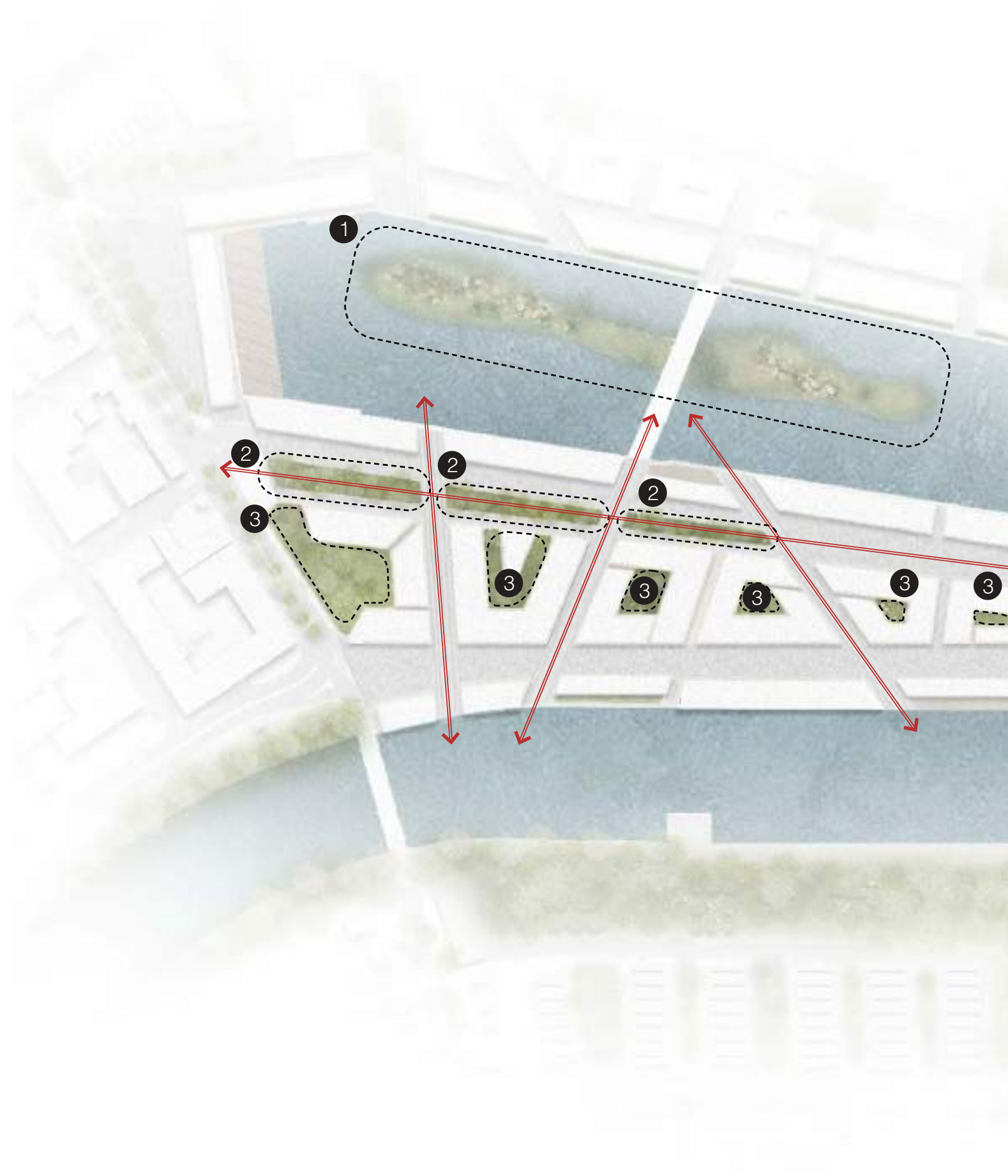


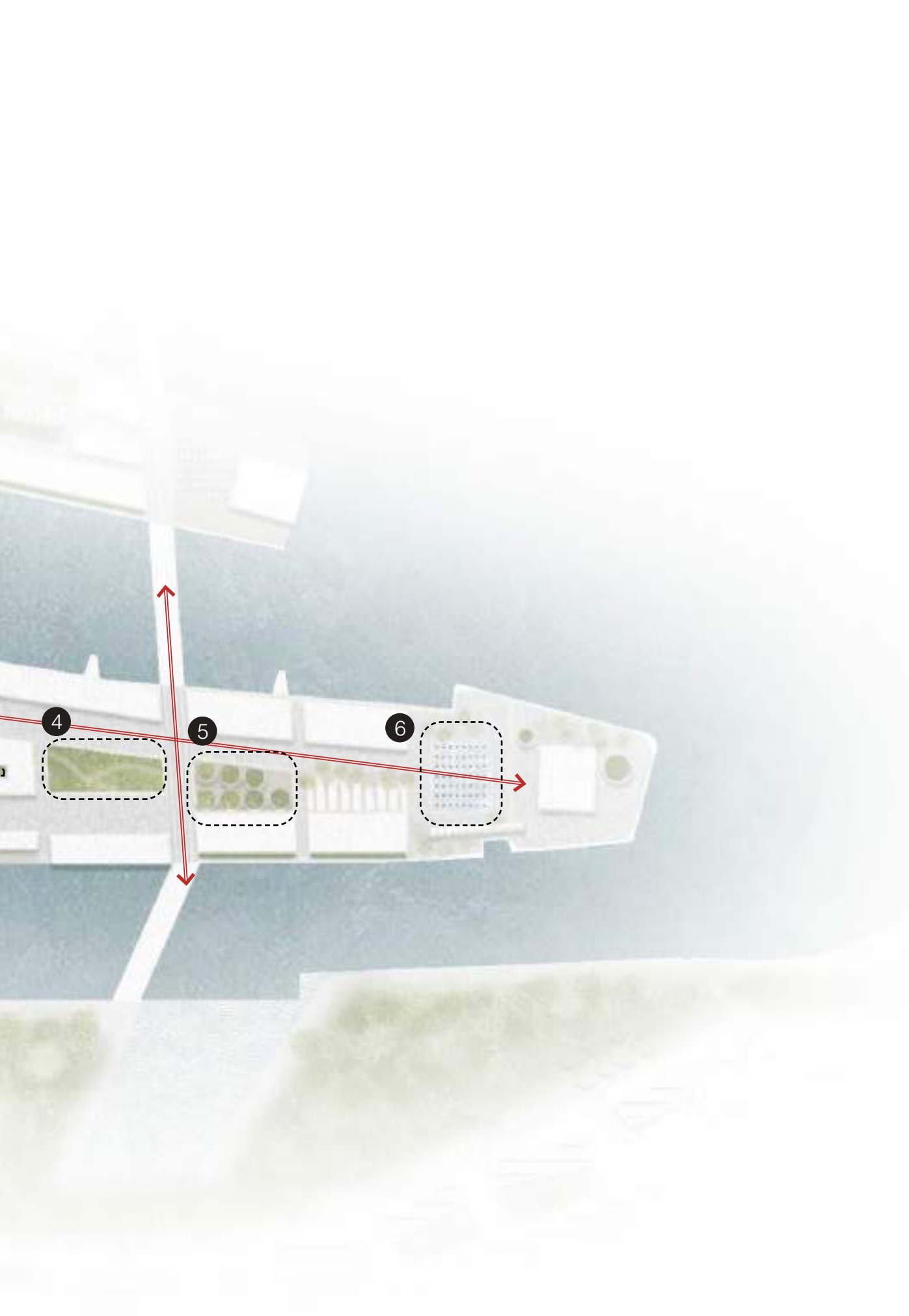
Landscape Catalogue

Mearn's Quay - Landscape

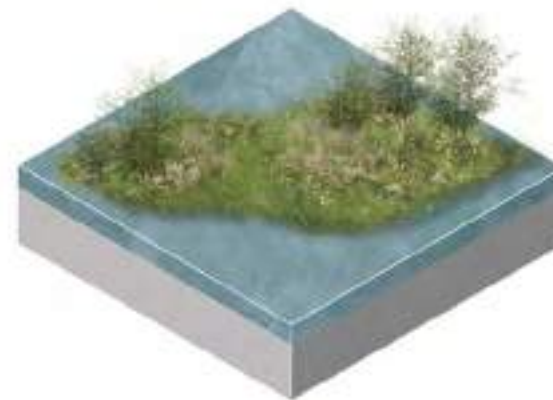


The landscape design aims to respond to the unique scale of the harbour and serve the residents as well as the public simultaneously. With a nod to the history of the harbour, different landscape typologies are introduced. Floating wetland aids water decontamination. Patches of tiny forests provide greenery comparable to the scale of the harbour. Various industrial heritage inspired landscape towards the tip of the Mearn's Quay function as public realm for communal activities and gathering.

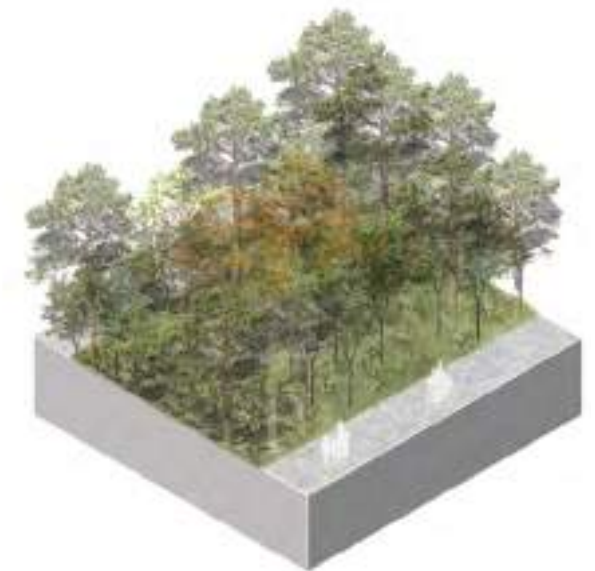




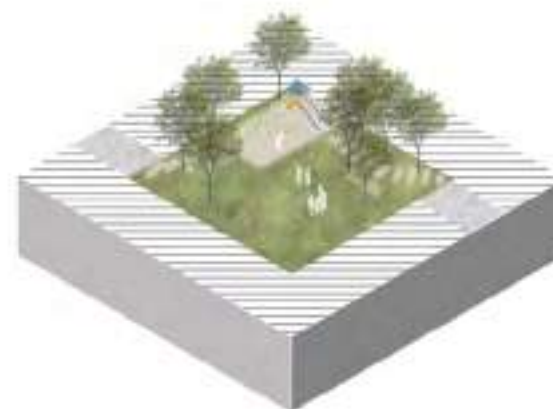
↔ Vista



1 Floating Wetland



2 Tiny Forest



3 Courtyard



4 Pocket Park



5 Industrial Heritage Park

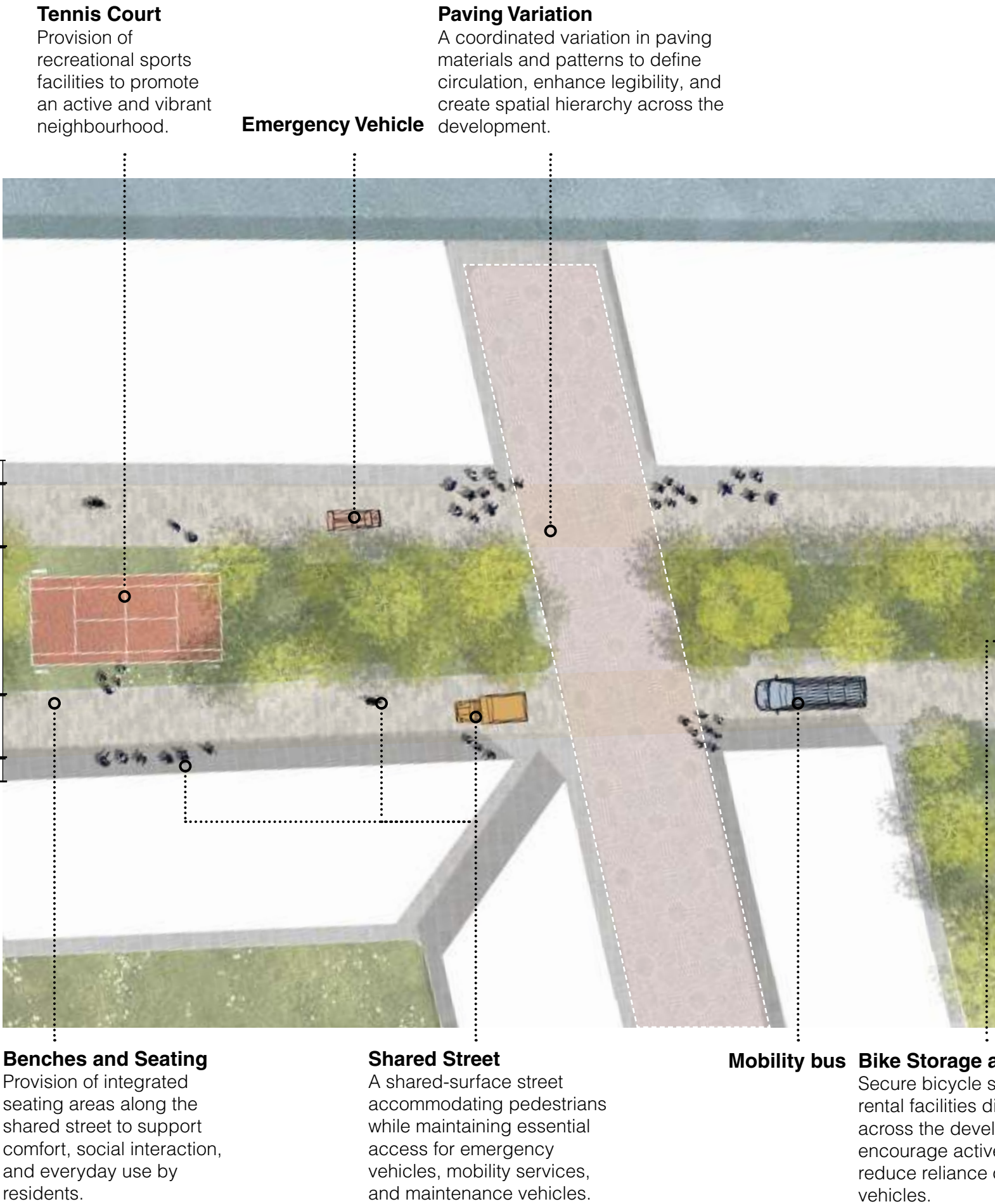


6 Water Fountain

Shared Street

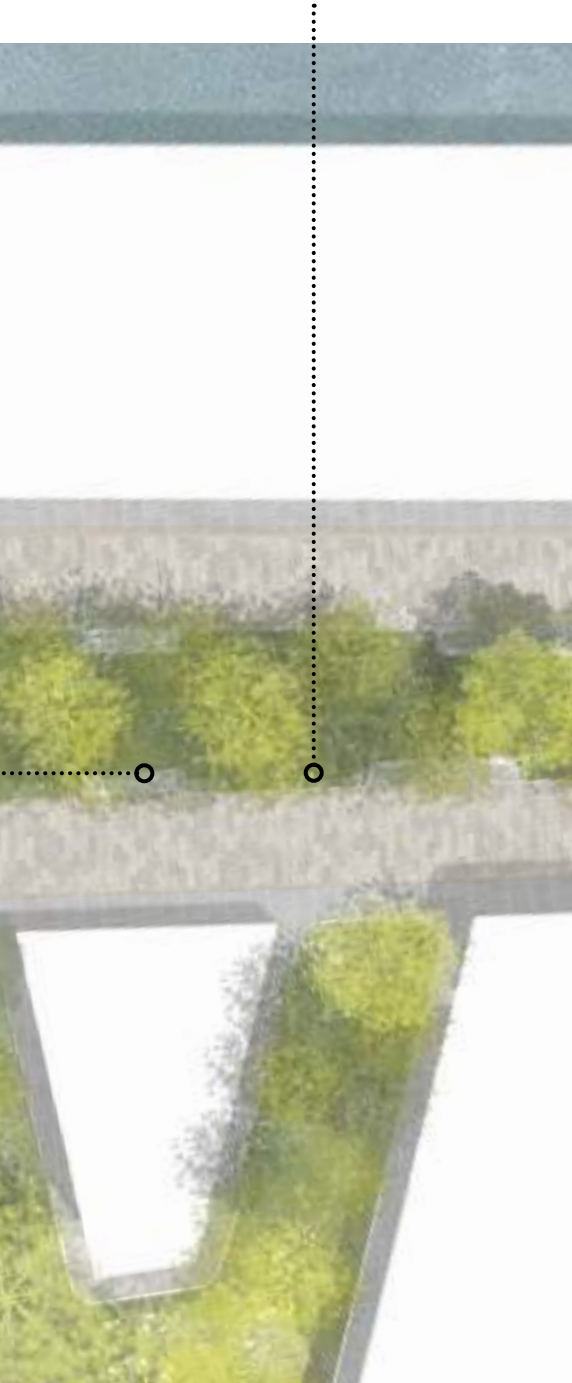
Mearn's Quay - Landscape

Shared street removes the strict separation between pedestrians, cyclists, and vehicles to create a more people-centred public realm. By reducing traffic speeds, and using surface materials, planting, and furniture to guide movement, shared streets encourage drivers to behave more cautiously and attentively. This approach fosters a safer, more inclusive environment that prioritises social interaction and walkability, transforming streets from purely circulatory spaces into vibrant, multifunctional public places.



Tiny Forest

A designated ecological zone where natural growth is prioritised and human access is restricted to support biodiversity and habitat creation.



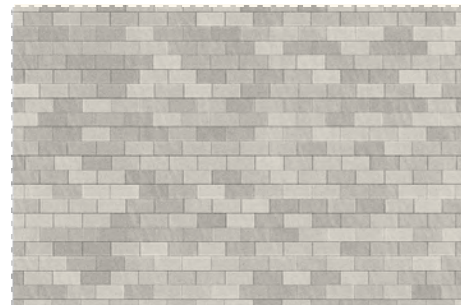
and Rentals

storage and
distributed
opment to
e travel and
on private



Paving - Vista

Different pavings used for direct users towards vista.



Paving - Shared Street

Shared by service vihecles, mobility buses, pedestrians and cyclists.



Paving - Safe Zone

2m paved safety area offset from buildings.



Diverse Public Realm

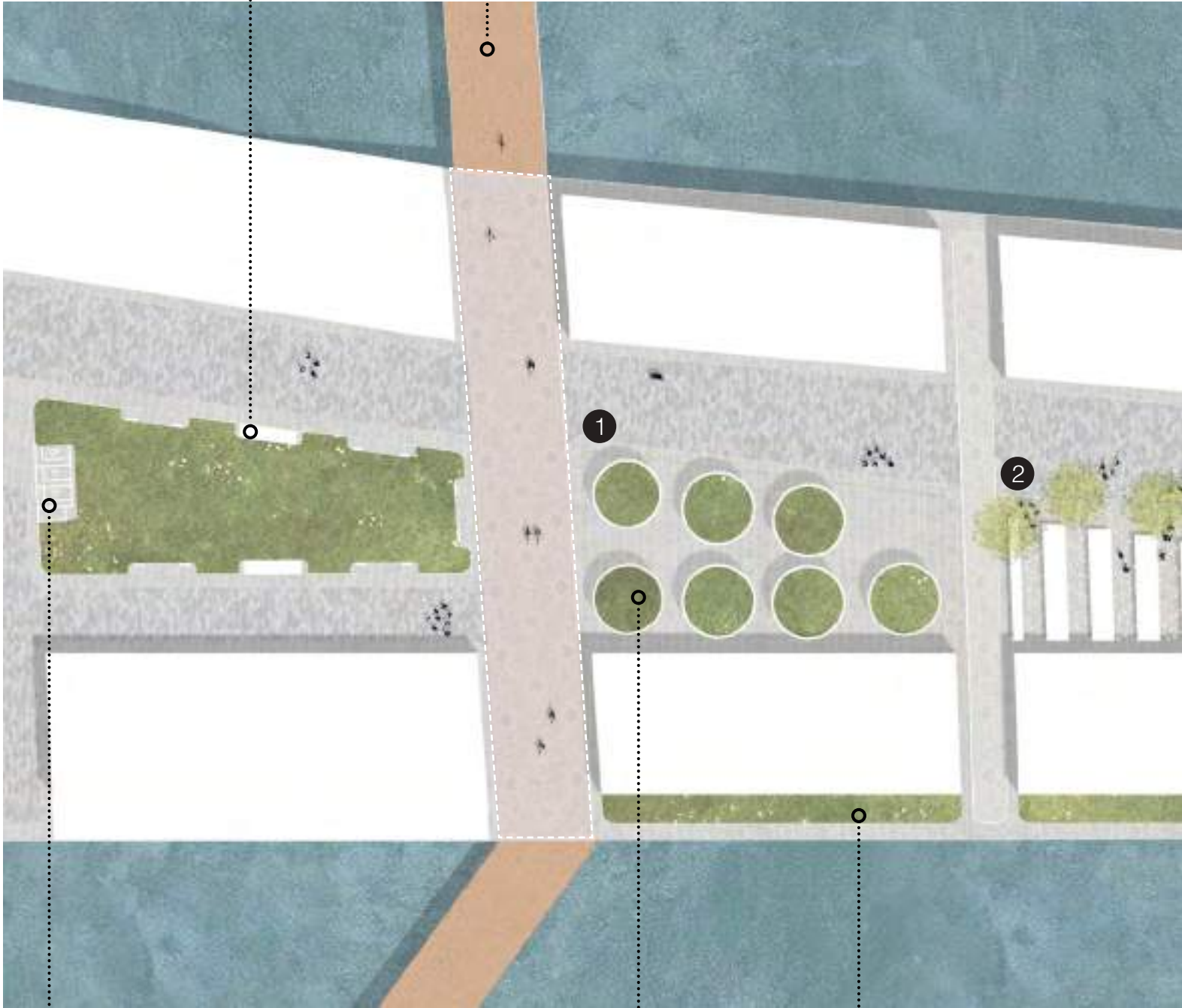
Mearn's Quay - Landscape

Inspired by industrial heritage and the historic fishing village that were a huge part of the identity of Aberdeen, the public realm is designed to celebrate the rich heritage, reintegrating the footprint of the industrial storage tanks into the

landscape and introducing fish market typology market stalls. The various public spaces allow different activities and promote community engagement.

Cycle Storage
Dedicated cycle parking integrated along the waterfront path, providing secure and accessible storage for daily commuters and visitors.

Bridge
A lightweight pedestrian-cycle bridge connecting both sides of the harbour, improving active-travel permeability across the site.



Disability and Service vehicle parking
Designated bays ensuring accessible drop-off, short-stay parking and convenient service access.

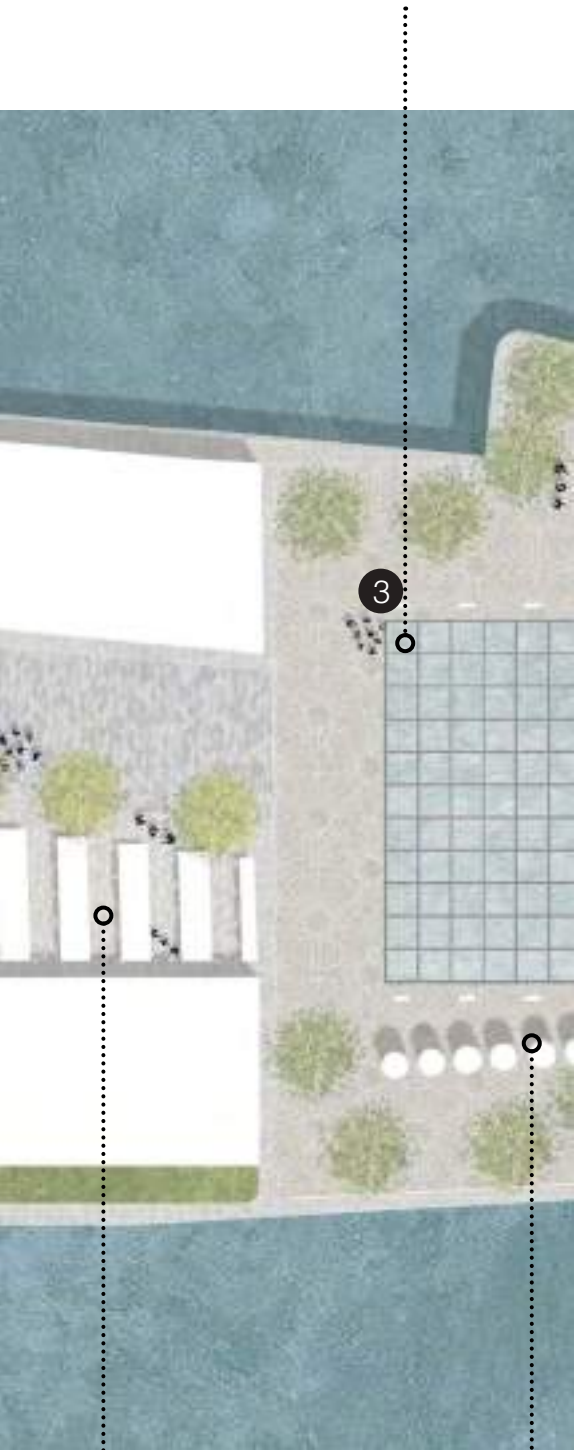
Retained Footprint of the Mud Tanks Turned Into the Landscape
Historic mud-tank geometries preserved as circular landscape inserts, creating a playful green pattern that celebrates the site's industrial past.

Private Garden
Private gardens for residents.

Market Stalls Mimic the Fish Market Typology
A row of modular stalls inspired by traditional market structures, providing flexible trading space for local producers, crafts, and seasonal events.

Dry Fountain Feature

A flexible dry-deck fountain plaza that acts as a focal gathering space, offering an interactive water feature that can also function as an event surface when inactive.

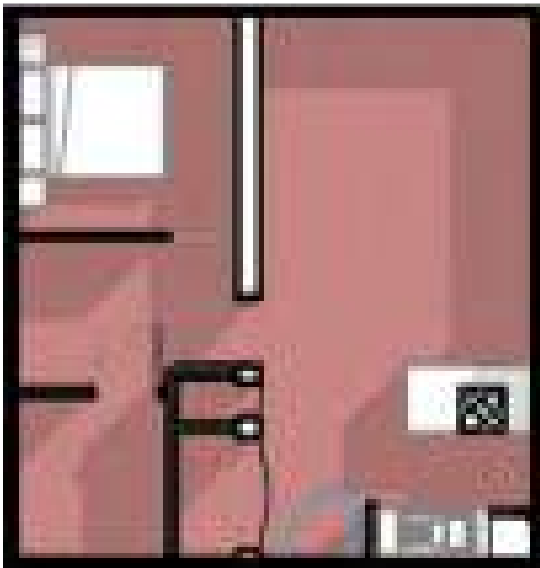


Picking Up the Fishery: Retained Cylinder Volumes for Heritage Nod and Wind Protection

Existing cylindrical forms re-used as sculptural elements along the promenade, referencing harbour heritage while offering wind-break zones for comfort.

Courtyard Block

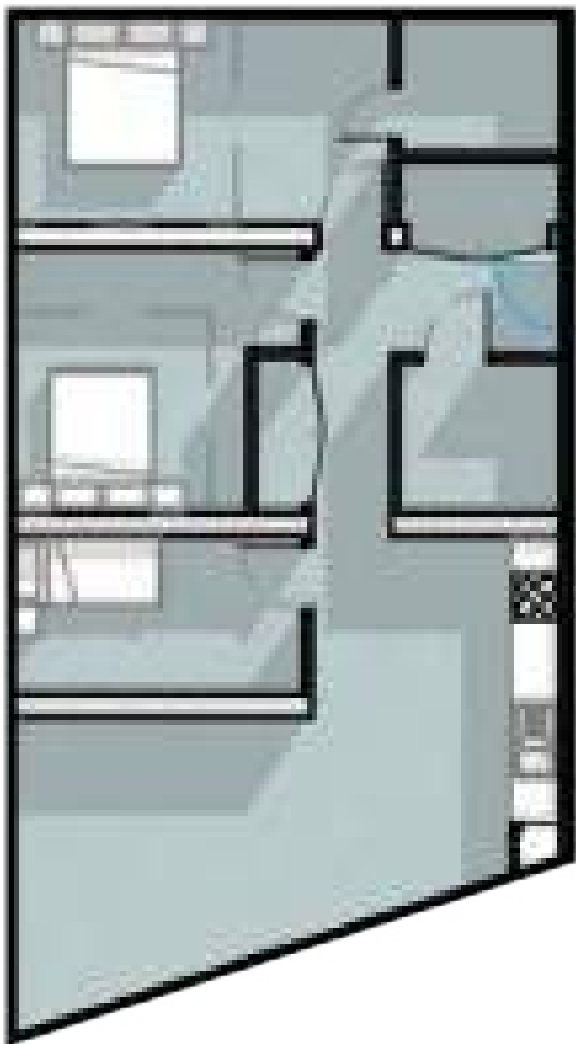
Mearn's Quay - Residential



Our first residential typology is a high density courtyard block, enclosing a greenspace and around 6-8 units per core. These units vary from 1B2P to 3B5P, suiting a range of residents, from Single adults to families.



- 3B5P Average area = 81.5 SQM
- 2B4P Average area = 73.6 SQM
- 1B2P Average area = 53.2 SQM
- Public Road
- Service Road



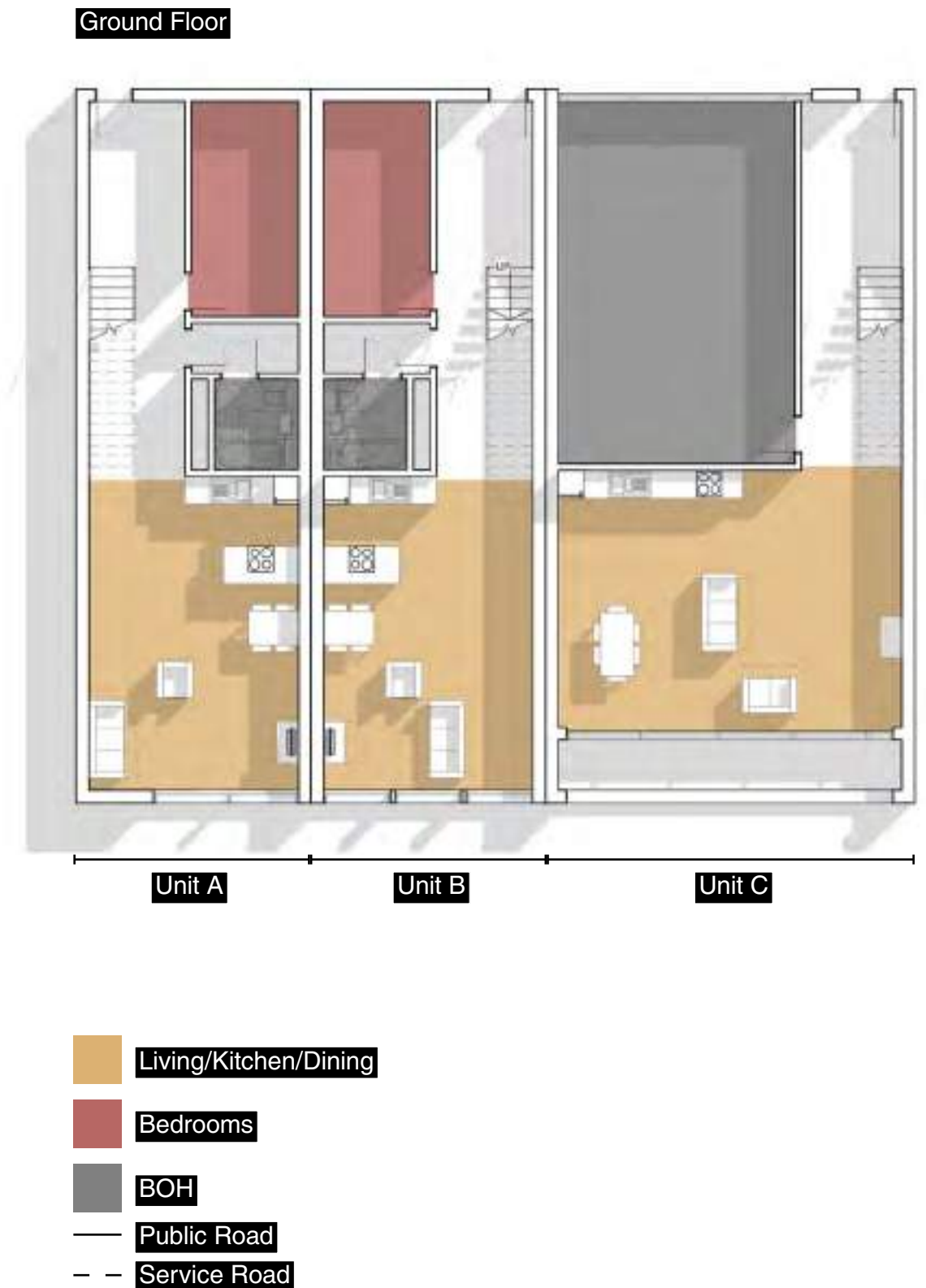


Dutch Waterfront

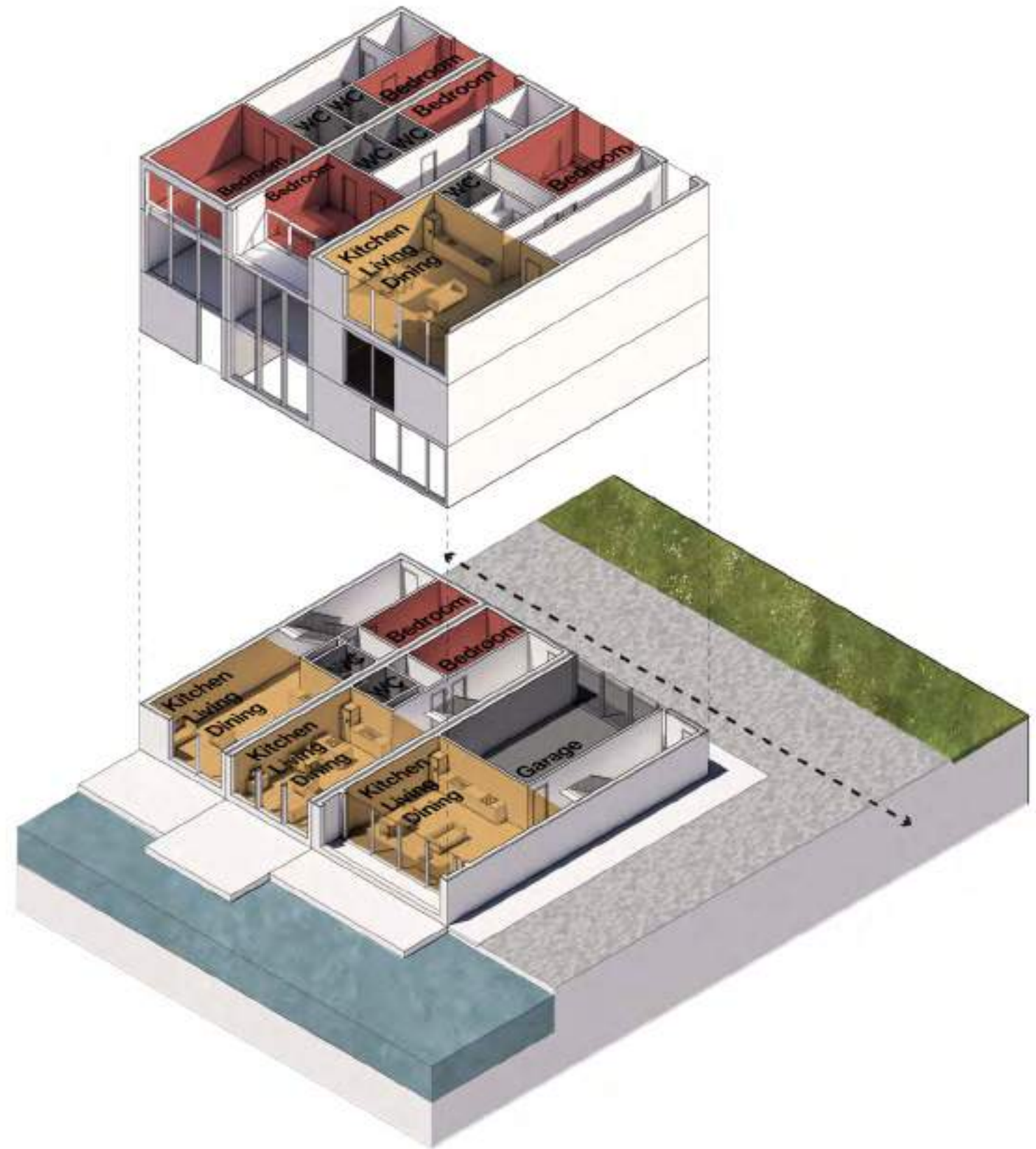
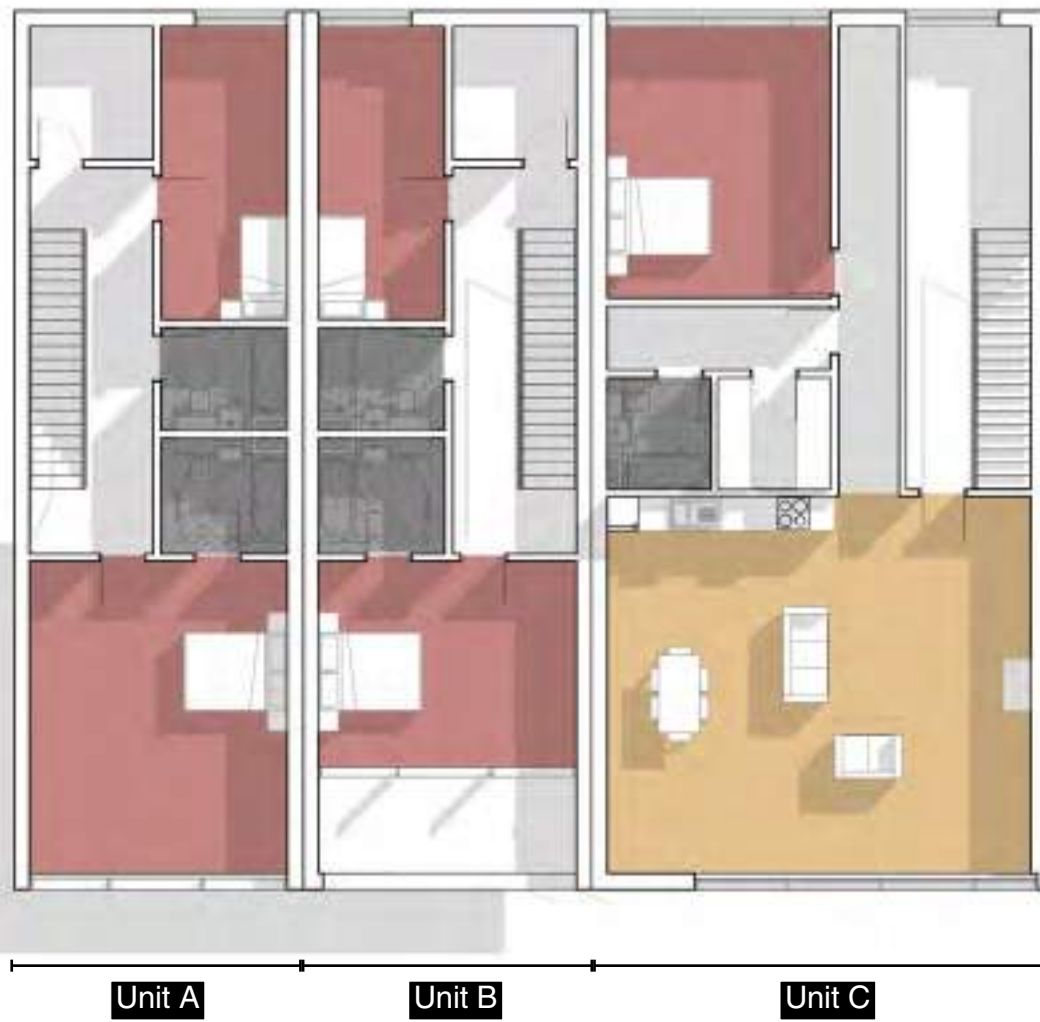
Mearn's Quay - Residential

The second residential typology is a collection of self build townhouses, similar to the type in Sporenburg in Amsterdam. Either full size or split into flats with a waterfront edge at the front and a service road at the rear.

Unit A and B are examples of the full town houses and Unit C is an example of a townhouse which has been converted into flats.



Typical Floors



Private Residential

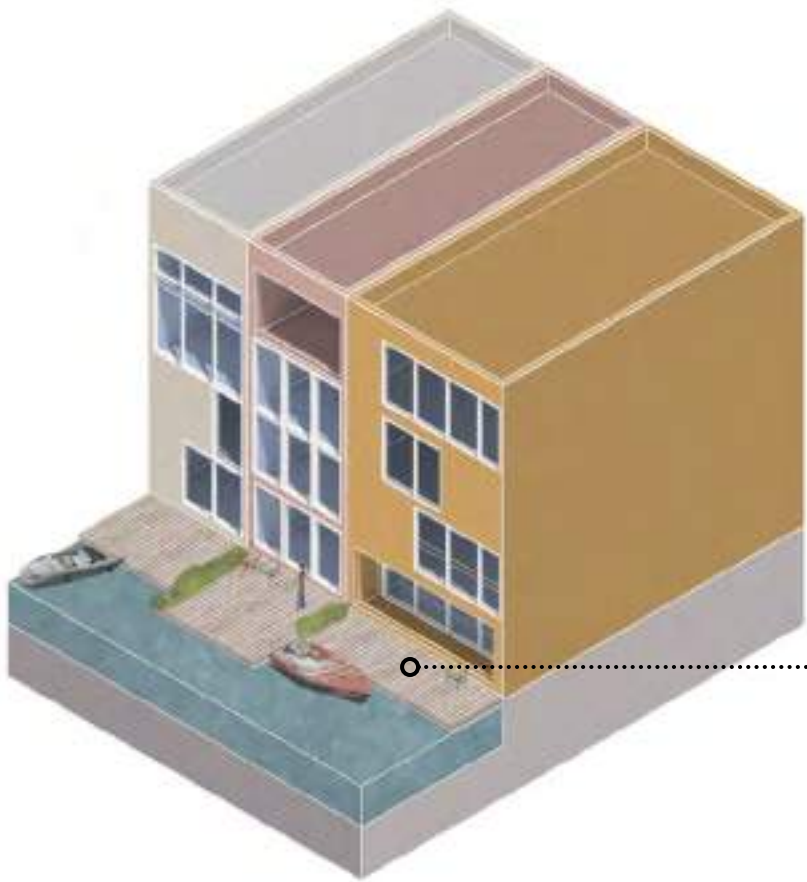
Waterfront Conditions 1 & 2



Privatised waterfront with loose design guide for a Dutch inspired waterfront living typology. Built directly against the harbour edge, the residents can have private boat parking and floating elements.

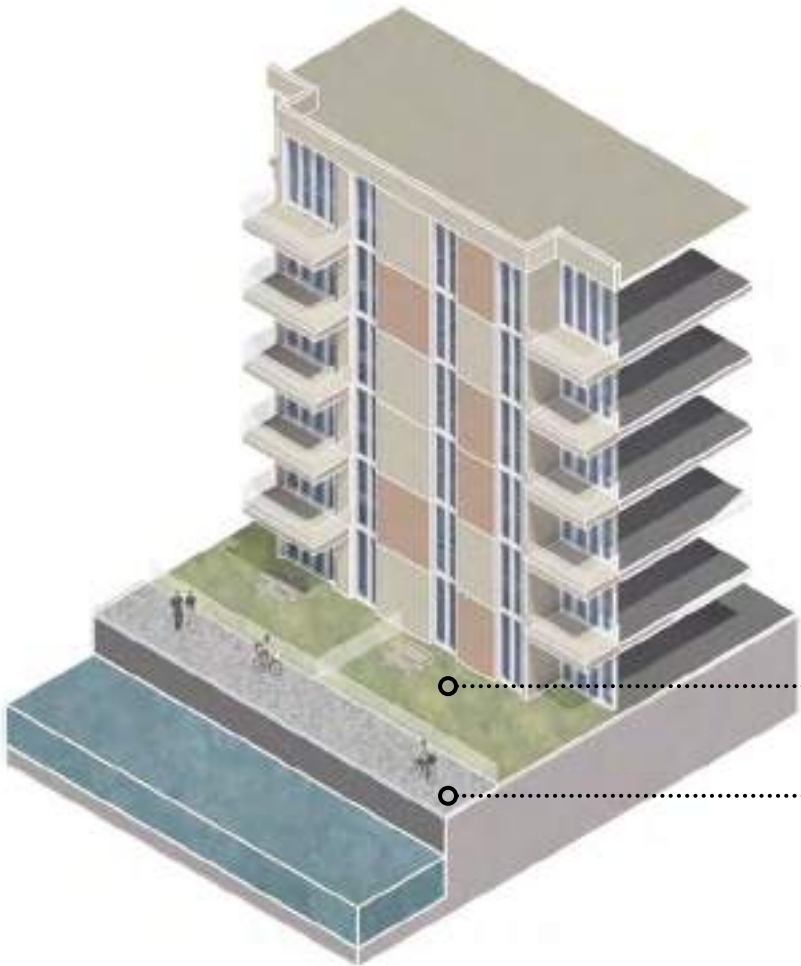


Typical flat blocks offer private gardens on the ground floor. With a narrow waterfront promenade accessible to public as well as residents.



Privatised Waterfront
Pontoons with boat parking space

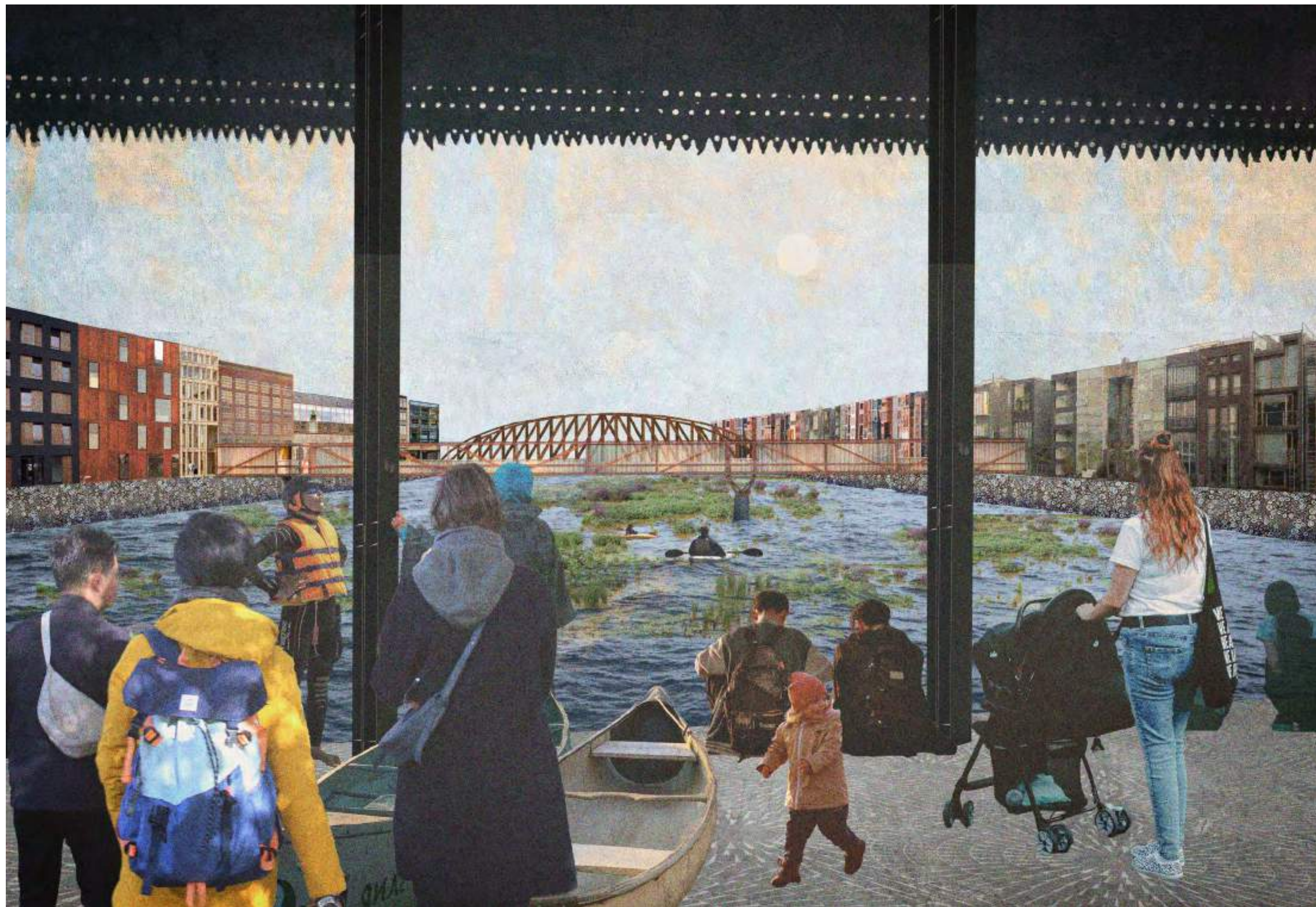
1 Dutch Waterfront



5m Private Garden
Private gardens for residents

Quiet Waterfront Edge
Narrower promenade compared to commercial edges

2 Residential Block with Private Gardens





Residential & Landscape, Phase 1



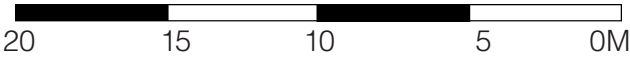
Inner Courtyard

Community Centre

Residential Service Road

Residential Block

Quiet Promenade



Phase 4

2050-2070

Locale Densification

Regional

- Green corridor linking suburban green-spaces

City

- Torry residential
- Tram phase 2
- City wide interventions

Locale

- Blaikie's Quay - Residential
- Blaikie's Quay - Landscape
- Blaikie's Quay - Promenade
- Northern Edge - Promenade
- Cultural institutions



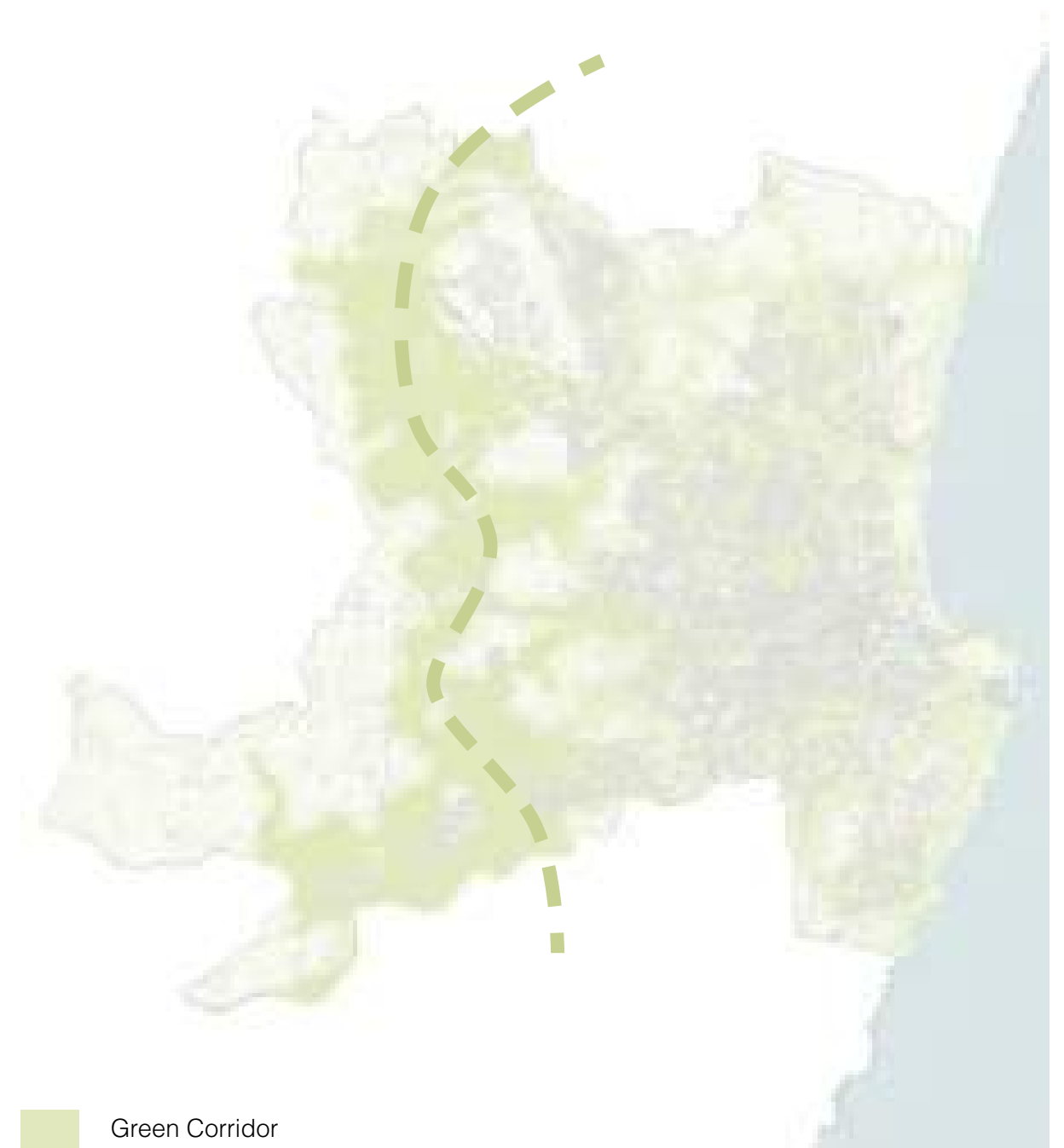
Contain Urban Sprawl

Green Corridor

Green corridors can function as strategic tools for containing urban sprawl by establishing clear ecological edges that limit outward expansion while strengthening environmental connectivity within the urban fabric.

Supported by policy change involving formally designating the corridors as protected landscape and preventing fragmenting urban developments, these corridors create a defined boundary that discourages development at

the city's periphery. At the same time, they enhance biodiversity, support active travel routes, and offer accessible recreational space for residents. In doing so, green corridors act not only as ecological infrastructure but also as spatial frameworks that guide more compact, sustainable patterns of urban growth.





Through the addition of the enhanced green network in this phase, there is a whole city regenerative attitude to improving the city's green and blue infrastructure.

Aberdeen City Local Development Plan 6.7-

Aberdeen's Green Space Network is a strategic city-wide network that connects natural green and blue spaces and habitats to each other.

The current Green Space Network has the harbour side as a significant break in the coastal green core from Broad hill and Queens links core to the River Dee core, establishing tiny forests and implementing Torry Riverside would strengthen the River Dee core and link it to the railway embankment core on the North side, and the Queens link core.

6.8

Assessing the city's blue features within the Green Space Network

Torry Riverside, the tidal lido and the floating wetlands in the harbour all play into improving the quality of the blue features in the city

NE4

Blocks development if it lies within an area designated at risk in a 1/200 flood event by 2080;

-Managing and reducing flood risk from rising sea levels, watercourses, storm and tidal surges, surface water and drainage systems
Rising sea levels and storm surges are mitigated through the use of extended dunes and a barrage.

Watercourses and surface water issues through the SUDS network and additional pocket parks

The Granite City Forest

Plan to expand woodland within the city boundary over the next 25 years

Through the tiny forest interventions and new nature reserve, this will help to meet the targets for 300 hectares by 2070

Peatland Action

National programme offering funding to restore peat land

Promoting the expansion of wetland, including peat land, as a flood remediation opportunity in phase 2

Aberdeen City Conservation Area Character Appraisal- Footdee & City Centre
References to the use of courtyards to create urban open space

Use of residential and retail in combination

Features from both conservation areas adjacent to each other reflect the existing typology and also help limit urban sprawl, which can be seen in the developed typologies across both quays.

Scottish National Planning Framework 4 Policy 3

Enhancing biodiversity and strengthening nature networks

Seen across phase 2&3 , (see Strategy for Scotland to be Nature Positive by 2030)

Policy 4

Protect and restore natural environments.

Seen particularly along the coast, where increased flood defences extend existing dunes to the North and protect Green Space Network cores all along the coast

Policy 6

Protect and expand woodland and tree cover.

Seen through the designation of tiny forests and the expansion of the green space network, with a design guide covering native plant species to adopt

Strategy for Scotland to be Nature positive by 2030

Phase 2 involves re-wilding the city centre in key areas and further investment in works to improve the quality of the river boundaries

Green Policy Summary

*Some of the key policy's used in establishing the kit of parts and locale approaches are provided here though they are not an exhaustive list.

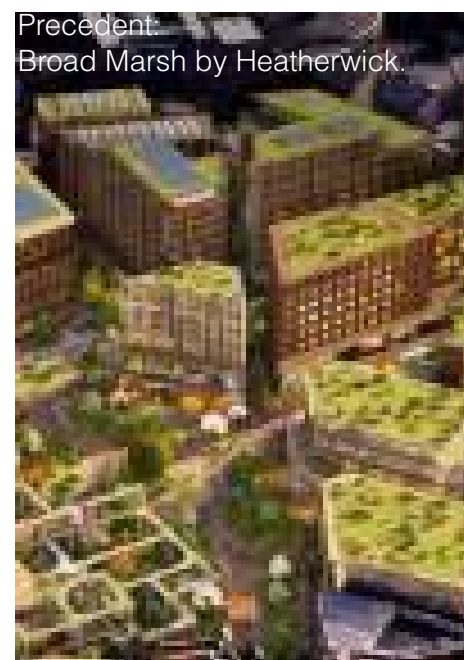
City Centre Regeneration

New First Impressions of Aberdeen

Curating new first impressions of Aberdeen which focus on better visual wayfinding and mixed use pockets of activity.



Inspiration from the city's history, regeneration of high streets by reintroducing old industries for a new audience. Supported by a new tram connection through King's Street, and Bon Accord stripped back to allow a continuous connection from George Street.



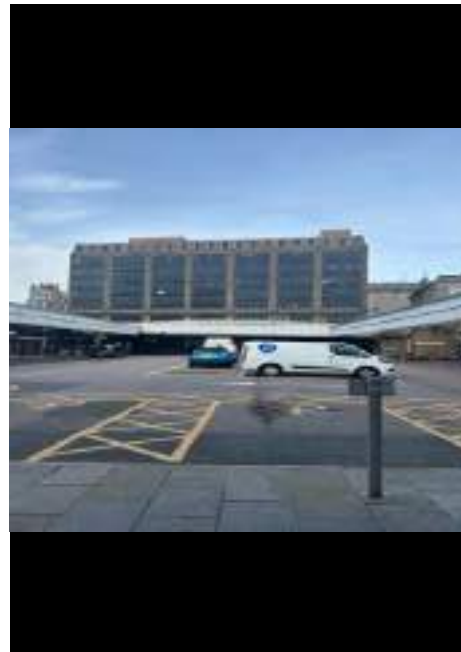
Precedent:
Broad Marsh by Heatherwick.



Retail Scene, Before & After



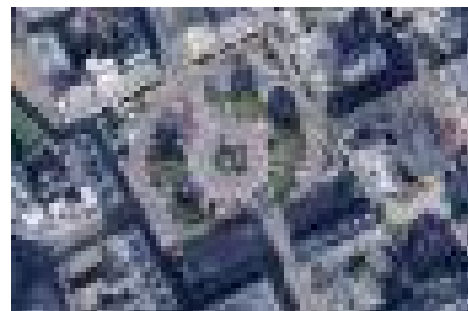
New views upon arriving to Aberdeen via train, current conditions see the station surrounded by car parks and run down shopping centres and poor introduction to way finding. We would like to propose a new introduction to Aberdeen, featuring a new mixed use campus cluster adjacent to the train station. These will cater to all kinds of people arriving into Aberdeen, business-people, students, locals or tourists.



Train Station Complex, Before & After



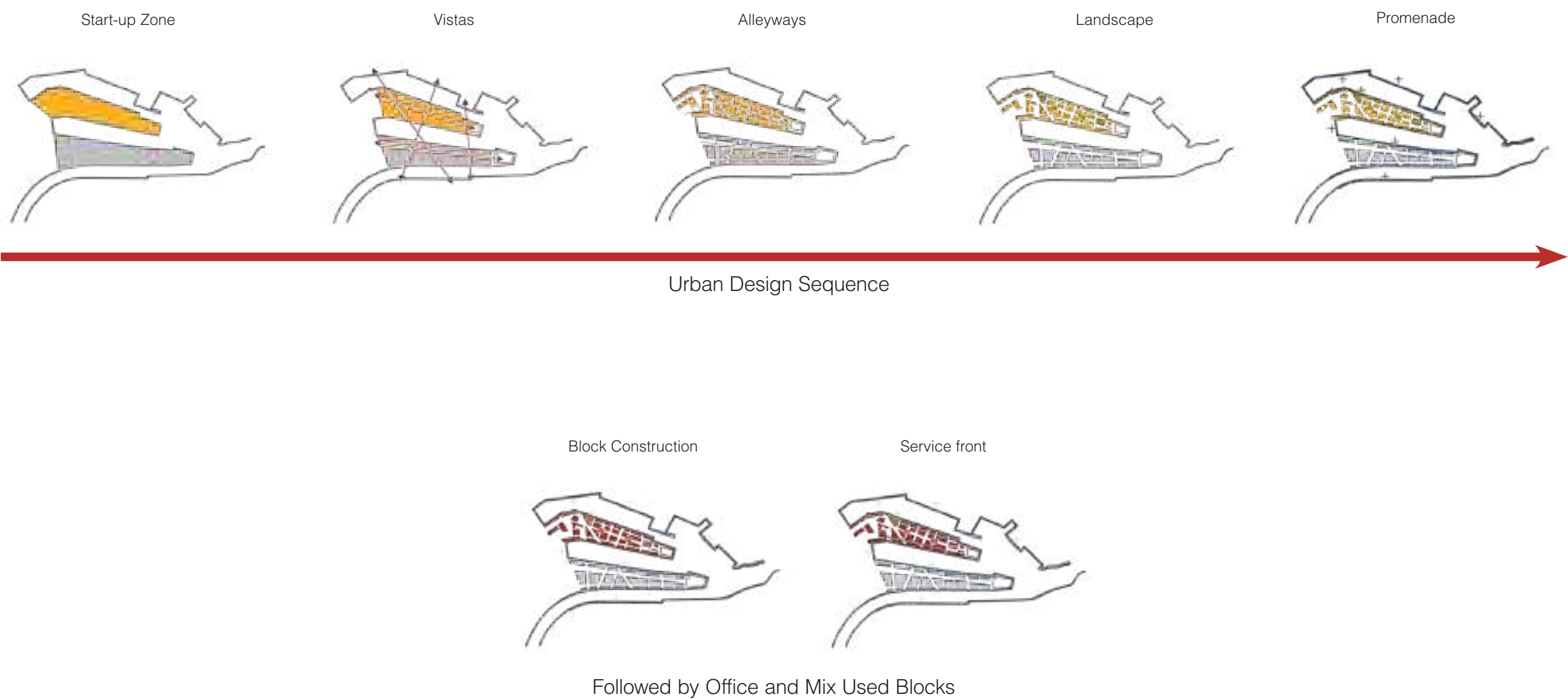
Pocket parks and pocket squares could be implemented to reclaim space used by cars, revitalise the high-street, and used as a surface water defence. We imagine precedents like Queens Square in Bristol, with a fresh air like that of Paley Park, New York.



Golden Square, Before & After

Locale - Blaikie's Quay

Development Sequence

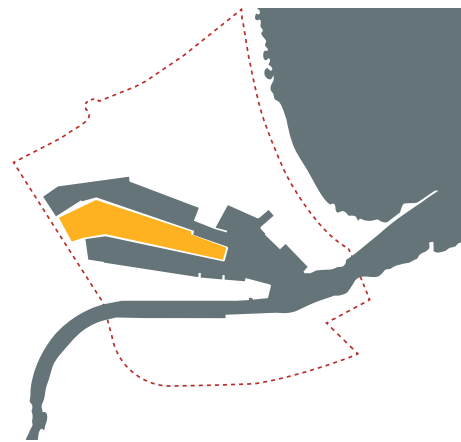




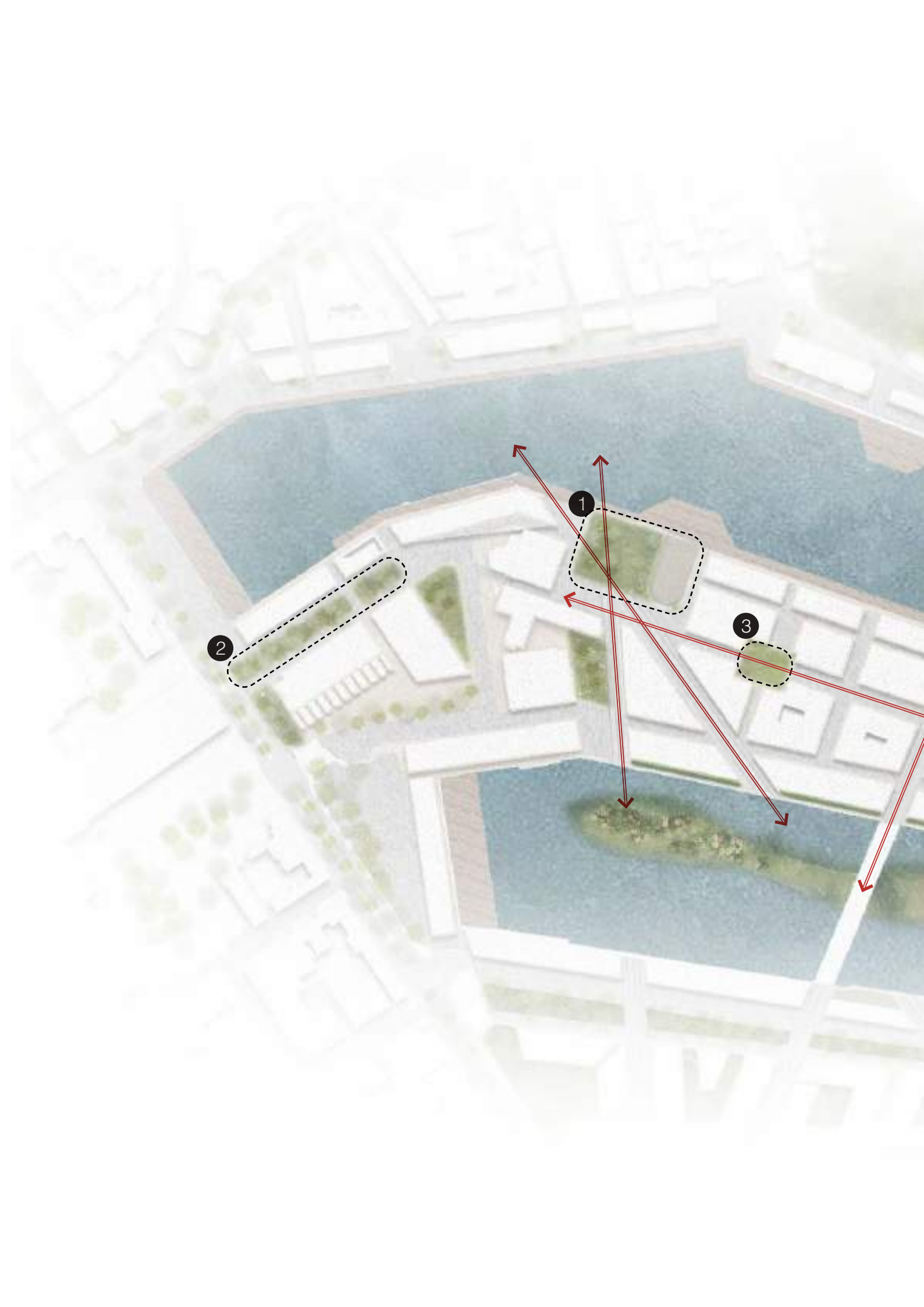
Blaikie's Quay
Within Phase 4

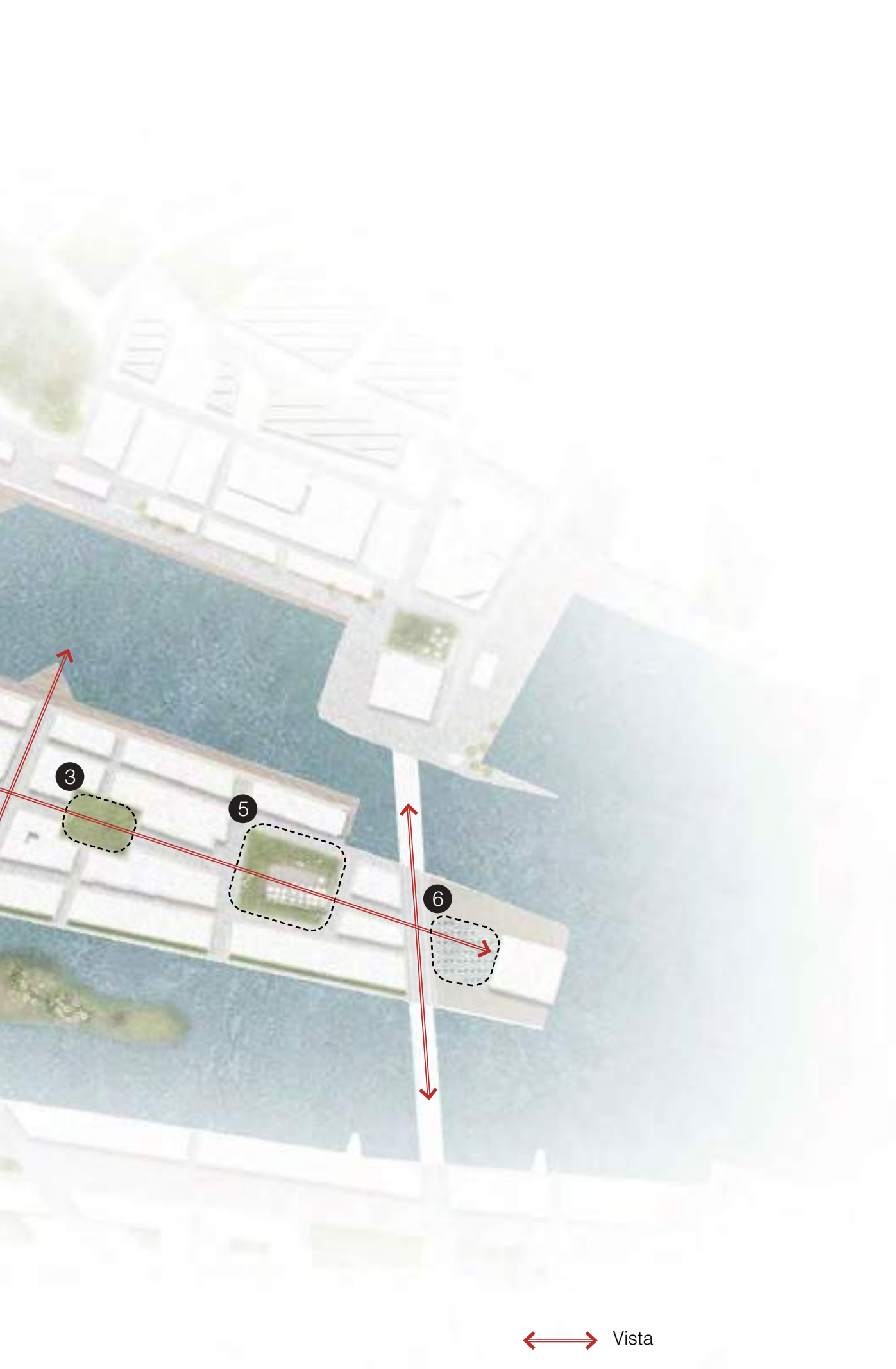
Landscape Catalogue

Blaikie's Quay - Landscaping

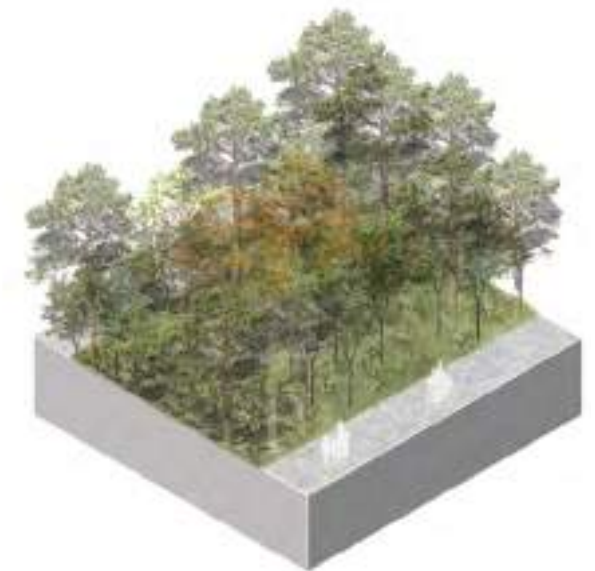


Suitable landscape design considering the larger scale buildings with more commercial functions is incorporated. Urban Courtyard positioned among the blocks serve as a breathing space embedded in the urban grain. Amphitheatre is introduced at the harbour waterfront, hosting activities and facilitate community gathering.





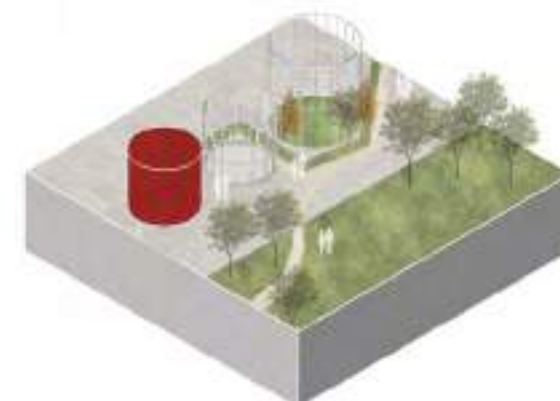
1 Sunken Plaza



2 Tiny Forest



3 Urban Courtyard



5 Industrial Heritage Park



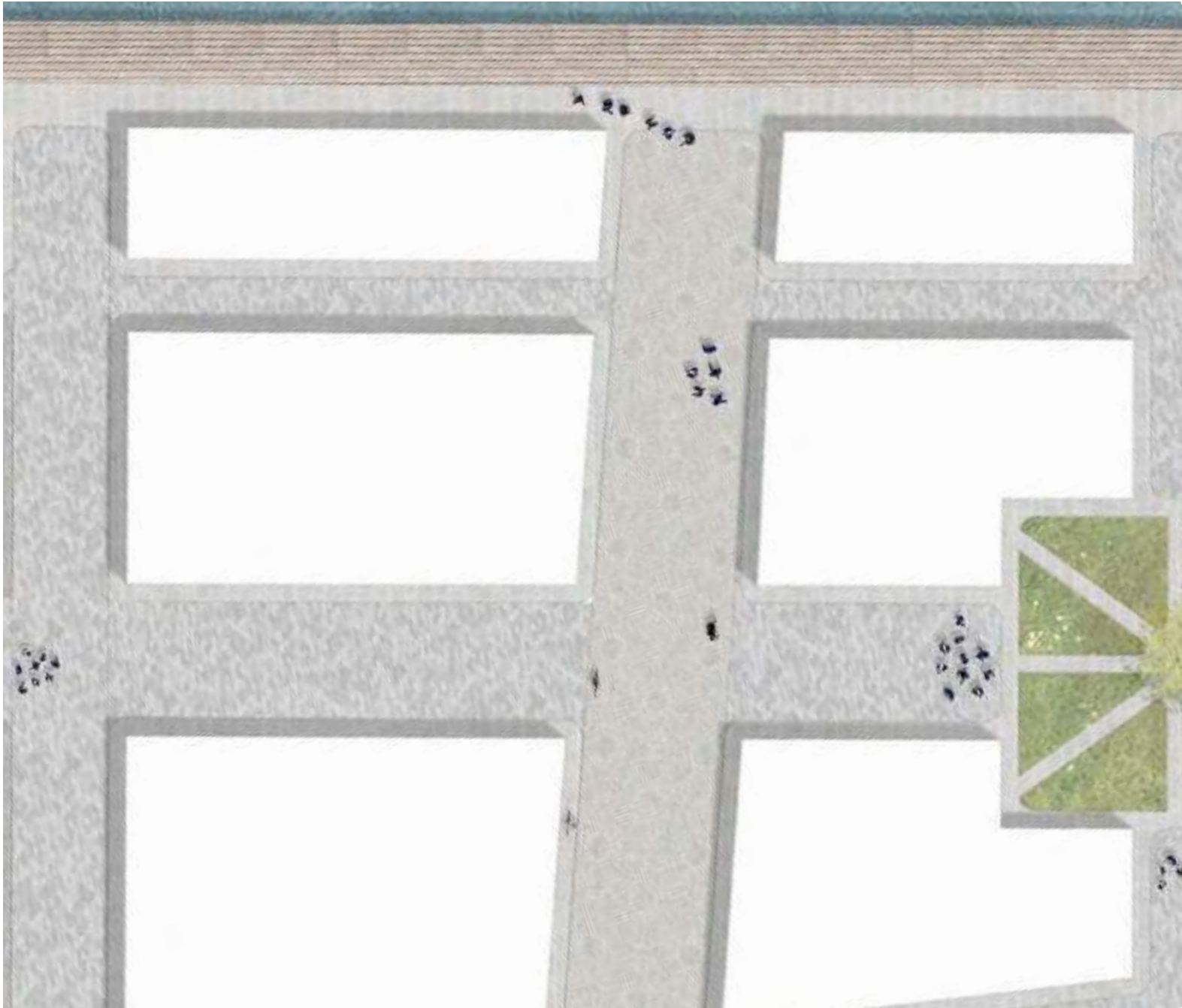
6 Water Fountain

Urban Courtyard

Blaikie's Quay - Landscaping

The urban courtyard responds to the more commercial nature of the Blaikie's Quay. It activates better views for the residents and the office users, offering a breathing space among the urban grain. It also accommodates convenient public amenities, such as cycle

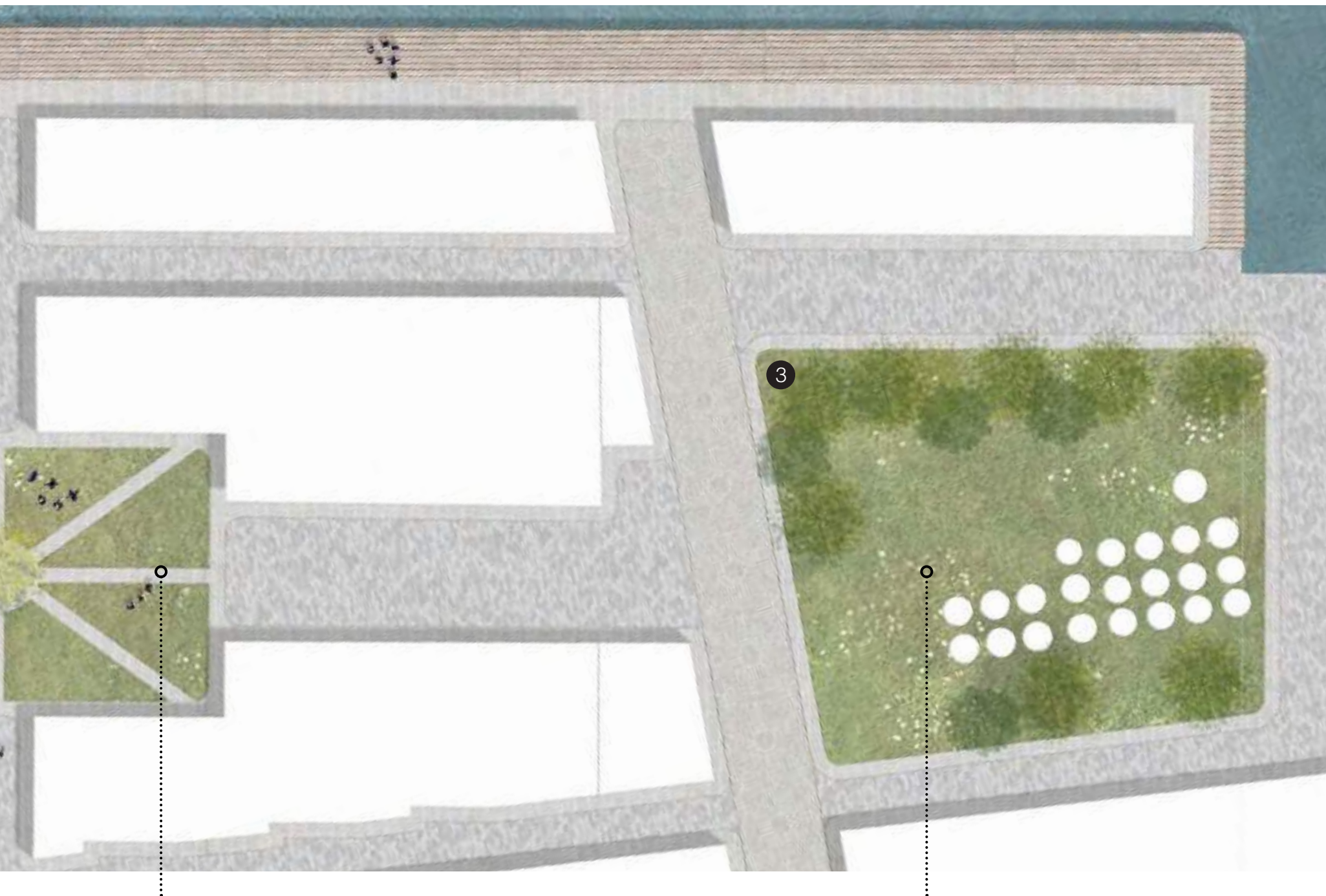
parking, rental points, and other shared facilities, ensuring that the space functions not only as a visual relief but also as an accessible, well-used community asset.



Promenade Deck Made Out of the Decommissioned Oil Structure
A new waterfront promenade constructed from repurposed oil-rig components, transforming industrial heritage into a durable public deck.

Paving Variations:
Subtle changes in paving patterns used to guide movement, define zones, and create visual interest across the public realm.

Commercial Buildings:
A series of mixed-use commercial blocks lining the pedestrian grid.



Central Courtyard for the Commercial Buildings and Offices:
 A shared green courtyard at the heart of the commercial cluster, offering social space, outdoor seating, and a landscaped break-out area for workers.

Industrial Heritage Inspired Public Green Space:
 Inspired by the industrial heritage.



Mansion Block

Blaikie's Quay - Residential

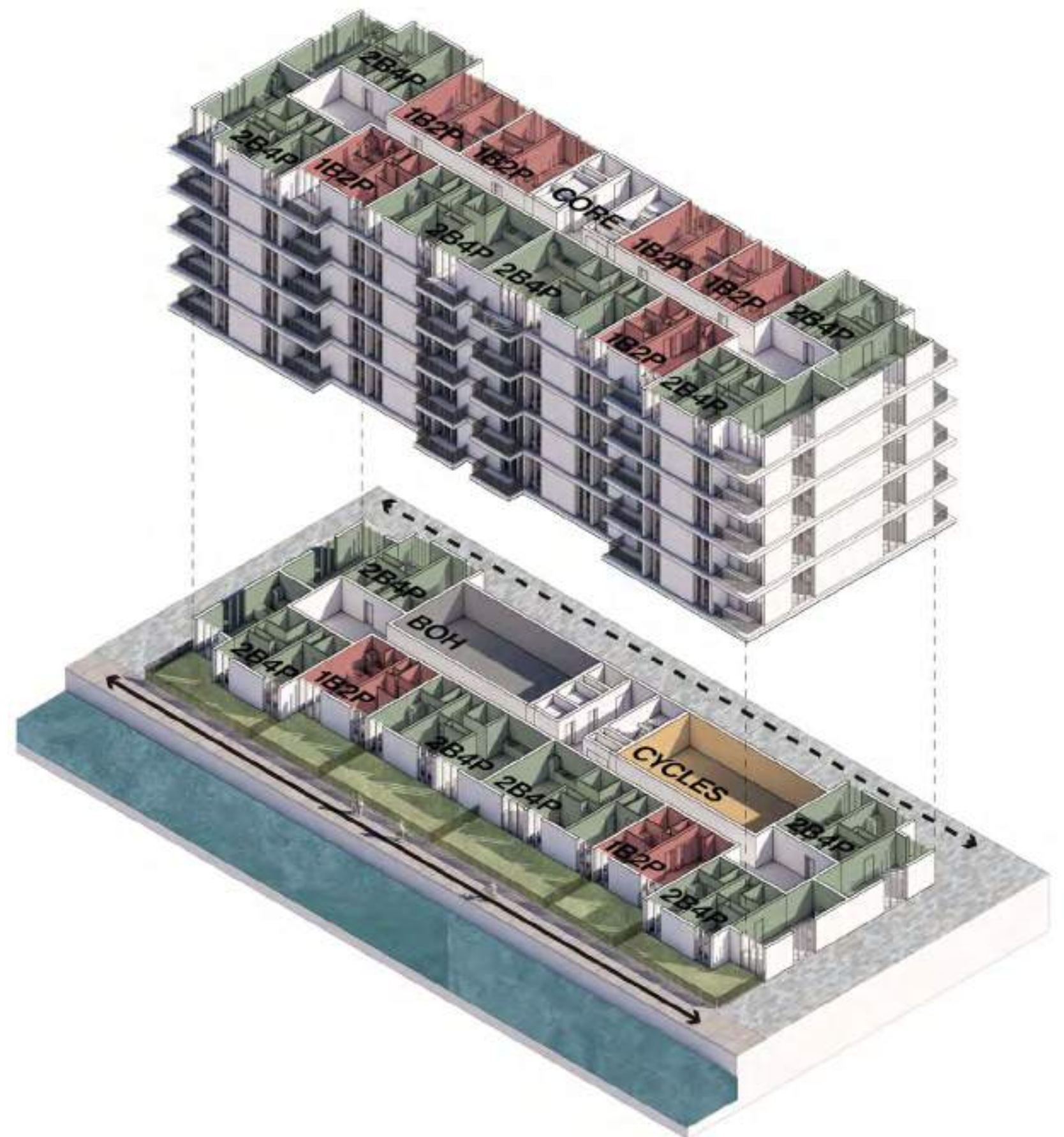
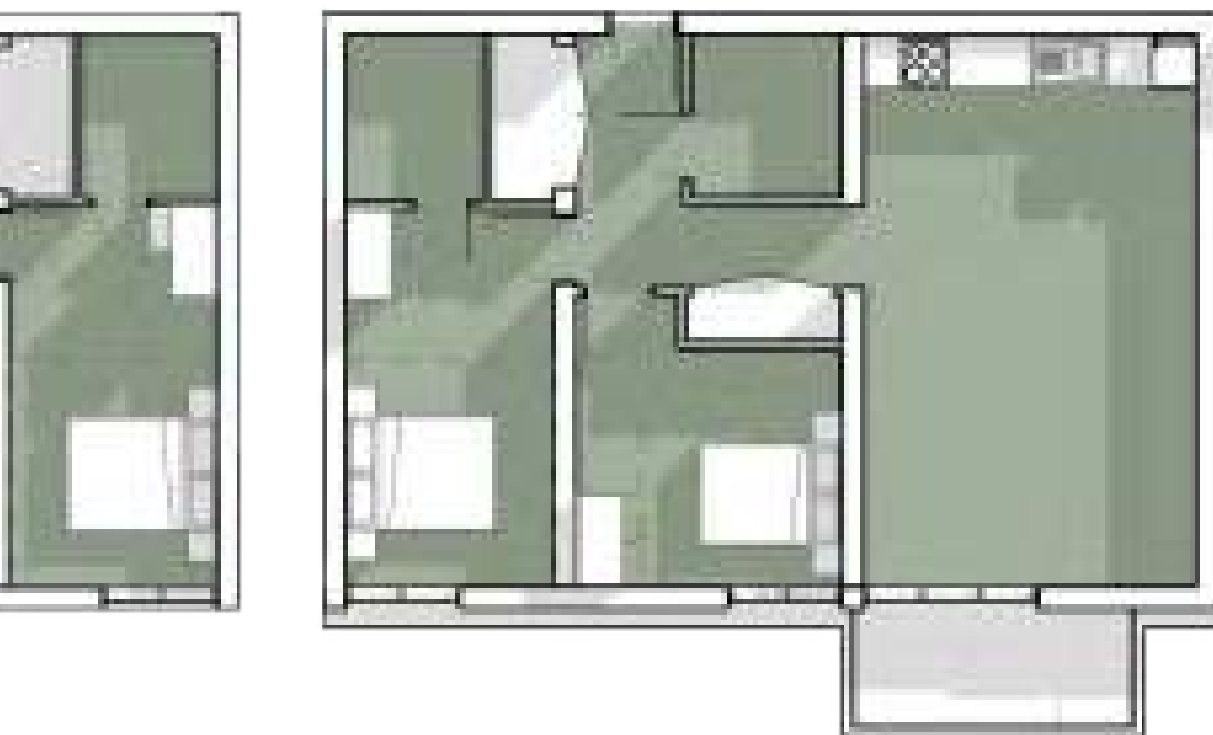
Our only typology introduced in residential phase 2 is a standard slab/mansion block that serves the Regents Quay with a mx of 1B2P and 2B4P to support professionals or young families.

The block has a clear front face

towards the waterfront with servicing being located along the service road to the rear



2B4P	Average area = 75.2 SQM
1B2P	Average area = 52.8 SQM



Northern Promenade

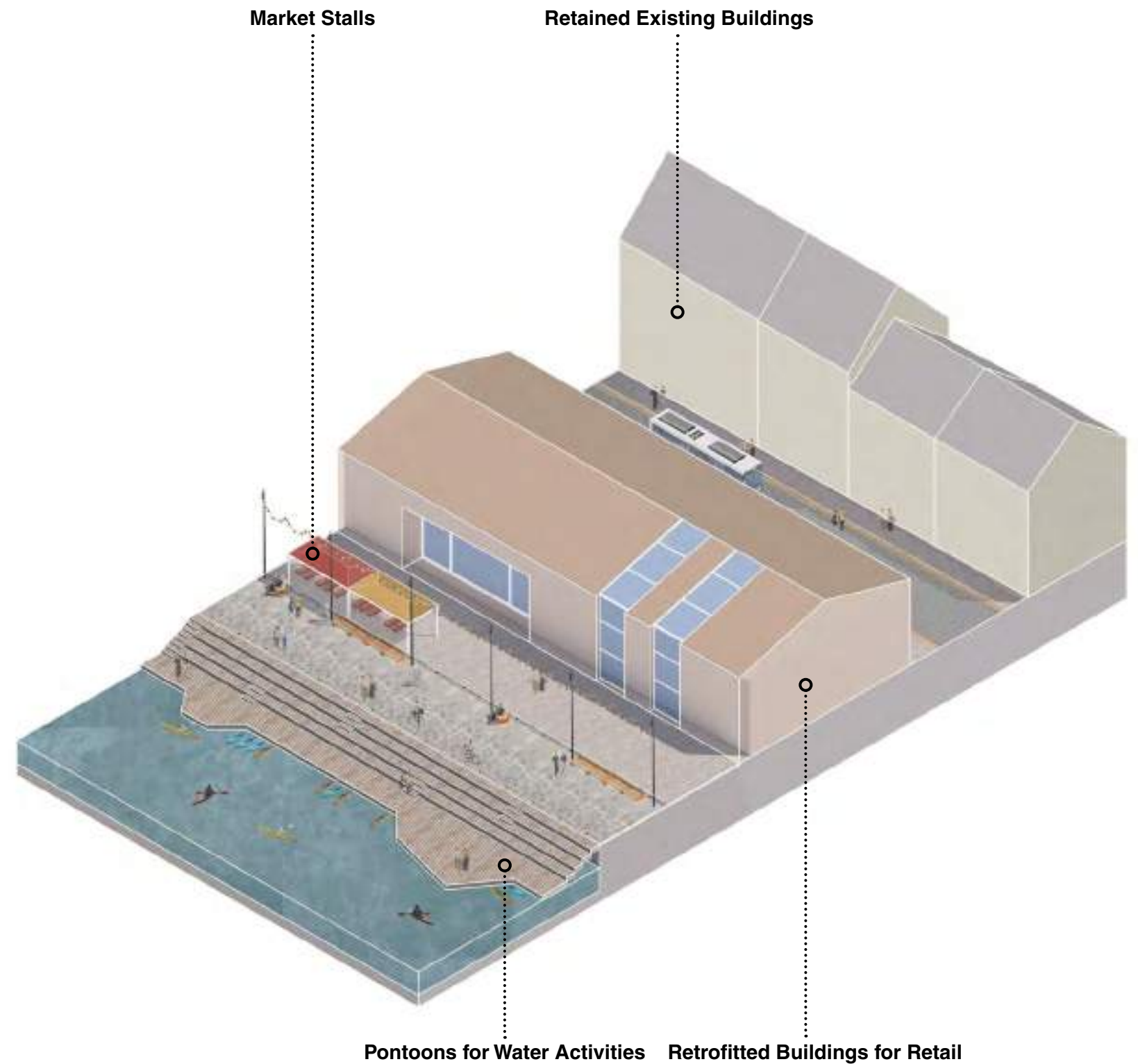
Waterfront Condition 3



As an extension of the main high street, the northern edge maintains the retail identity of a spacious shopping promenade with recreational elements that encourage people to pause and stay. Terraced steps gently descend toward the water,

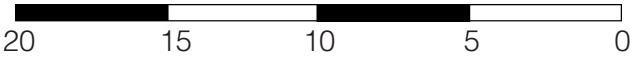


while floating pontoons support a range of water activities. These features will transform the northern edge into the most vibrant waterfront destination within the masterplan, blending retail, leisure, and public life along a lively, accessible urban waterfront.





Waterfront Condition 3, Northern Promenade





Blaikie's Quay

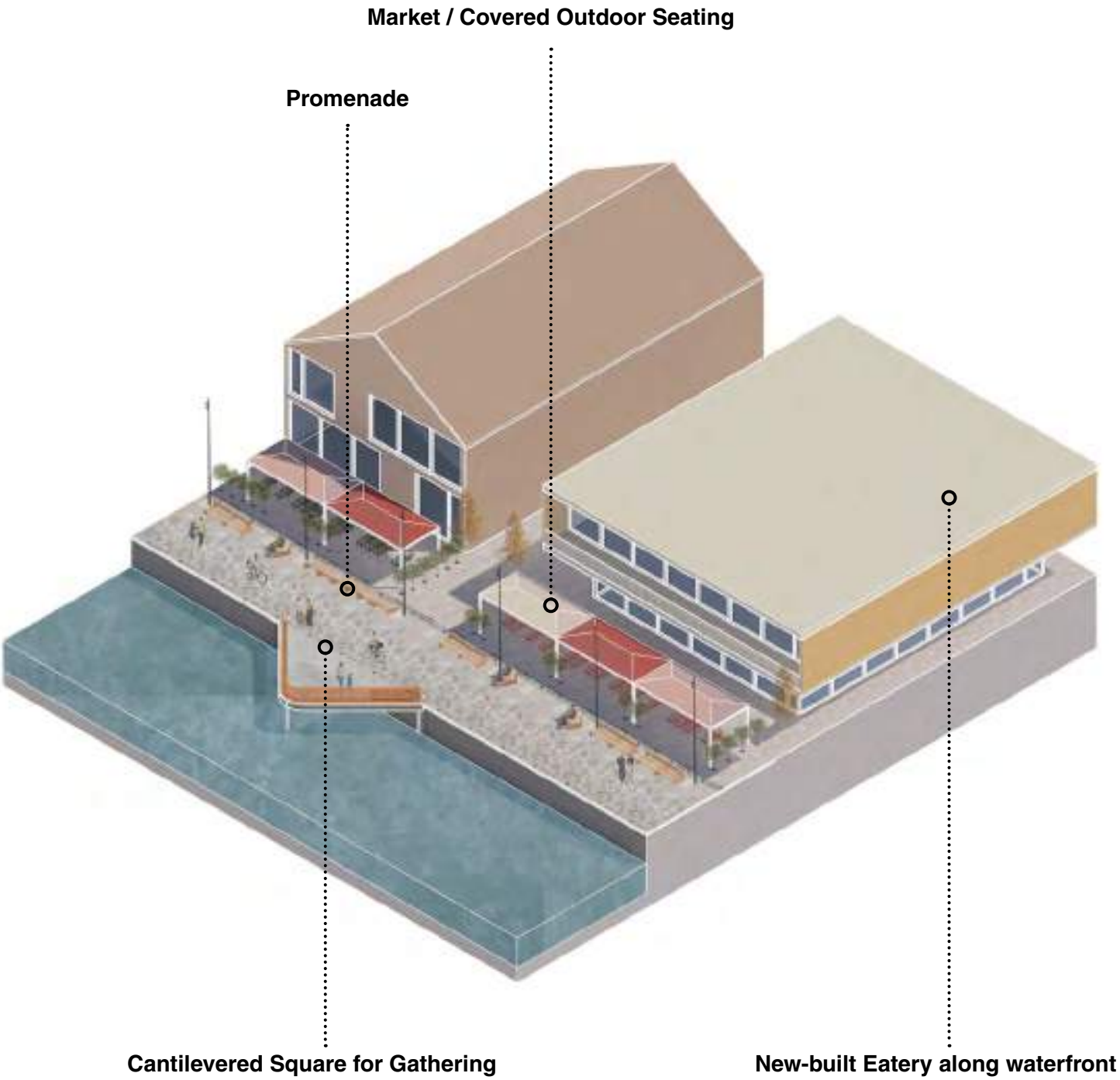
Waterfront Condition 4

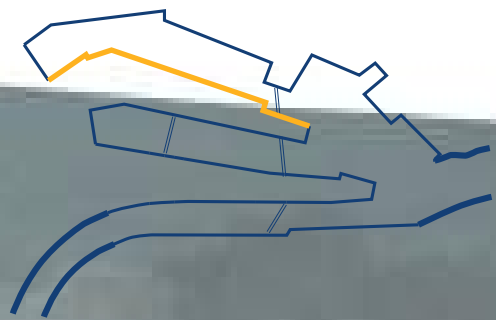


With the reinstated fish market typology, a series of sheltered seating areas and flexible market spaces now line Regents Quay, reinforcing its historic identity while supporting contemporary public use. Restaurants spill out onto the promenade, animating



the waterfront with outdoor dining and pedestrian activity. Cantilevered platforms are positioned at key vista points, punctuating the linearity of the promenade and offering elevated gathering spaces that frame views across the harbour.





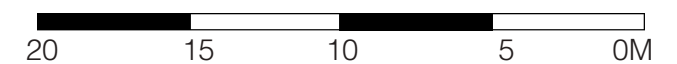
Pontoon
attached to
promenade

Mixed use blocks along
the promenade

Amphitheatre for cultural
performance

Oil tanks repurposed to house social
and forum spaces

Waterfront Condition 4, Commercial Promenade





Torry Riverside

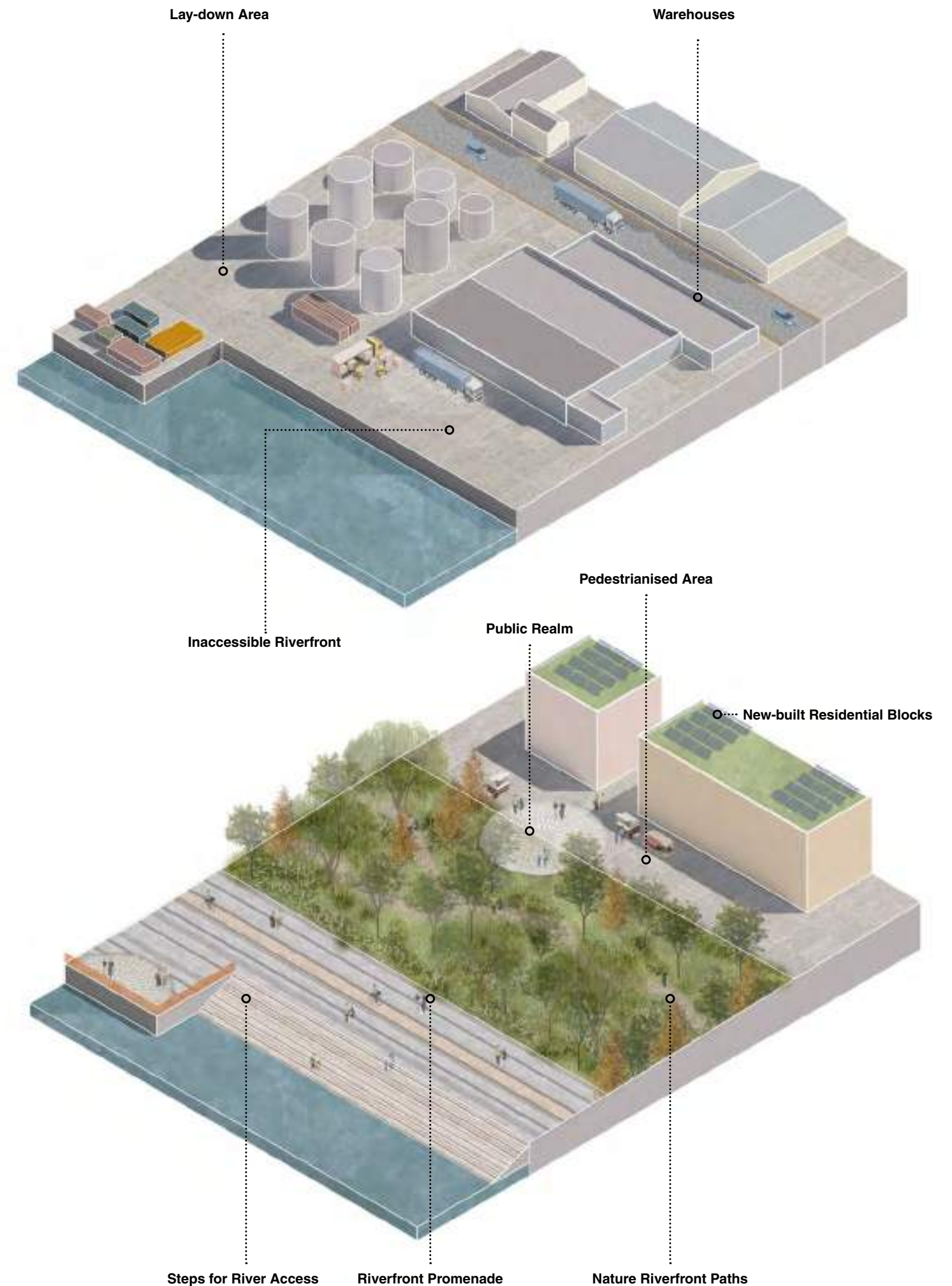
Waterfront Condition 5



Currently lined with industrial uses, this section of the River Dee has been inaccessible to the public. The proposed rewilding of the riverfront restores a more natural edge condition, creating an ecological walking route that reconnects



the area to the existing riverside path. As the riverfront also forms an interface with the residential neighbourhood of Torry, new pedestrianised public spaces are introduced.





Waterfront Condition 5, Torry Riverside

20 15 10 5 0M

Cultural Institutions

Celebrating History

Across the scheme, we have found ways to offer integrated cultural centres that reflect Aberdeen's past, present, and future.

Welcoming you into the site with reinstated fish markets and the relocated Cordona amusement park, you find Aberdeen's history side by side with its future. A new water sports centre and an outdoor performance space, in the form of an amphitheatre modelled on a dry dock, suggest potential diversification of the cultural offerings in Aberdeen.

Meanwhile, across the River Dee, Torry is made a focus across the harbour, with vistas highlighting a new community centre on the border of its reinstated natural waterfront and proposed Lido. Torry Riverside features bird-watching towers constructed with repurposed parts of decommissioned oil rigs.

The cultural cluster of proposed schemes at the tips of the both quays brings users right the way across the site, and frames the harbour.



Birdwatching towers



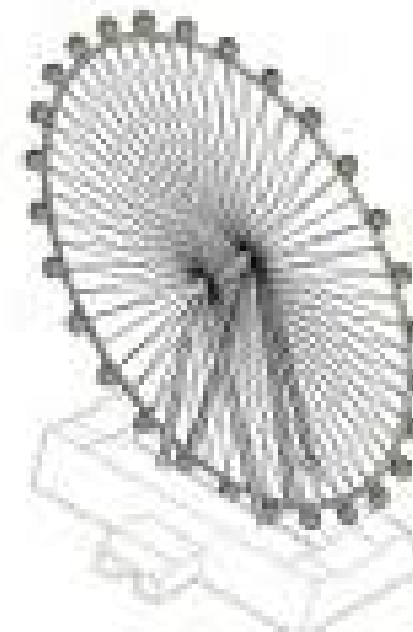
Oil drum think tanks



Reinstated historic markets



Recycled granite benches



Relocated Cordona



Amphitheatre



- Cultural Cluster
- Joint Community Campus
- Integrated Residential District
- Relocate Cordona
- Integrated Transport Hub

Phase 5

2060-2080

Sustained Growth

City

- Barrage

Locale

- University Research Cluster





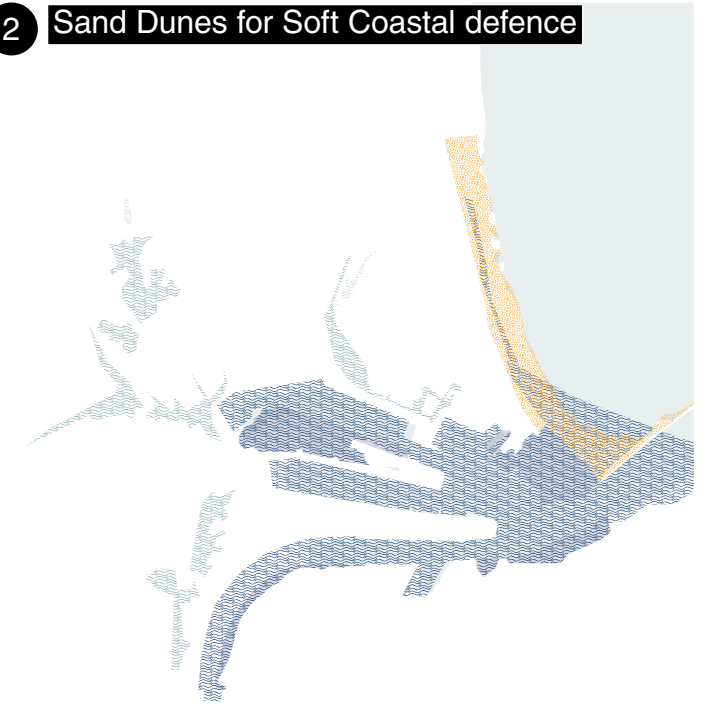
The walkable barrage is the key element of our coastal flood-defence strategy, protecting Footdee and the harbour from extreme tidal surges alongside softer measures like sand dunes and pocket parks. It creates a continuous pedestrian link between Torry and Footdee and will only close during major flood events (around 1-in-200 years), ensuring everyday public access while providing long-term protection.



1 Existing Condition



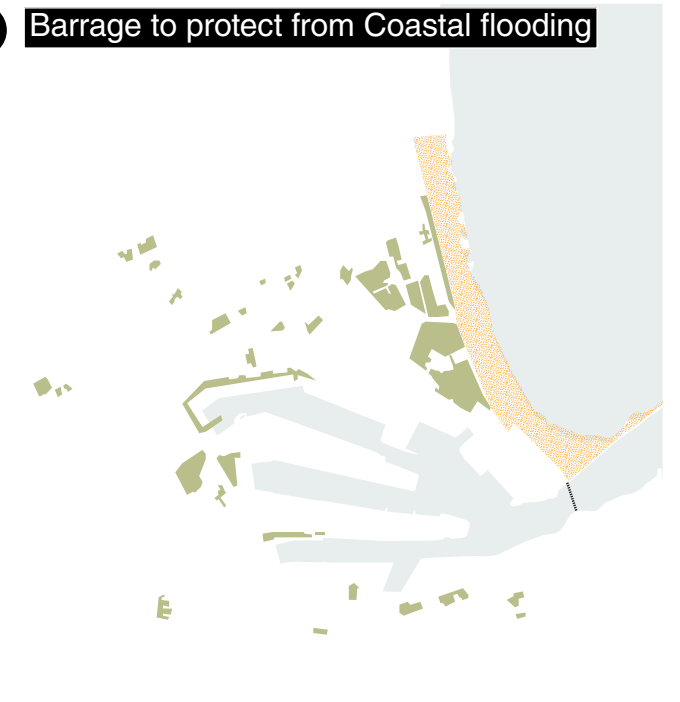
2 Sand Dunes for Soft Coastal defence



3 Pocket Parks for Surface flooding



4 Barrage to protect from Coastal flooding



Flooding Defence, Barrage

A joint campus between Gray's School of Art (RGU) and the University of Aberdeen is proposed beside the train station, forming a mixed-use innovation hub focused on energy transition and low-carbon industries. It creates a new gateway into the city for students, researchers and businesses, drawing inspiration from ID Manchester and the UAL campus at King's Cross.

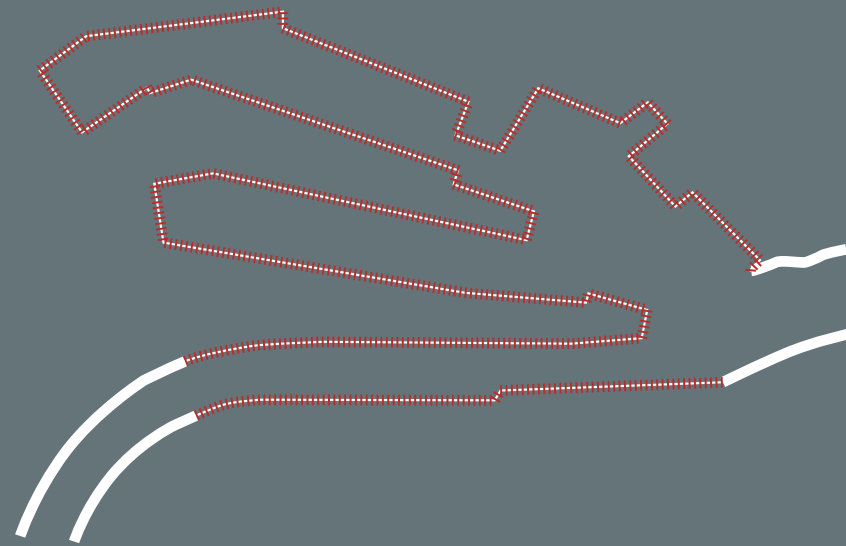


University Culture Cluster



Masterplan Overview

Existing



Proposed



- Proposed Bridge
- Privatised Waterfront
- Public Waterfront

Waterfront Conditions

1 Dutch Waterfront



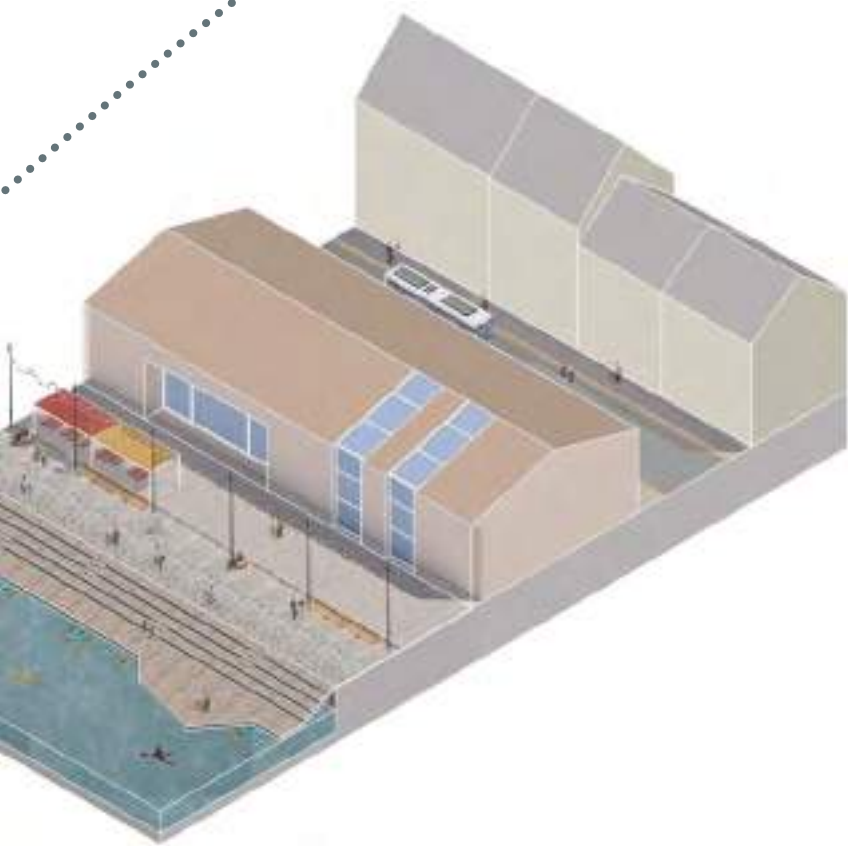
2 Residential Frontage with 5m private garden



Private Edge

3 Shopping

Public Edge



g Promenade with Water Activities



4 Promenade with Eatery



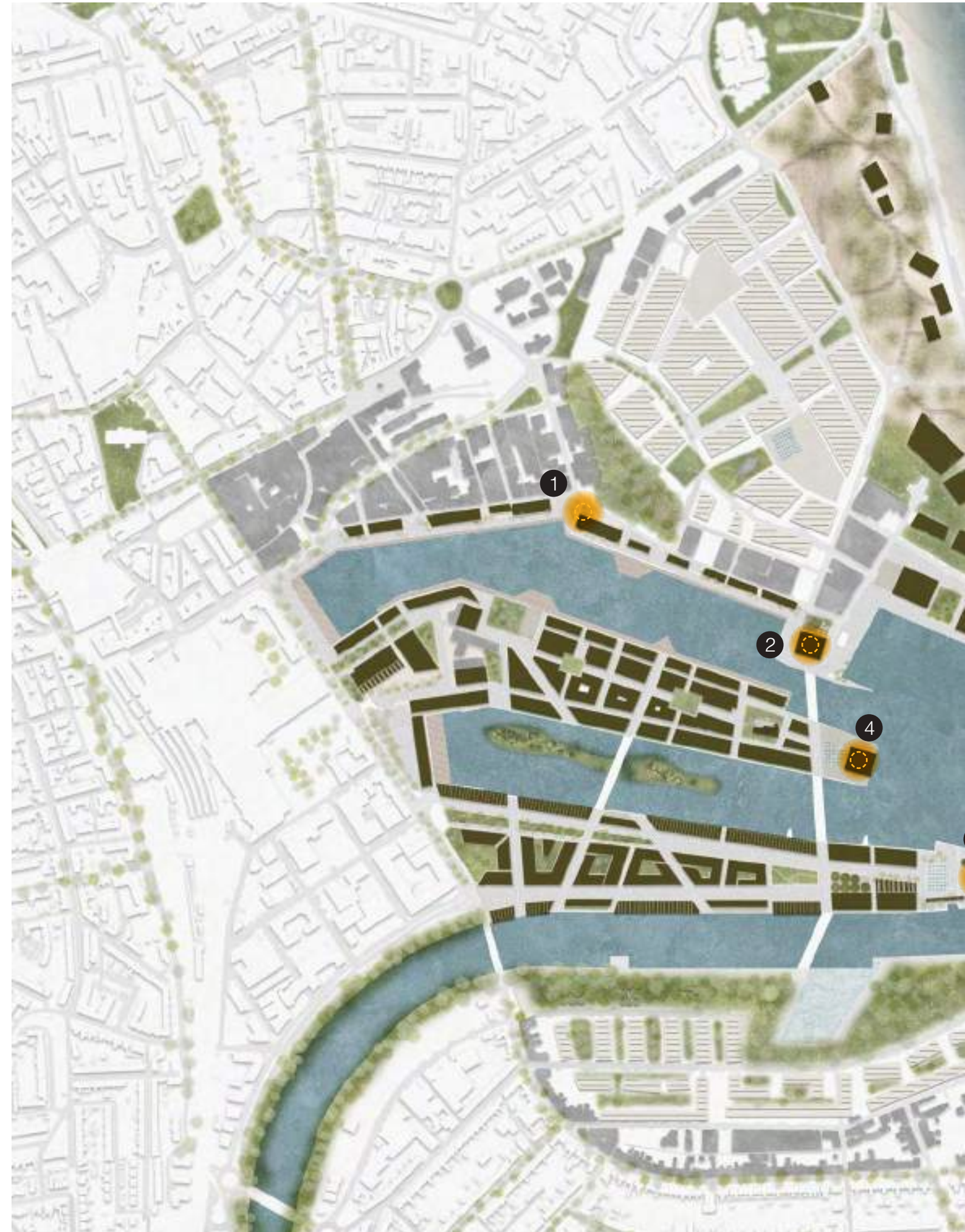
5 Riverfront Nature Walk

Proposed Plan

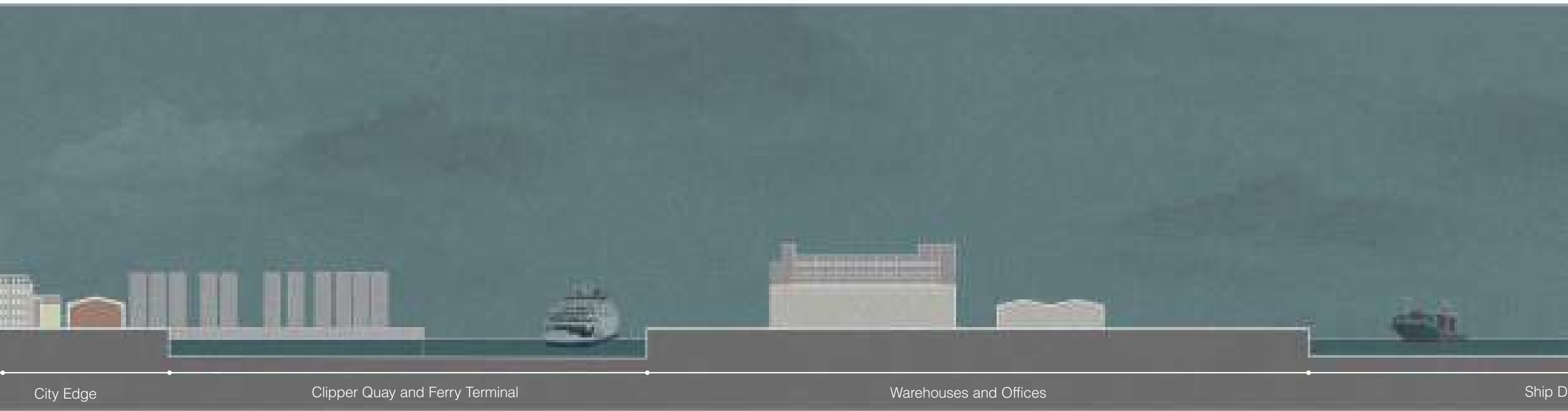
A new heart for Aberdeen

- 1 Daria - Art Gallery Space
- 2 Saffron - Prints Studio / School
- 3 Becca - Music School
- 4 John - Museum
- 5 Gabriella - Wellbeing Centre
- 6 Hera - Training Centre

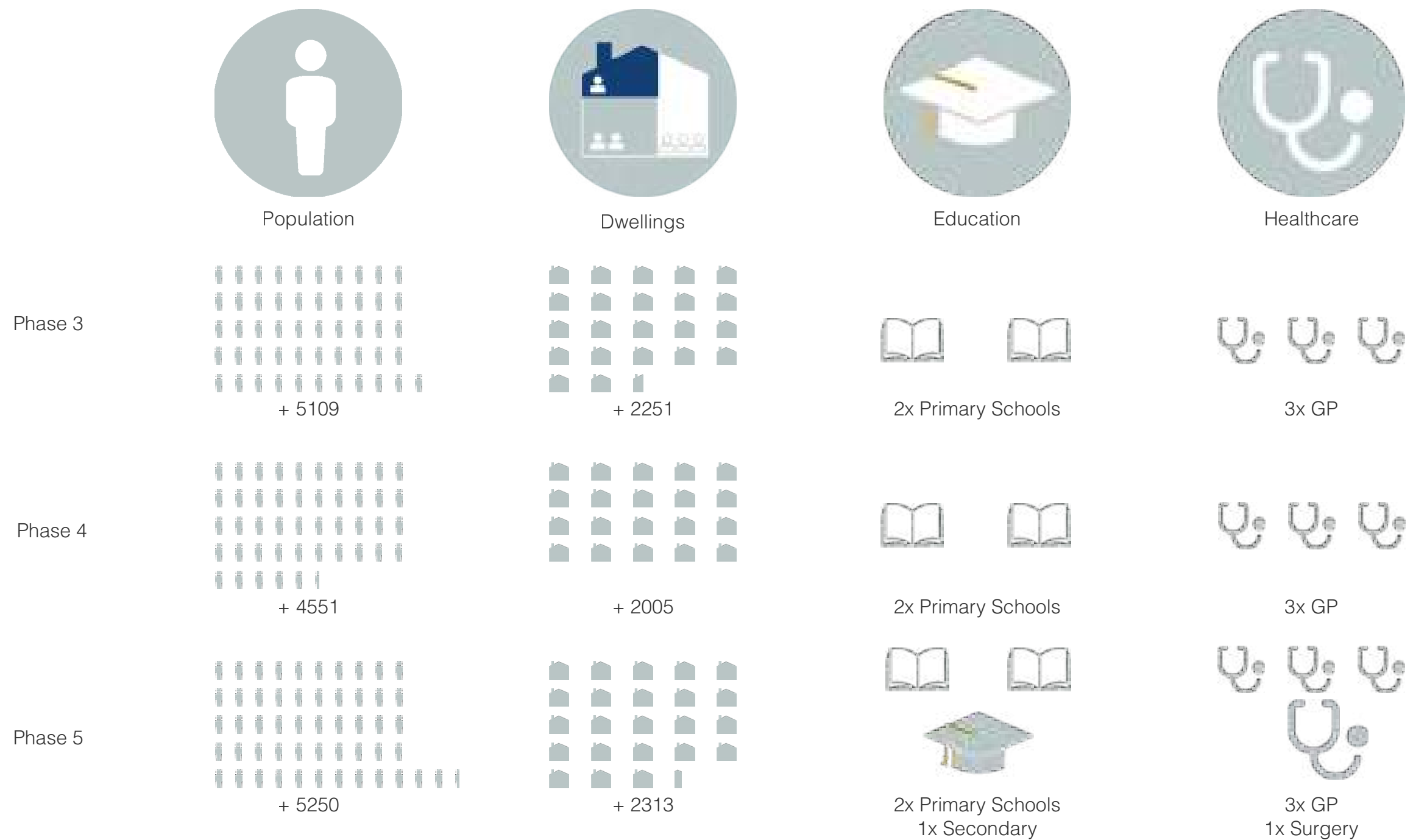
This new heart for Aberdeen will act as a catalyst and anchor for future growth whilst the City transitions into new industries. Supporting this will be our projects in the coming semester, centred around enriching a new vibrant urban neighbourhood adjacent to the city proper.











Building Use Class

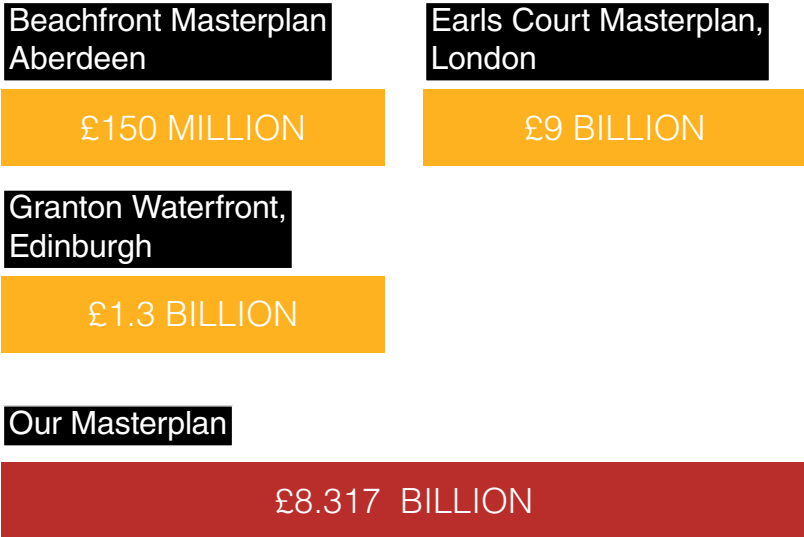






Viability

Our masterplan represents a major opportunity for Aberdeen’s regeneration, directing significant investment into upgrading existing infrastructure, delivering new strategic interventions, and modernising the site’s energy network. While comparable in cost to major schemes such as London’s Earls Court, this proposal uniquely embeds substantial infrastructure investment, delivering wider and longer-term economic benefits. By prioritising transport, public realm, flood resilience and future-ready energy networks, the project embeds public benefit within private development. It stimulates construction and knowledge-based employment, strengthens the city’s energy-transition economy, and reconnects the waterfront, generating sustained economic and social value beyond the site itself.



Our Masterplan involves introduction of a new tram and railway links, which makes it more expensive than others.

	CARBON KG CO ₂ EQ/SQM	COST (£)	PROFIT (£)
LOCALE: DEMOLITION	3 089 340	261 049 230	
LOCALE: BUILDINGS	573 405 505	5 659 359 540	2 670 240 960
RETROFIT	3 266 640	102 173 240	67 474 264
NEW-BUILT	573 405 505	5 659 359 540	2 670 240 960
INFRASTRUCTURE	335 639 575	29 369 621 656	£2.737 BILLION
LINKING THE GOLF COURSES	7 081 096	18 835 715	
TRAMS SYSTEM	271 904 305	219 635 444	
RAIL SYSTEM EXTENSION	47 761 520	88 467 313	
OTHER	77 477 750	425 530 000	
BRIDGES	82 800 00	41 400 000	
ENERGY	5320000	4 597 600	
PUBLIC REALM & LANDSCAPE	8 152 000	36 607 500	
TOTAL	917 251 080	£8.317 BILLION	

Financing

The masterplan is funded through a balanced mix of public and private investment, reflecting Aberdeen's shift away from oil dependence. Public bodies, led by Aberdeen City Council, Transport Scotland, the Scottish Government, the Heat Network Fund, Creative Scotland, and the city's universities, provide core funding, policy alignment, and long-term stability. Private partners add targeted capital and expertise: FirstGroup and Stagecoach support transport; Bancon Group contributes to housing; Harbour Energy, Equinor,

and the Net Zero Technology Centre strengthen the energy-transition infrastructure; and Aberdeen Inspired supports cultural initiatives. Additional community-focused investment comes from organisations such as Aberdeen Football Club.

	TRANSPORT	HOUSING	ENERGY	CULTURE	UNIVERSITY
	ABERDEEN CITY COUNCIL				
PUBLIC	TRANSPORT SCOTLAND	THE SCOTTISH GOVERNMENT	HEAT NETWORK FUND (HNF)	CREATIVE SCOTLAND	UNIVERSITY OF ABERDEEN
			UK GREEN HEAT NETWORK FUND		RGU
PRIVATE	FIRSTGROUP	BANCON GROUP	HARBOUR ENERGY	ABERDEEN INSPIRED	NET ZERO TECHNOLOGY CENTRE
	STAGECOACH NORTH SCOTLAND	FIRST ENDEAVOUR LLP	EQUINOR		HARBOUR ENERGY
OTHER	ABERDEEN FOOTBALL CLUB				

Sustainable Design

UN Goals and BREEAM targets

BREEAM targets have been addressed alongside the UN Sustainable Development Goals. Certain goals have been excluded where they catered for the construction industry.

A selection of proposed strategies is outlined below to illustrate how these targets are being met.



BREEAM: Delivery of assets that are resilient and use resources efficiently to reduce operational costs to built environment

Proposal: Alleviate fuel poverty by designing residential complying to Passivhaus Standard, and provide affordable transport mode



BREEAM: Encourages the provision of comfortable, healthy and safe internal and external environments for asset users and other within the society

Proposal: Bonding with nature via board walks, promenade designed to accommodate cycling and water activities, and addition of GP



BREEAM: Provision of education and training opportunities to local communities

Proposal: Training centres and schools provided at the locale, making sure opportunities available for all



BREEAM: Provision of energy efficient systems, minimise energy consumption and carbon dioxide emissions, whilst maintaining a reliable supply

Proposal: Site is powered by wind turbines, supplemented by on-site solar energy generation



BREEAM: Create opportunities within locale area for upskilling and professional training

Proposal: The masterplan will generate a diversity of jobs in the retail, energy, and transport sectors



BREEAM: Promotes delivery of sustainable and resilient buildings and infrastructure, improve sustainability performance of assets

Proposal: Design start-up zone with innovative environmental features such as chilled beams



BREEAM: Encourage measures that results positive social impact

Proposal: Liberating waterfront so it is accessible to all, implementation of communal parks provide public spaces across the city.



BREEAM: Promote sustainable transport, affordable safe housing, flood resilience, reduced air pollution, and access to public and green spaces

Proposal: Green edge and parks retain urban sprawl, locale is transformed to regenerative neighbourhood



BREEAM: Sustainable procurement and use of construction materials by recognising materials that are reused or recycled.

Proposal: Reuse de-commissioned oil rigs, demolished warehouse parts. Waste management is considered



BREEAM: Sustainable use of energy, minimising greenhouse gas emissions, climate change adaptation and flood resilient measures

Proposal: Flood resilient measures at river, beachfront and city



BREEAM: Prevention and control of pollution and surface water runoff associated with the location and use of assets

Proposal: Tree shading lowers river temperature, and reducing chemical runoff improves water quality to preserve habitats.



BREEAM: Sustainable land use, protection and creation of ecological features. Improvement of long term biodiversity for asset sites and surrounding land

Proposal: Wetland cultivation, green corridor linkage and implementation of parks improve and create habitats for life on land

Reflection

The Journey & Further Opportunities

The fundamental aim for the masterplan is to create a vibrant new heart of Aberdeen, revive the long-lost connection with the waterfront, reinstate the legacy of Aberdeen’s rich industrial heritage and support the energy transition towards renewables. The masterplan envisions an inclusive, sustainable environment for both residents and visitors.

By densifying the city centre by offering affordable housing, providing local employment opportunities and activating the waterfront for recreation, the proposal positions the Locale as a catalyst for urban regeneration. It also celebrates the potential of the strategic location of the locale to bridge the north and south of Aberdeen, strengthening ecological and social connectivity.

Reflecting on the journey, as a group, we found the experience both challenging and rewarding. Learning a city in depth and working at a city scale offered a rare and valuable experience in our architectural education. Despite our differences, our varied interests and strengths are celebrated in the collaborative groupwork environment, resulting in a proposal shaped by diverse perspectives.

We love our locale. It is a place layered with history, memories and identity. It has brought prosperity to generations of Aberdonians. We hope our proposal does it justice by reimagining a waterfront that feels welcoming, people-centred, and resilient to the challenges ahead.



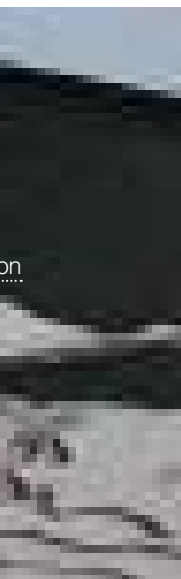
1 Innovation Hub

The Innovation Hub is envisioned as a major catalyst for the energy transition, attracting research activity and investment. Its integration with the proposed transport hub is therefore essential and we will continue to explore the users journeys from the transport terminal. The public realm will be designed with a mix of hard and soft landscaping to support a range of activities and foster a lively environment. With the addition of student accommodation, this emerging district is poised to become the vibrant heart of research, learning, and innovation within the masterplan.



2 Residential Area - Torry

The frontage facing Torry requires careful consideration. A new transport hub is proposed to better integrate the area with the rest of Aberdeen through accessible public transport. A series of pocket greens will be introduced along the waterfront to offer much-needed recreational areas for the community. Additionally, a clear hierarchy of access points will be established to organise movement, improve pedestrian safety, and ensure that the riverfront becomes a more inclusive and welcoming edge for Torry residents.



3 Beyond Phase 5 - Building on Floating Islands

consideration. A
ate Torry with
transport. A
the frontage
community.
oints will be
permeability,
nclusive and

As an opportunity identified during the final review, the masterplan proposes exploring the introduction of additional Dutch waterfront typologies on an island beyond Phase 5. This expansion aims to deepen public engagement with the water while providing flexible housing options to accommodate Aberdeen's growing population. In doing so, the later phases of the masterplan continue to reinforce a water-oriented urban identity and support adaptable living environments for emerging community needs.



Appendix

	General					Carbon			Sequestra	
	Length (m)	Width (m)	Area (per unit / m²)	Quantity	Area (Total)	kg CO₂ / m²	Total CO₂		tion (50	To
									years)	
Demolition										
Building area					154467		20		3089340	
Locale: Buildings										
Retrofit									3266640	
Commercial					36296		90		3266640	
Proposed					1436386				573405505	
Commercial					246075		400		98430000	
Residential					509113		250		127278250	
University					633875		500		316937435	
Arts / Culture					47323		650		30759820	
Infrastructure										
Linking the Golf Courses									7081096	
Tarmac removal	14162.2	12	169946.3				20		3398926.08	
Fence removal	14162.2	1	14162.2				20		283243.84	
Greenery	14162.2	12	169946.3				20		3398926.08	
Trams / Bus System									335,693,575.20	
Tram									271904305.2	
Aberdeen Hrbour to Blackdog Mobility hub	12874.7	7			90123.04		1000		90123040	
Portlethen to Aberdeen harbour	11555.1	7			80885.4284		1000		80885428.4	
Aberdeen harbour to Westhill Mobility hub	14162.3	7			99135.8368		1000		99135836.8	
Tram Stations			200	22	4400		400		1760000	
Railway									47761520	
Railway to the airport leg	3218.7	14			45061.5		1000		45061520	
Railway stops			1,500	3	4500		600		2700000	
Other									7747750	
Building Mobility Hubs (20x40; 4 storey)				4	800		600		480000	
Guided Bus Infrastructure	4153	7			29071		250		7267750	
Bridges									8280000	8
	1	72	20	1440			1000		1440000	
	2	108	20	2160			1000		2160000	
	3	104	20	2080			1000		2080000	
	4	130	20	2600			1000		2600000	
Freight siding redeveloped into cycling	3000	4	12000							
Energy										
Ambient Loop and Infrastructure connected					53200		1000		53200000	
Public Realm & Landscape									8152000	
Trees			1100/ha	27225						
Soft Landscaping			222800				20		4456000	
Hard landscaping			61600				60		3696000	

Total	Cost						Profit (+10%)	
	Land (£)/m2	Land value	Percentage	Built (£) /m2	Built Value	Total	Land (£)	Value
	£1,690.00					£261,049,230.00		
								£67,474,264.0
	£1,690.00	61340240		£1,125	£40,833,000	£102,173,240	£1,859.0	£67,474,264.0
		£2,427,491,782				£5,012,985,988		£2,670,240,960.53
	£1,690.00	£415,866,750		£1,800	£442,935,000	£858,801,750	£1,859.0	£457,453,425.00
	£1,690.00	£860,400,970		£1,800	£916,403,400	£1,776,804,370	£1,859.0	£946,441,067.00
	£1,690.00	£1,071,248,530.30		£1,800	£1,140,974,766	£2,212,223,296	£1,859.0	£1,178,373,383.33
	£1,690.00	£79,975,532		£1,800	£85,181,040	£165,156,572	£1,859.0	£87,973,085.20
								£2,737,715,224.53
		£2,910,596,491.76						
		18835715.36						
	10	1699463.04						
	10	141621.92						
	100	16994630.4						
		£2,369,621,656.40						
		£1,894,530,136						
	£7,000	£630,861,280						
	£7,000	£566,197,999						
	£7,000	£693,950,858						
	£800	£3,520,000						
		£49,561,520.0						
	£1,000	£45,061,520.0						
	£1,000	£4,500,000.0						
	£	425,530,000.00						
	£	800.00	£		384,000,000.00			
	£	10,000.00	£		41,530,000.00			
3280000		£			42,450,000.00			
	£	5,000.00	£		7,200,000.00			
	£	5,000.00	£		10,800,000.00			
	£	5,000.00	£		10,400,000.00			
	£	5,000.00	£		13,000,000.00			
	£	350.00	£		1,050,000.00			
		£			4,597,600.00			
	£	700.00	£		4,597,600.00			
		£			31,162,500.00			
	100	£			2,722,500.00			
	100	£			22,280,000.00			
	100	£			6,160,000.00			

